

SUMNER – BONNEY LAKE SCHOOL DISTRICT BONNEY LAKE HIGH SCHOOL EXPANSION VALUE ENGINEERING REPORT NOVEMBER 19TH, 2025

Updated with Implementation Data 1/8/2026



Value Engineering Team

Timothy Buckley, AIA, CVS, LEED AP BD+C, Team Facilitator

Preeti Pustake, Cost Estimator / Construction Management

Mark Van Wormer, PE, Civil / Landscape

Melissa White, PE, Structural

Brett Cato, LEED AP, Architectural

Doug Smith, PE, MBA, VMA, CCP, LEED AP, cSBA, Mechanical

Carrie Taylor, Electrical

Sarah Partap, MBA, VMA, VE Project Manager

Kara White, VE Project Coordinator, Co-Facilitator

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I. EXECUTIVE SUMMARY

INTRODUCTION

This value engineering (VE) report is presented to the Sumner-Bonney Lake School District and the design team to support decision making at the conclusion of the schematic design phase for the Bonney Lake High School Expansion project.

The goals for this study were to review the current design concepts and identify potential opportunities for design cost effectiveness and efficiency. The VE team sought to identify site development, planning, and building system alternatives that may offer first cost or life cycle cost benefits and/or improve project quality and reduce construction risks.

Value Engineering Team and Process

This value study followed the SAVE® International eight-phase Value Methodology and Job Plan. The Value team was comprised of a multidisciplinary group of subject matter experts and was led by a Certified Value Specialist facilitator. The team included professionals representing the Architectural, Civil, Structural, Mechanical, Electrical, and Cost Estimating/Construction Management disciplines.

At the initial kickoff meeting, the District and the design team presented their project requirements and basis of planning and design. Following the kickoff, the VE study team worked together using the formal value methodology. The essential and secondary functions of the project components were identified along with their associated costs, design alternatives were generated, and the most viable alternatives were further developed.

SUMMARY:

VALIDATION OF PROGRAM AND DESIGN ELEMENTS

The Value Methodology is not a cost-cutting exercise but is about finding opportunities to improve overall value. It is also useful for Districts and design teams in validating decisions that have been made by the District and design team. In the process of comparing alternative concepts with the current design, the VE team noted the following planning and design components, as well as owner project requirements merit strong continued support:

- Location of the addition footprint, and 3-story footprint
- Accommodating plaza and well-defined primary entrance
- Expressed entry awning providing significant cover with transparency over the "Welcome Center"

- Expressed entry stair “beacon” (great visibility from the road and middle school)
- Efficient structure and plan area

Value Engineering Proposals

Key proposals include:

- Stormwater strategy
- Enhanced site accessibility and circulation
- Enhanced independent study
- Improved energy efficiency
- Power transformation

Success of the formal VE process is not merely measured in terms of the value of cost reductions, but rather in the accepted implementation of all VE proposals and their contributions toward performance improvements in the project as a whole. In recognition of such additional benefits and to assist Washington State in representing the results of its VE program to the state legislature, performance measures have been developed and standardized by the Cascadia Chapter of SAVE International. The following table summarizes the VE team's proposals relating to these performance measures:

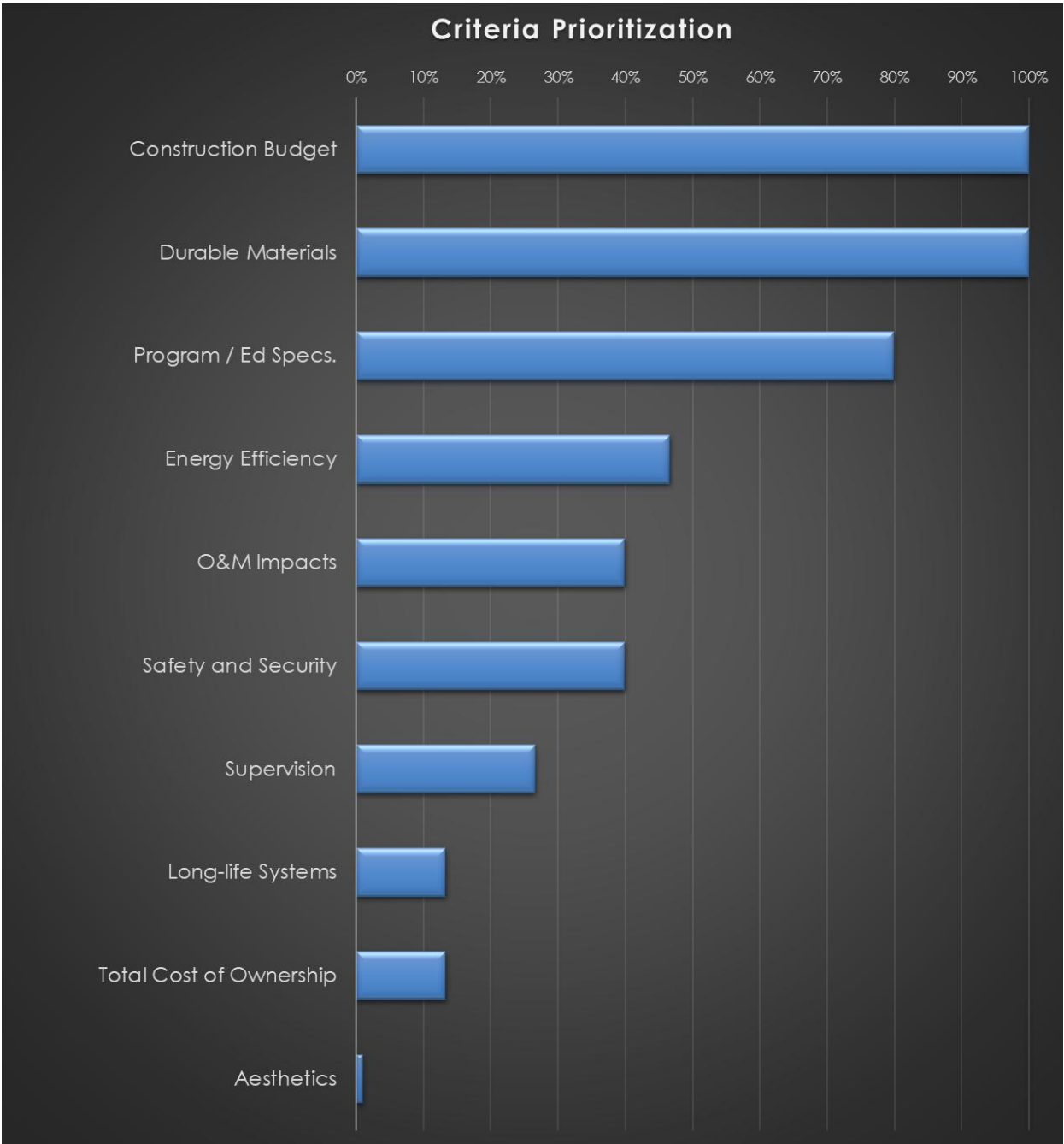
Performance Measures	Number of Proposals
Program	1
Aesthetics	5
Facility Preservation	1
Total Cost of Ownership / LCCA	8
Environmental Sustainability	8
Schedule	3
Constructability	4
Occupant Comfort, Safety & Performance	3

CRITERIA PRIORITIZATION

Members of the Sumner Bonney Lake School District leadership were led through a criteria prioritization exercise. This exercise identified criteria that are important to measure the success of the project and allows stakeholders to vote on which criteria are most important or of lower priority to them.

Criteria	Definition
Aesthetics	Prominent front entrance, general appearance, expression, welcoming.
Construction Budget	First cost (including development and construction costs).
Durable Materials	Long-life materials that will withstand the test of time.
Energy Efficiency	Improves energy performance of the facility.
Long-life Systems	Systems that will last a long time (but may come at a first-cost premium).
O&M Impacts	Support of ongoing operations and maintenance, including maintenance cost.
Program / Ed Specs.	Strict adherence to the program or Educational Specifications.
Safety and Security	Design for a safe and secure environment.
Supervision	Design for "eyes on the street" by administration and staff.
Total Cost of Ownership	The total long-term / life cycle cost.

The following criteria prioritization graph shows how the various project criteria were weighted by the stakeholders. The VE team uses this graph when evaluating and developing proposals to gauge the performance of proposals against project goals.



II. PROJECT DESCRIPTION

PROJECT INFORMATION

Cost:	Addition	\$19,314,491	\$440.87/SF
	Renovation	\$4,138,961	\$182.17/SF
	Site	<u>\$9,192,253</u>	
	Total	\$32,645,706	
Size:	New	41,905 SF	
	Renovation	16,782 SF	
Location:	10920 199 th Ave Ct. East Bonney Lake, WA 98391		
Schedule:	Construction	26 months (June 2026 – September 2028)	
Delivery Method:	GC/CM		
Building Construction Type:	II-B, Fully Sprinklered		
Occupancy:	Education	Students:	1,600+

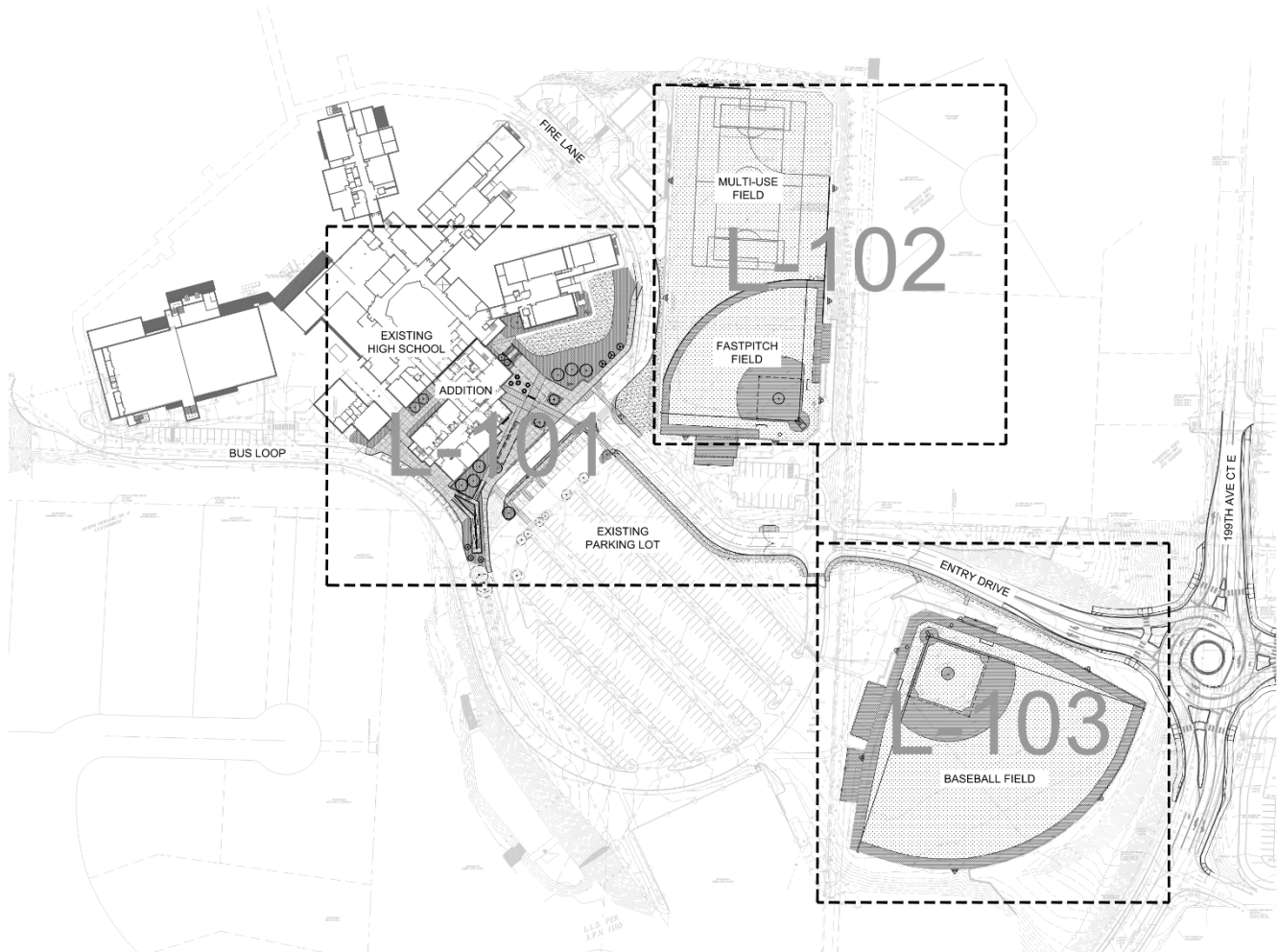
Description (transposed from BLHS Schematic Design Booklet, Dated 10/27/25)

The Bonney Lake High School Expansion project is designed to relieve severe overcrowding at a campus originally built for 1,260 students but is now serving more than 1,600. The addition and renovation will provide twelve new classrooms, expanded administration areas, and new program space for the Developmental Learning Center, bringing displaced programs and portables back into the main building.

The design creates a new main entrance that offers a stronger identity for the school and improved wayfinding, while emphasizing safety, flexibility, and efficient circulation throughout the site. Other key upgrades include improved traffic flow on 199th Avenue Court East, new pedestrian pathways, modernized signage and exterior branding, and field improvements with new turf and lighting.

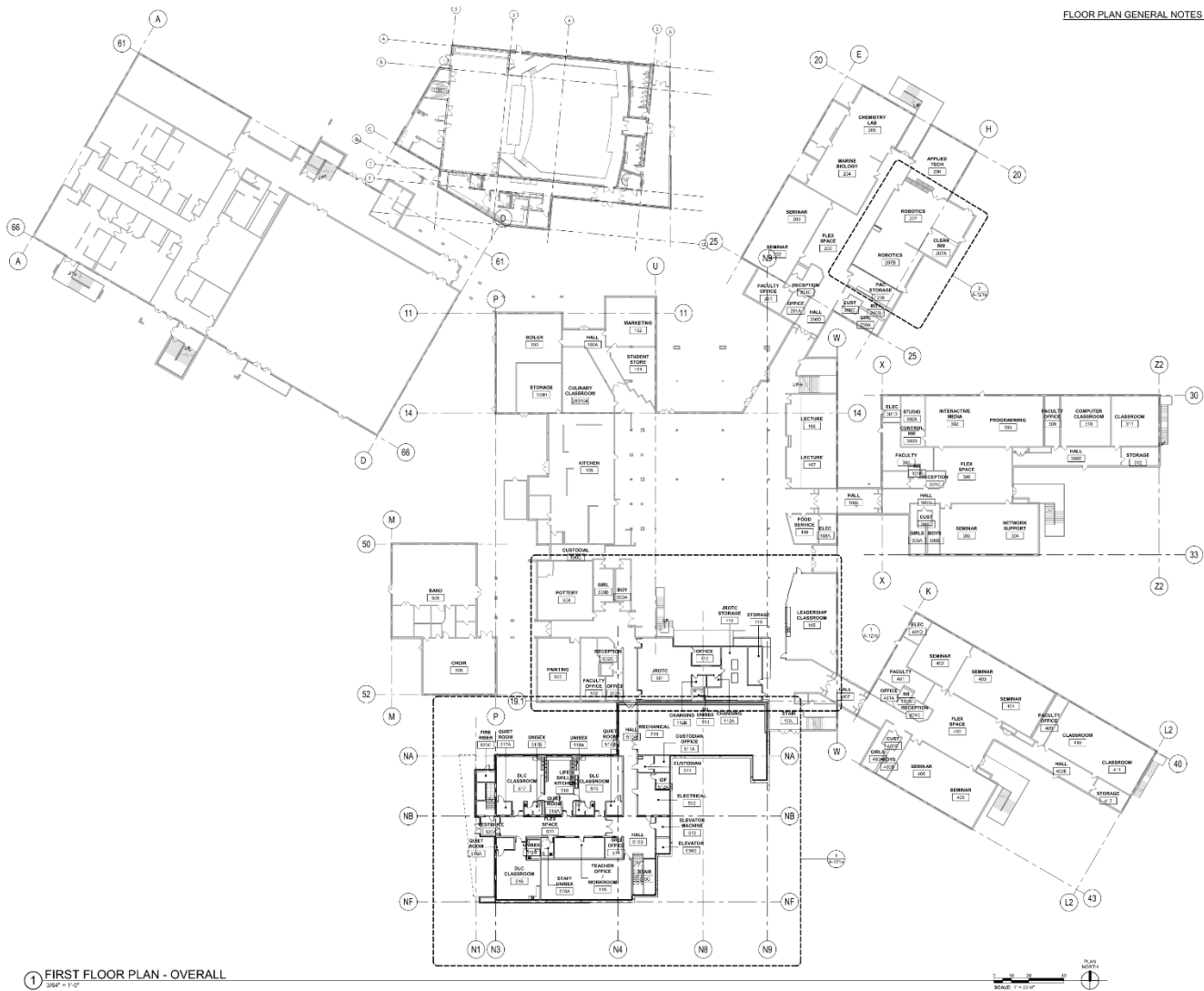
The project will be delivered under the GCCM method and constructed in phases on an occupied site to maintain school operations throughout.

SITE PLAN

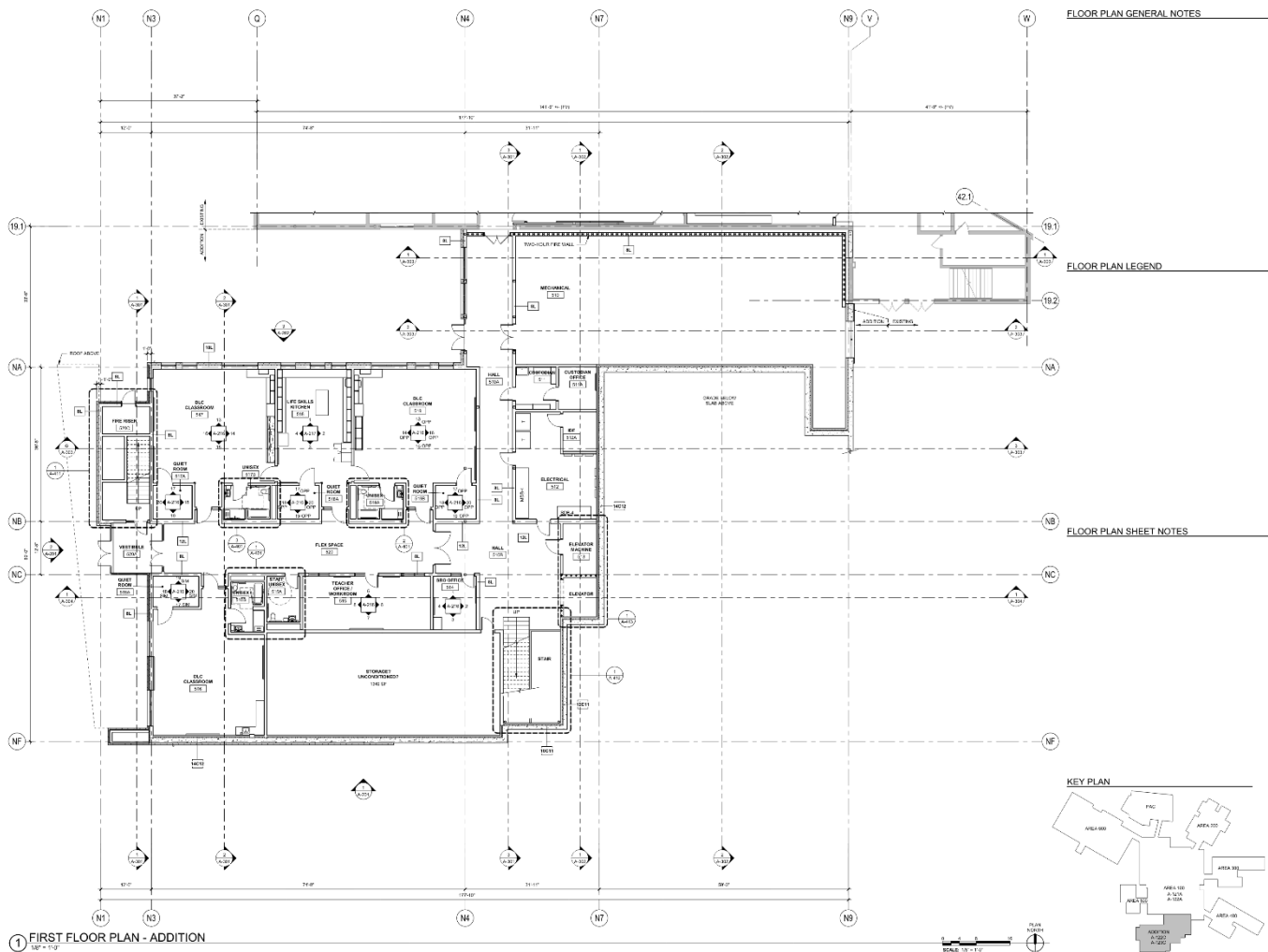


FLOOR PLANS

First Floor

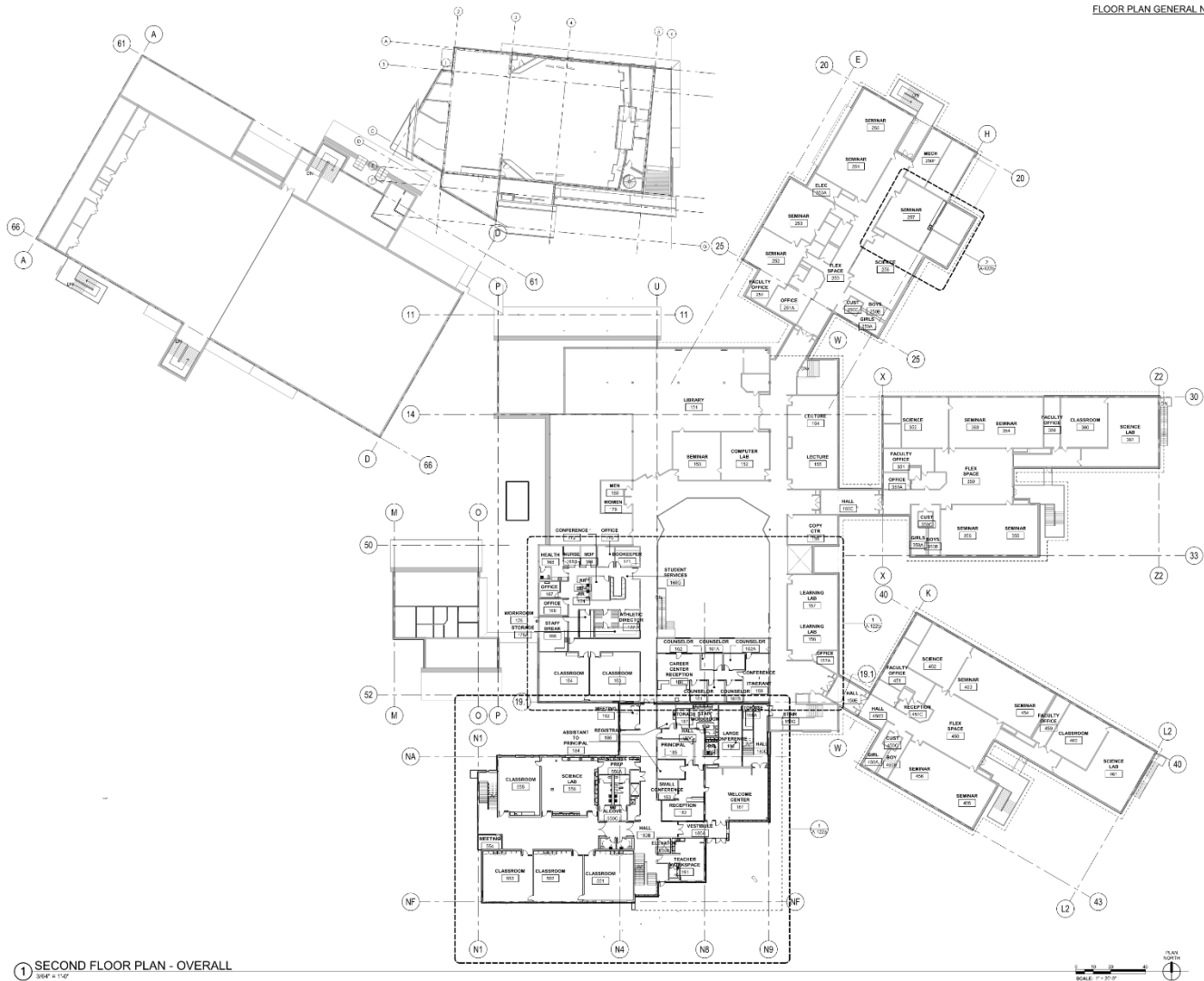


First Floor Addition

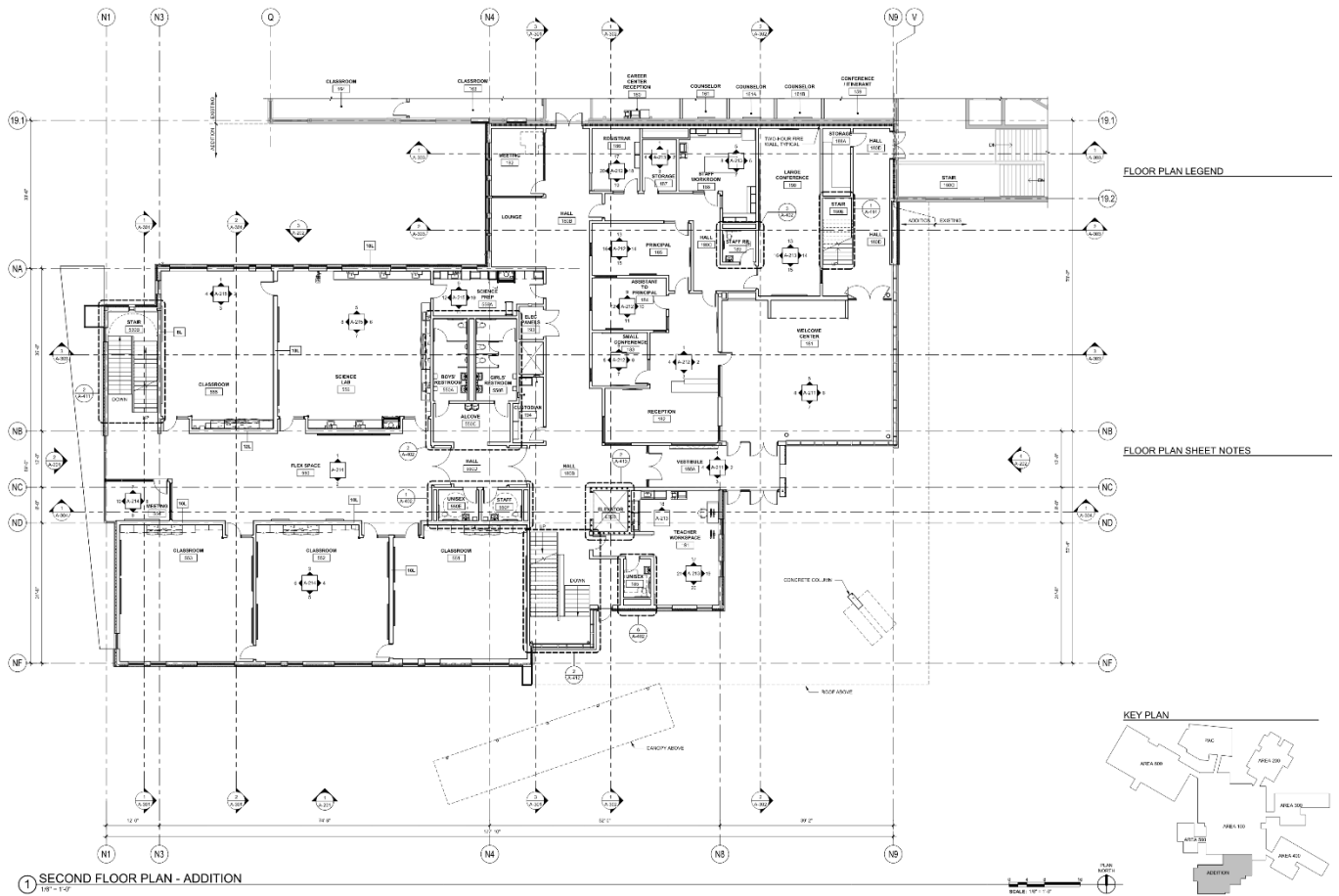


Second Floor

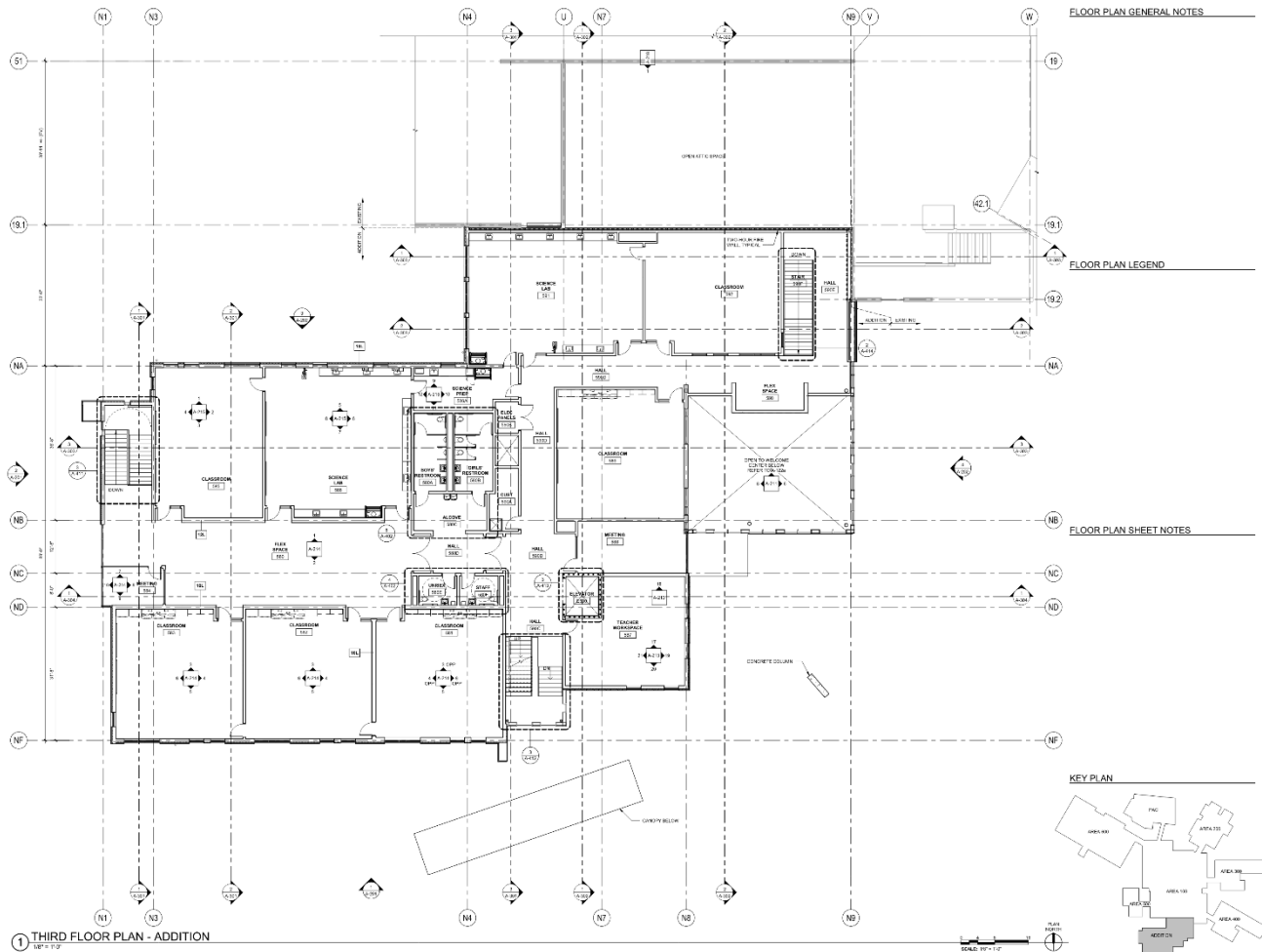
FLOOR PLAN GENERAL NOTES



Second Floor Addition



Third Floor Addition



RENDERING



III. VALUE ENGINEERING PROPOSALS

The following section presents the developed VE proposals. Each proposal describes the current concept, then compares it to the VE concept. Order-of-magnitude cost estimates are included for each alternative, comparing the current design to the estimated VE concepts.

To evaluate proposals beyond simply budget considerations, the VE team ranked each proposal based on how well it addresses Sumner-Bonney Lake School District's ten identified priorities. Proposals were also evaluated for functional performance (how well the proposal meets its intended function) and implementation practicality (the relative difficulty of implementation).

In addition to the formal VE proposals (with expanded summary, backup and cost estimates), the value team has also included a number of line-item "Revision Proposals" indicating various additional opportunities to improve value.

Not all proposals are first cost savings or "Quantitative" ideas. Some are "Qualitative" options that may add first cost, but add value to the project with potential life cycle cost savings over time or address other quality and functional improvements.

ID	Name	Current Cost	VE Proposal Cost	Difference	Criteria Scored Ranking	Note*	LCCA Difference
C1	Storm Vault	\$190,000	\$97,000	\$93,000	4		
C2	ADA Ramp	\$197,000	\$274,000	(\$77,000)	6		
S1	Floor Framing	\$572,000	\$352,000	\$220,000	8		
S2	Lateral System	\$145,000	\$173,000	\$199,000	9		
A1	Stair Tower	\$175,000	\$91,000	\$84,000	10		
A2	Study Nooks	\$4,000	\$15,000	(\$11,000)	5		
M1	Integrated Design	\$8,224,000	\$8,015,000	\$209,000	2	LCCA	\$418,000
M2	Thermal Generation	\$3,625,000	\$910,000	\$2,715,000	3		
M3	Energy Source	\$3,625,000	\$3,508,000	\$117,000	1	LCCA	\$409,000
E1	Transfer Power	\$38,000	\$87,000	(\$49,000)	7		
E2	Store Power	\$152,000	\$197,000	(\$45,000)	12		
E3	Electrical Service	\$220,000	\$94,000	\$126,000	11		

SUBTOTAL VE PROPOSALS	\$3,581,000
SUBTOTAL QUANTITATIVE* VE PROPOSALS	\$3,763,000

* Quantitative Savings Per US Army Corps of Engineers VE Standard tally excludes Qualitative (cost add) proposals

NIC – Indicates Not Included in totals (cost or portions are included in a different idea)

LCCA – Indicates Life Cycle Cost Analysis savings potential

REVISION PROPOSAL ITEMS

No.	Description	ROM Difference	Note*
R1	Add traffic control at two intersecting driveways in front of school (stop signs and paint).	(\$8,000)	
R2	Designated carpool only parking.	(\$5,000)	
R3	Place excavated soils from the addition basement at the two sports fields; place the sports field drainage network at existing grade and cover with the excavated basement soil.	\$180,000	
R4	Spread excess earth on sports field (no haul off site and dump).	\$40,000	
R5	Add more seating in plaza entrance area.	(\$15,000)	
R6	Provide smart irrigation controls with weather sensors.	(\$30,000)	
R7	Temporary irrigation in beds ILO permanent.	\$15,000	
R8	Drip irrigation systems ILO high volume heads.	(\$25,000)	
R9	Install tree bags for new tree establishment irrigation ILO permanent irrigation system.	\$10,000	
R10	Add banner supports to light poles along the entry drive.	(\$5,000)	
R11	Add storage sheds at ball fields.	(\$50,000)	
R12	Add prefab concession stand at baseball field + hydration station (for parity with SHS fast pitch).	(\$150,000)	
R13	Bid drainage and artificial turf fields as separate ballfield add alternates.	(\$1,884,383)	
R14	Eliminate landscape irrigation booster pump (see L3.05).	\$20,000	
R15	Truncated dome and bollards ILO curb at bus drop for enhanced barrier free access.	(\$35,000)	
R16	Provide 1 parent drop zone at DLC.	(\$45,000)	
R17	Steel frame and storefront ILO curtain wall.	\$50,000	
R18	Omit holes in roof overhang for the planters.	\$10,000	
R19	Consider TK Elevator or Skyline systems elevator as cost effective options (see https://www.primariuselevator.com/).	\$50,000	
R20	Reduce window-to-wall ratio from 19% to 15% to reduce heating & cooling load.	\$100,000	

No.	Description	ROM Difference	Note*
R21	Replace metal panels at accessible areas and use brick for improved impact durability.	(\$50,000)	
R22	Channel ILO downspout (see BCRA Olympia example on 2-story).	(\$10,000)	
R23	Add roof transfer ladder over the 2-hr parapet.	(\$8,000)	
R24	Reconfigure roof to eliminate large cricket on west end and allow single sheet flow roof plane.	\$25,000	
R25	Install skylight cap over opening at entry precast concrete (thwart rain staining of expressed element).	(\$16,000)	
R26	Project images on the plaza under the main entry canopy.	(\$10,000)	
R27	Install 3M or other security film on all glass within accessible areas to protect from scratch, acid, other graffiti, or breakage.	(\$25,000)	
R28	Use exposed concrete ILO rubber flooring at stair treads.	\$18,000	
R29	Remove at-grade exit door at main stair in hall 180B (is not required and adds security risk).	\$5,000	
R30	Provide mechanical equipment access panels in both drop and hard-lid ceilings. Sesame Exposed Grid Ceiling Hatches are a proven product.	(\$20,000)	
R31	Enhance acoustics of teacher prep room assemblies for better confidentiality and less disruption.	(\$22,000)	
R32	High-efficiency equipment and fixtures.	(\$120,000)	
R33	Open-source HVAC controls ILO sole-source (Honeywell).	\$70,000	
R34	Non-metallic (PEX) ILO metallic hydronic distribution run-out piping.	\$30,000	
R35	Fan coil units in corridor ceiling space ILO classroom ceiling space to facilitate maintenance.	\$15,000	
R36	Semi-ducted ILO fully-ducted ventilation (DOAS) air distribution (transfer grill to corridor).	\$100,000	
R37	High-efficiency DOAS ventilation system ILO code minimum (C406).	(\$75,000)	LCCA
R38	High-efficiency air-source heat pumps ILO code minimum at central plant.	(\$150,000)	LCCA
R39	High-efficiency science room fume hoods with reduced exhaust air requirement.	(\$20,000)	LCCA
R40	Increased ventilation (by 30%) throughout ILO code minimum ventilation.	(\$75,000)	

No.	Description	ROM Difference	Note*
R41	Pre-programmed special ventilation modes including: 1) Airborne contagion (more ventilation), 2) Wildfire smoke (less ventilation), and 3) Attack (no ventilation).	(\$30,000)	
R42	New small generator just for addition emergency lighting and other loads including MDF room etc. Leave existing to continue to serve existing building.	(\$80,000)	
R43	Ultra-violet (UV) air treatment to eliminate indoor airborne contagion.	(\$40,000)	
R44	Radiant floor heating at main entry/welcome center public space.	(\$60,000)	
R45	Velux or similar auto venting skylights (with integral wind/rain lockout controls) for fresh air.	(\$25,000)	
R46	Energy dashboard at main entry lobby for education.	(\$20,000)	
R47	Use flat plate fixed core heat recovery ILO heat wheel for air quality (avoid cross contamination).	(\$40,000)	
R48	Use demand-controlled ventilation (DCV): Integrate CO ₂ sensors to adjust fresh air intake based on occupancy, reducing energy use.	(\$50,000)	LCCA
R49	Two of four air-source heat pumps to be simultaneous heating & cooling capable ILO just one, for increased redundancy.	(\$30,000)	
R50	Provide fast-acting air flow control valves (dampers) to science rooms to match performance of the fume hood and general exhaust fast-active valves (dampers).	(\$25,000)	
R51	PEX ILO copper water pipes.	\$35,000	
R52	Non-metallic (ABS or PVC) drain, waste & vent (DW&V) piping ILO cast-iron (wrap with carpet pad for sound deadening).	\$45,000	
R53	Eco-power faucets (charge hands-free lav. battery by water pressure in-line micro turbines).	(\$12,000)	LCCA
R54	Relocate restrooms and other plumbing fixtures for back-to-back or immediately adjacent space configuration to simplify the plumbing system and reduce cost.	\$22,000	
R55	Provide automatic trap primers at all floor drains.	(\$5,000)	
R56	Aluminum feeders ILO copper.	\$10,000	
R57	Re-routing of power conduits outside in lieu of through kitchen area and avoid internal disruption	(\$30,000)	

No.	Description	ROM Difference	Note*
	with pull boxes, (2) 4" conduits and (2) 2" conduits, and avoid risk and fit and conflicts inside. Finish with painted sheet metal on exterior to hide utilities.		
R58	Add PV on covered walkways and canopies in addition to the building roof .	(\$125,000)	
R59	Add low pedestrian lights in ramp stairway area.	(\$20,000)	
R60	Increase solar panel energy generation by 20%.	(\$70,000)	
R61	Show future PV zone and provide capped conduit pathways routed to space reserved for additional future inverters.	(\$20,000)	
R62	Add large format video/digital wall display in welcome area (like Sumner HS in commons but smaller).	(\$45,000)	
R63	Upgrade existing fire alarm system to all voice.	(\$120,000)	
R64	Add conduit to existing building intercom system.	(\$10,000)	

SUBTOTAL REVISION PROPOSALS	(\$2,860,383)
<i>SUBTOTAL QUANTITATIVE* REVISION PROPOSALS</i>	<i>\$850,000</i>

* Quantitative Savings Per US Army Corps of Engineers VE Standard (tally excludes Qualitative - cost add proposals)

NIC – Indicates Not Included in totals (cost or portions are included in a different idea)

LCCA – Indicates Life Cycle Cost Analysis savings potential

TOTAL PROPOSAL SUMMARY

GRAND TOTAL OF ALL PROPOSALS	\$720,617
<i>GRAND TOTAL OF ALL QUANTITATIVE PROPOSALS</i>	<i>\$4,613,000</i>

	PROPOSAL	C1
COMPONENT: Storm Vault	AUTHOR	MSV
BASELINE CONCEPT: Stormwater from the underground drainage system at the ballfields is collected and discharged to the existing storm pipe system. The existing 8" storm pipe system may not have the capacity to accommodate the additional drainage from the ballfields and could cause backups on the school grounds.		
VALUE PROPOSAL SYNOPSIS: Provide an underground storage tank that can store stormwater draining from the ballfields and discharge it to the existing storm pipe system at a slower, more acceptable rate in lieu of replacing the existing storm pipe to accommodate the anticipated flow volume.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Improved	Degraded	Improved
FUNCTIONS:		
Store Fluid	Collect Fluid	Control Fluid
ADVANTAGES: - Minimizes existing pipe replacement - Reduces flooding - Minimizes pavement repair - Minimizes disruption on campus		DISADVANTAGES: - Involves additional stormwater modeling - Requires tank maintenance - Requires additional control structures

DISCUSSION AND JUSTIFICATION

The planned synthetic ballfields located on the east side of the school site will have an underground drainage system. The existing detention pond located on the far west side of the school site is reported to be sized to accommodate full buildout of the school property. However, the existing storm pipe system may not have enough capacity to accommodate additional discharge from the planned ballfields. Additional stormwater detention may be needed at the ballfields to slow down the discharge rate to the existing conveyance system.

An underground tank with a control structure can be a good way to provide additional detention without obstructing activities on the ground surface. Higher level stormwater modeling will be needed to determine the detention volume needed at each ballfield and to determine the best control structure design to work in conjunction with the existing detention pond system. This approach will likely be more cost effective than replacing existing pipes and repairing the resulting disturbed surfaces within the school campus.

Also, weather events are becoming increasingly severe. It will be important to include provisions in the plan to monitor weather carefully and to provide additional Temporary Erosion and Sediment Control measures to address a potential major weather event.

SCHEDULE SAVINGS:

Maintained Schedule. This work should not be on the critical path.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Improved Reliability. Additional detention at the source will relieve capacity issues downstream.

Maintenance & Operations:

Degraded Maintenance and Operations. The additional detention systems will require more routine maintenance.

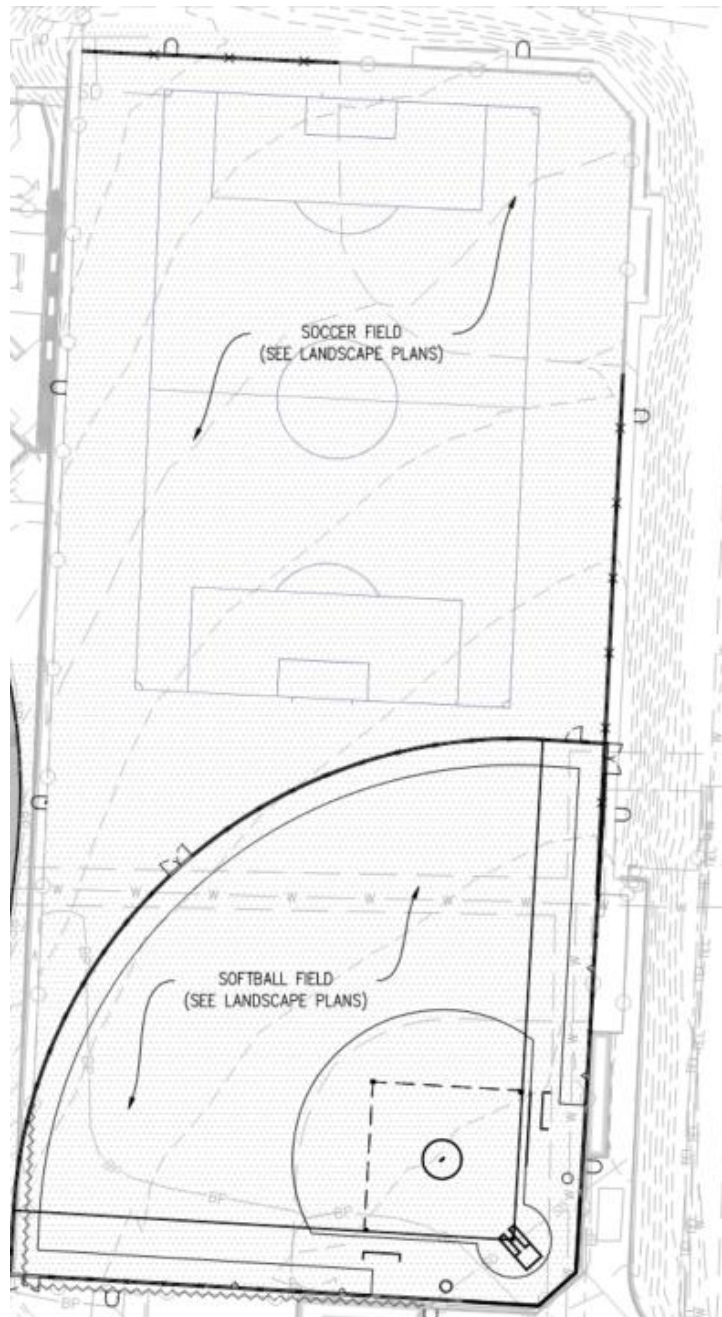
Functionality:

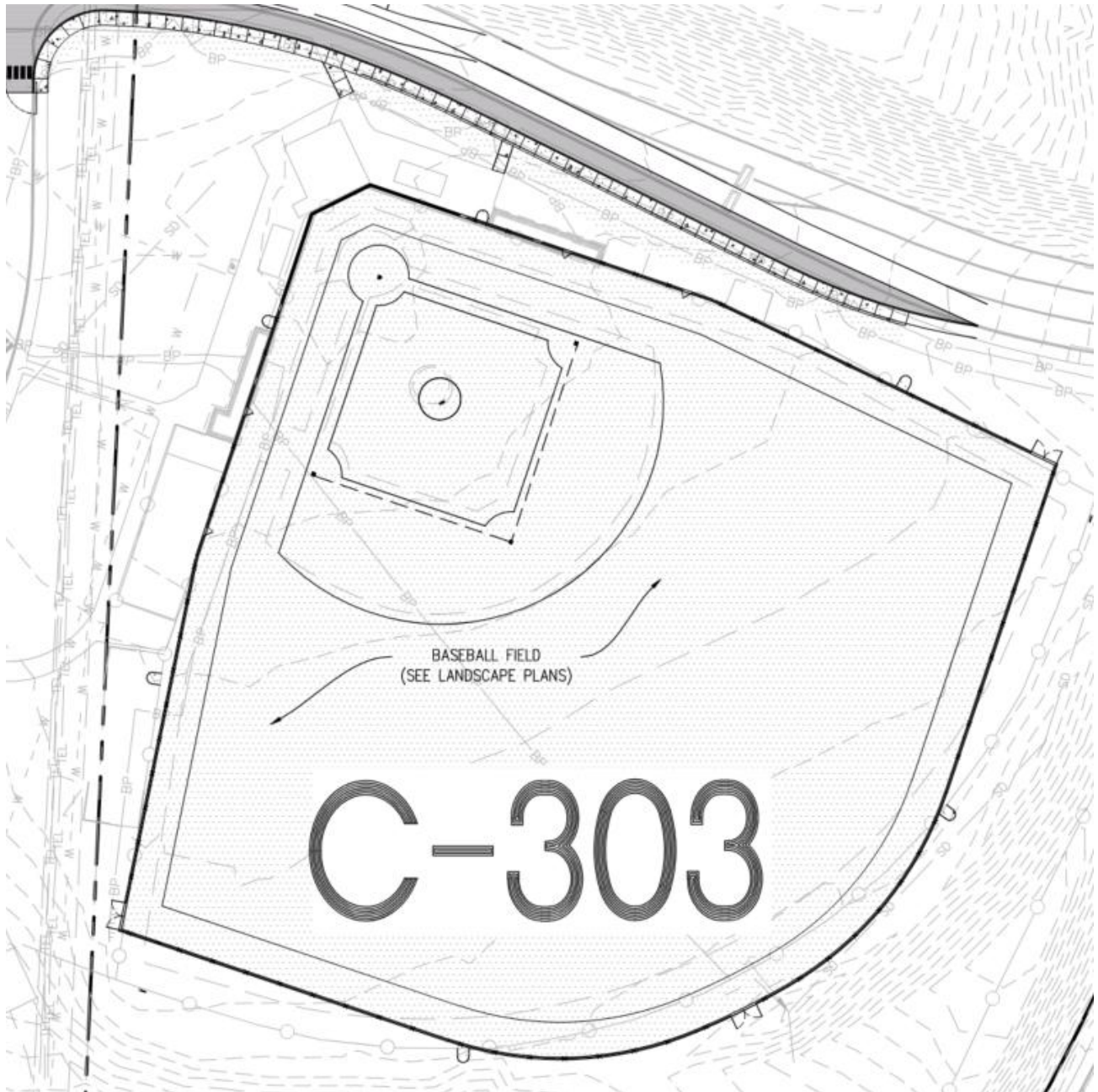
Improved Functionality. The detention systems will enable the existing stormwater conveyance system to continue functioning normally.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea: The design team was concerned about the 8" size of the existing conveyance system to accommodate additional stormwater from the ballfield improvements. Storm events have evolved in recent years, and the design team requested that the VE team take a look at identifying a solution.

SKETCHES/DIAGRAMS:

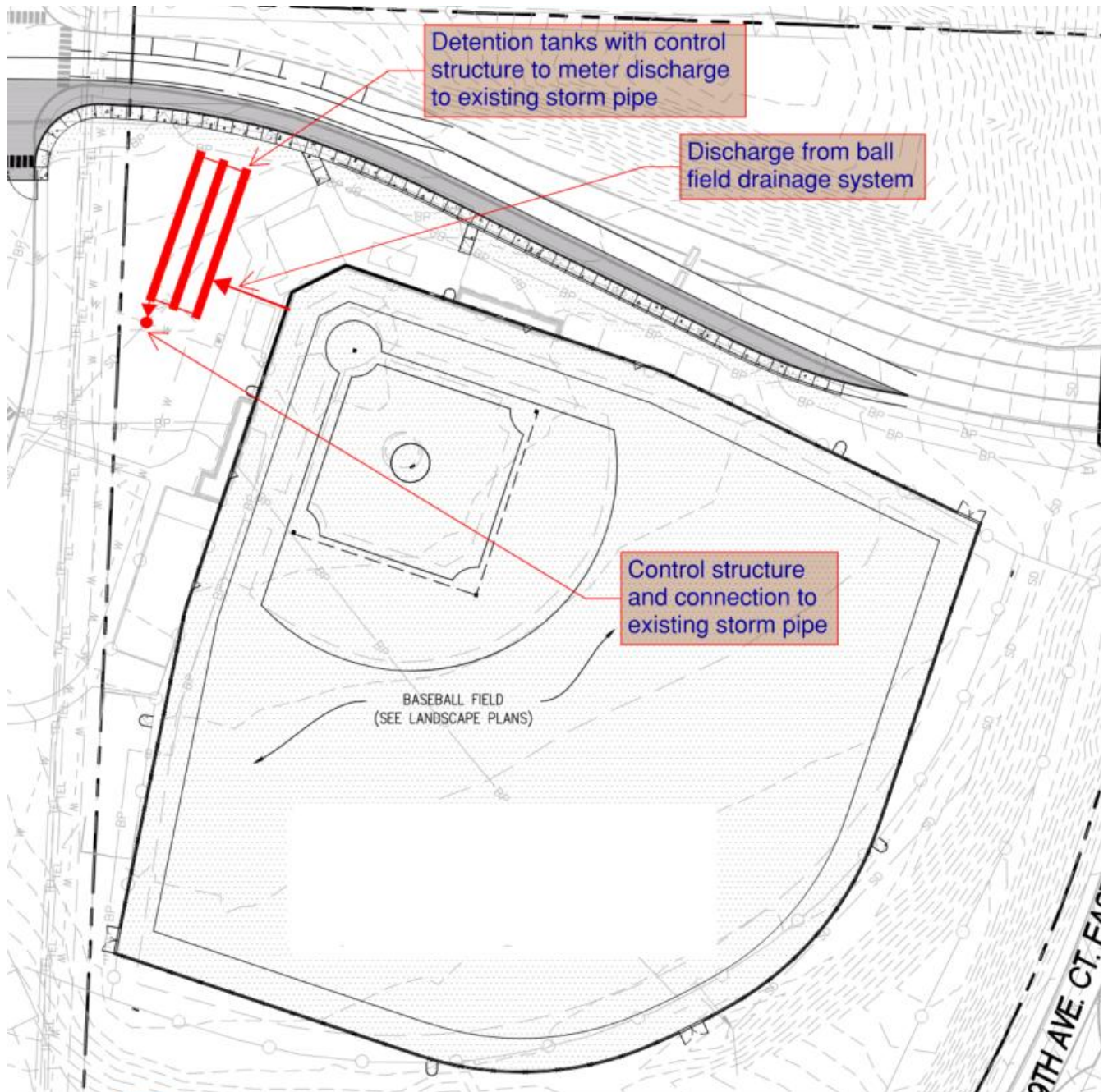
BASELINE CONCEPT





VALUE PROPOSAL CONCEPT





Examples of detention systems



Corrugated metal pipe detention system



HDPE pipe detention system

ASSUMPTIONS AND CALCULATIONS:

This proposal assumes that the existing campus stormwater conveyance system cannot accommodate the additional discharge from the planned ball field drainage system. For the cost estimate, we compared replacing existing pipes with installing detention systems at the ballfields. We estimated that 1,000 feet of existing 8" pipe would be replaced with 12" pipe if detention systems are not built for the ball fields. This proposal also assumes that there is enough elevation difference between the planned ballfield drain system and the existing pipe system to fit a detention tank system.

COST ESTIMATE FORM

COMPONENT:

Storm Vault

C1

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Replace 1000lf of existing stormwater pipe and repair surface	1,000	LF	150	\$150,000	Softball, baseball and soccer fields	42,544	SF		\$0
					Add 24" corrugated metal pipe (CMP)	300	LF	119	\$35,620
					Connection to existing system	2	EA	5,690	\$11,380
					Control structure	2	EA	15,000	\$30,000
Subtotal				\$150,000	Subtotal				\$77,000
General Contractor Markup	26.41%			\$39,615	General Contractor Markup	26.41%			\$20,336
Total to nearest \$1000				\$190,000	Total to nearest \$1000				\$97,000
					Difference				\$93,000

MENG Analysis

Proposal

C1

	PROPOSAL	C2
COMPONENT: ADA Ramp	AUTHOR	MW
BASELINE CONCEPT: ADA ramp planned at the southwest corner of the new building is squeezed into a tight area combined with a new stairway. The stairway and the ramp are both narrow and may be perceived as uninviting.		
VALUE PROPOSAL SYNOPSIS: Provide a wider, more inviting ramp that accommodates all users. This can be accomplished by eliminating the stairs, shifting the DLC drop-off area north, and shifting the drive lanes west.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Improved	Improved
FUNCTIONS:		
Support People	Circulate People	Define Space
ADVANTAGES: • More inviting ramp • Accommodates two-way traffic • Easier to maintain • Mixes users together • Improves safety		DISADVANTAGES: • Involves significant re-design • Involves realignment of existing drive • Increases distance between DLC drop-off and side entrance • Requires additional concrete • Increases cost

DISCUSSION AND JUSTIFICATION

Providing a wide, meandering ramp to connect the DLC drop-off with the front entrance of the school presents a more inviting and inclusive travel way for all users. The wide ramp will also help reduce congestion. Shifting the access drive westerly and shifting the DLC drop-off zone about 30 feet north will allow more room to accommodate this wider design. Elevating the drive and the walk in this area as much as feasible will also reduce the length of the ramp needed. This design approach can improve aesthetics with space and shape, including low-voltage lighting, more trees, and shrubbery will further enhance aesthetics. Shifting the DLC drop-off zone further north increases the travel distance to the building entrance minimally and will increase construction costs. This design approach improves safety by eliminating stairs. Furthermore, a ramp is easier to maintain than stairs.

SCHEDULE SAVINGS:

Maintained Schedule. This work would not affect the critical path of the schedule.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained Reliability. Concrete ramps are durable.

Maintenance & Operations:

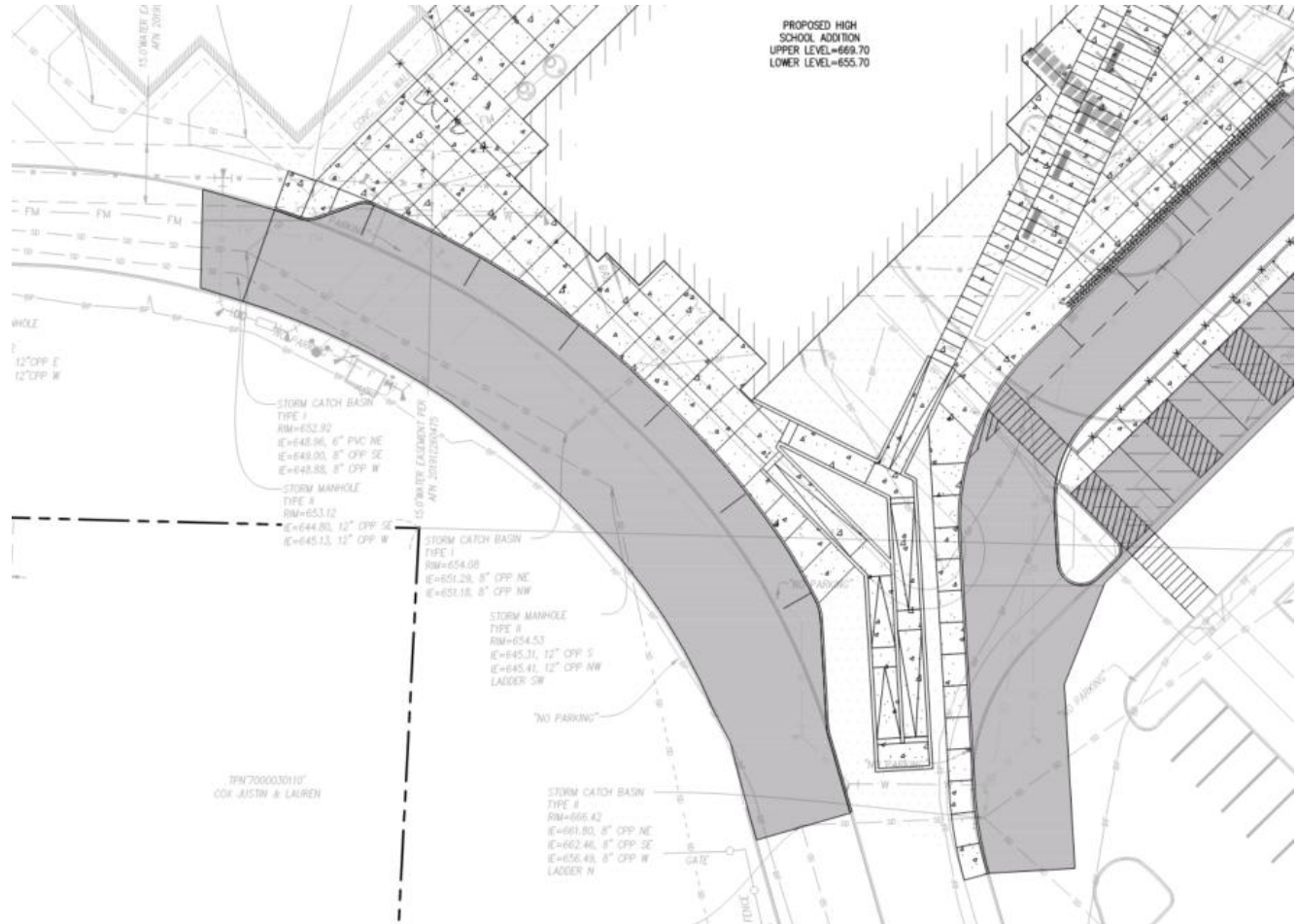
Improved Maintenance and Operations. Wide ramps are easier to maintain than stairs.

Functionality:

Improved Functionality. Wide ramps are safer than stairs and promote inclusiveness and help reduce congestion.

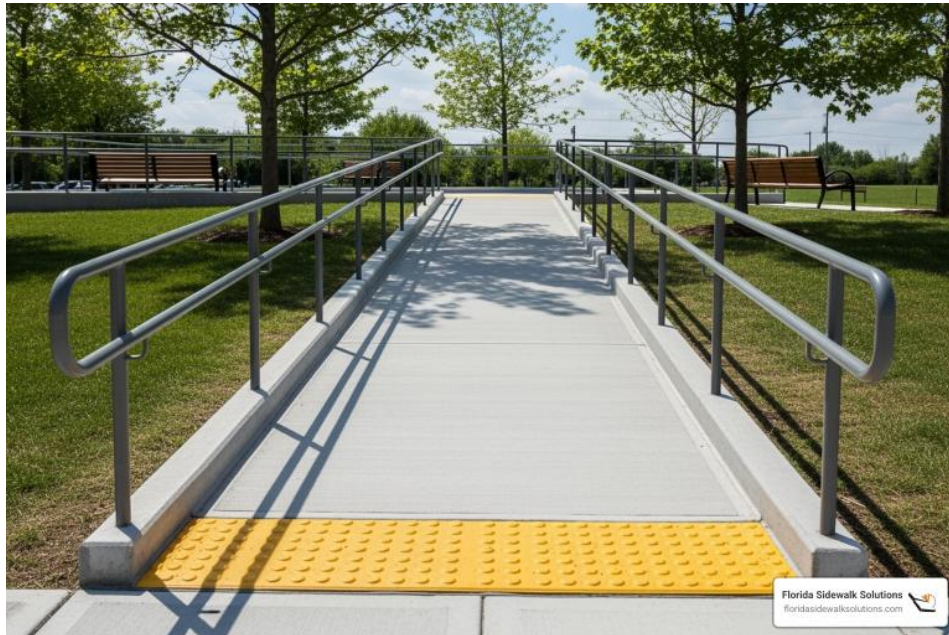
REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

Stakeholders were interested in the idea of eliminating the stairs and having a wider ramp that could improve function.



VALUE PROPOSAL CONCEPT





This example shows railing and edge treatment for a ramp.



This example shows how a wide ramp can be more inviting for all users.

ASSUMPTIONS AND CALCULATIONS:

This proposal assumes that there is enough room on school property to shift the roadway to enable more flexibility in the ramp design. No other obstructions appeared to prohibit this proposal idea.

COST ESTIMATE FORM

COMPONENT:

ADA Ramp

C2

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Concrete Stairs	134	LF	100	\$13,400	Concrete Stairs	0			\$0
Pedestrian Concrete Paving	2,600	SF	12	\$31,200	Pedestrian Concrete Paving	3,000	SF	12	\$36,000
Concrete Ramps	1,000	SF	25	\$25,000	Concrete Ramps	2,730	SF	25	\$68,250
Handrails at Ramps and Stairs	364	LF	135	\$49,140	Handrails at Ramps and Stairs	364	LF	135	\$49,140
Heavy Duty Asphalt Pavement (4" AC Paving over 8" CSBC)	278	SY	56	\$15,560	Heavy Duty Asphalt Pavement (4" AC Paving over 8" CSBC)	278	SY	56	\$15,560
DLC dropoff	12	SY	30	\$370	DLC dropoff	12	SY	30	\$370
Green spaces plated with trees and shrubs	2,000	SF	11	\$21,220	Green spaces plated with trees and shrubs	4,500	SF	11	\$47,750
Subtotal				\$155,890	Subtotal				\$217,070
General Contractor Markup	26.41%			\$41,171	General Contractor Markup	26.41%			\$57,328
Total to nearest \$1000				\$197,000	Total to nearest \$1000				\$274,000
					Difference				(\$77,000)

MENG Analysis

Proposal

C2

	PROPOSAL	S1
COMPONENT: Floor Framing	AUTHOR	MKW
BASELINE CONCEPT: The design structural floor assembly is reinforced concrete slab on metal deck supported with steel wide-flange beams spaced at 8 feet on center, typically.		
VALUE PROPOSAL SYNOPSIS: We propose a structural floor assembly with reinforced concrete slab on metal deck supported on 22" to 24" deep open web bar joists spaced at 2'-8" on center to 4' on center.		
SCHEDULE SAVINGS: Maintained (Similar)		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Maintained	Maintained
FUNCTIONS:		
Support Load	Transfer Load	Resist Deterioration
ADVANTAGES: • Lighter individual members • Allows more space for non-structural elements • Reduced labor for installation • Cost savings		DISADVANTAGES: • More members • Some WF Still Needed

DISCUSSION AND JUSTIFICATION

Open web bar joists are lighter structural elements which can ease the installation process. The bar joists provide more open space in the floor cavity, that can permit wiring and some pipes to pass through the structure, rather than having to be supported fully below the structure. Open web bar joists are being used in the baseline concept for a portion of the roof, so this would not introduce additional framing concepts to the building.

SCHEDULE SAVINGS:

Maintained. Open web bar joists have a similar fabrication and delivery schedule to wide-flanged steel beams, depending upon supplier and fabricator. There will be increased installation activity because there need to be more trusses overall than WF beams, however the trusses could be expected to be simpler to install.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained. Common structural systems, both steel wide-flange beams and steel trusses, provide good reliability and durability for the lifetime of a building.

Maintenance & Operations:

Maintained. Neither system is anticipated to require maintenance once constructed. Both systems would be designed to support code-required loading. WF beams would provide improved space between elements of the structural system in the floor cavity, while trusses would provide improved access for penetrations through the elements of the horizontal system.

Functionality:

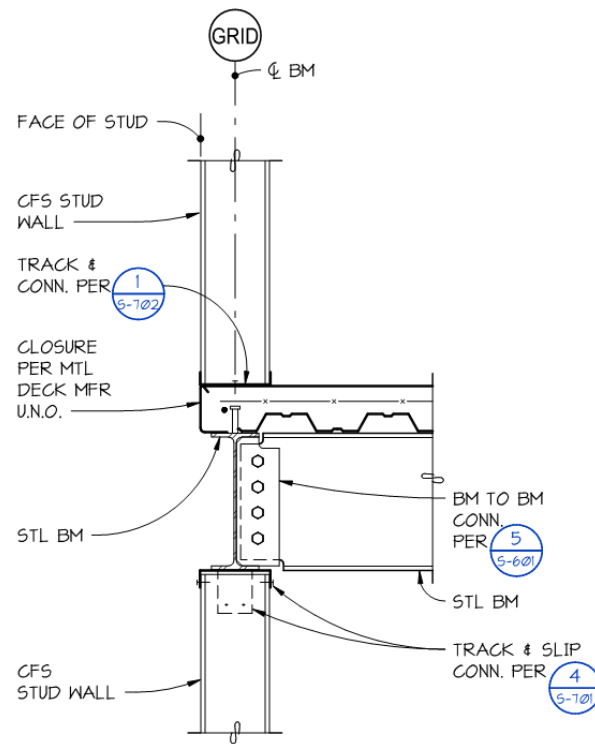
Maintained. Both systems adequately serve the principal function.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

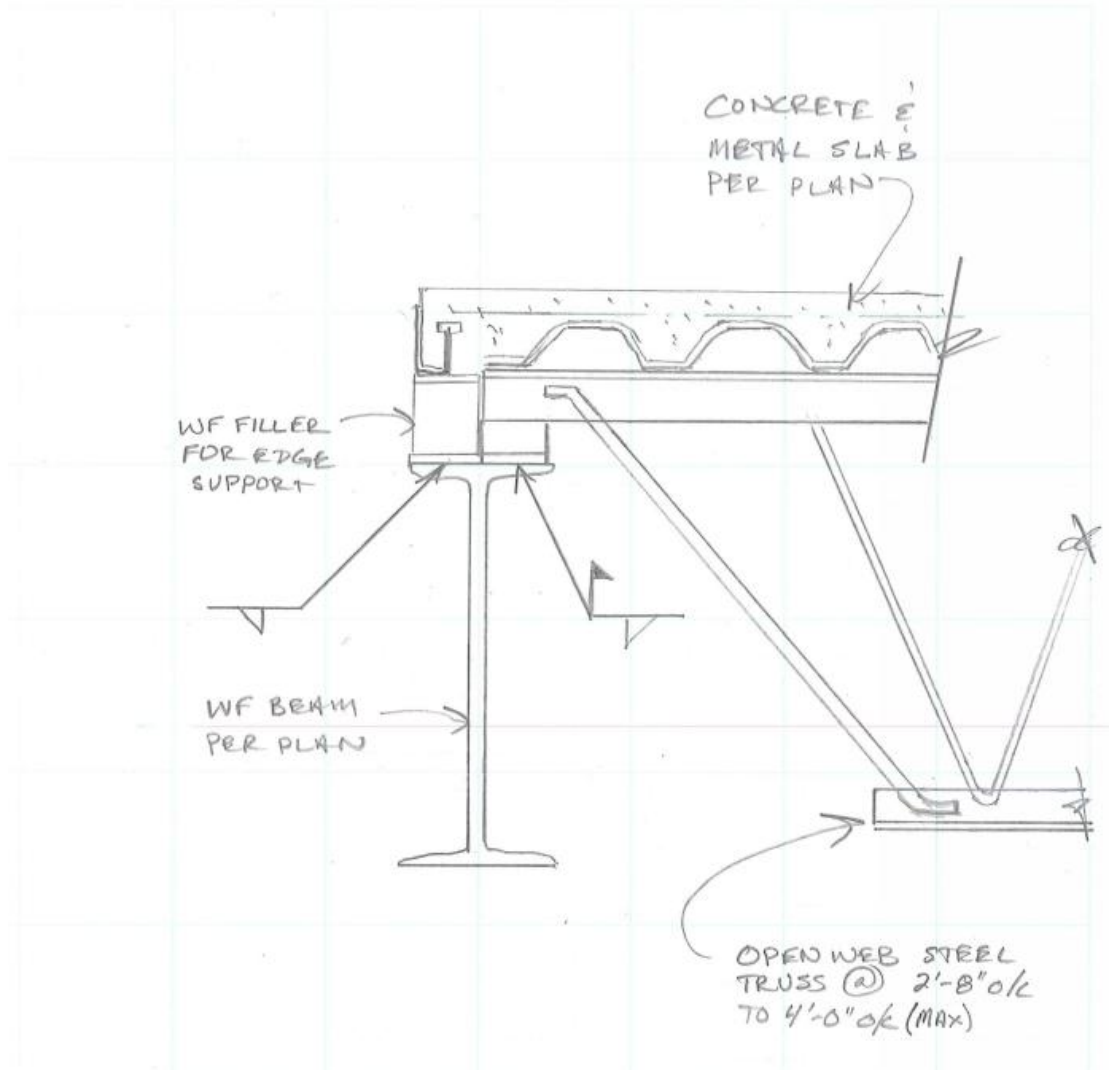
They would be interested in exploring this alternative.

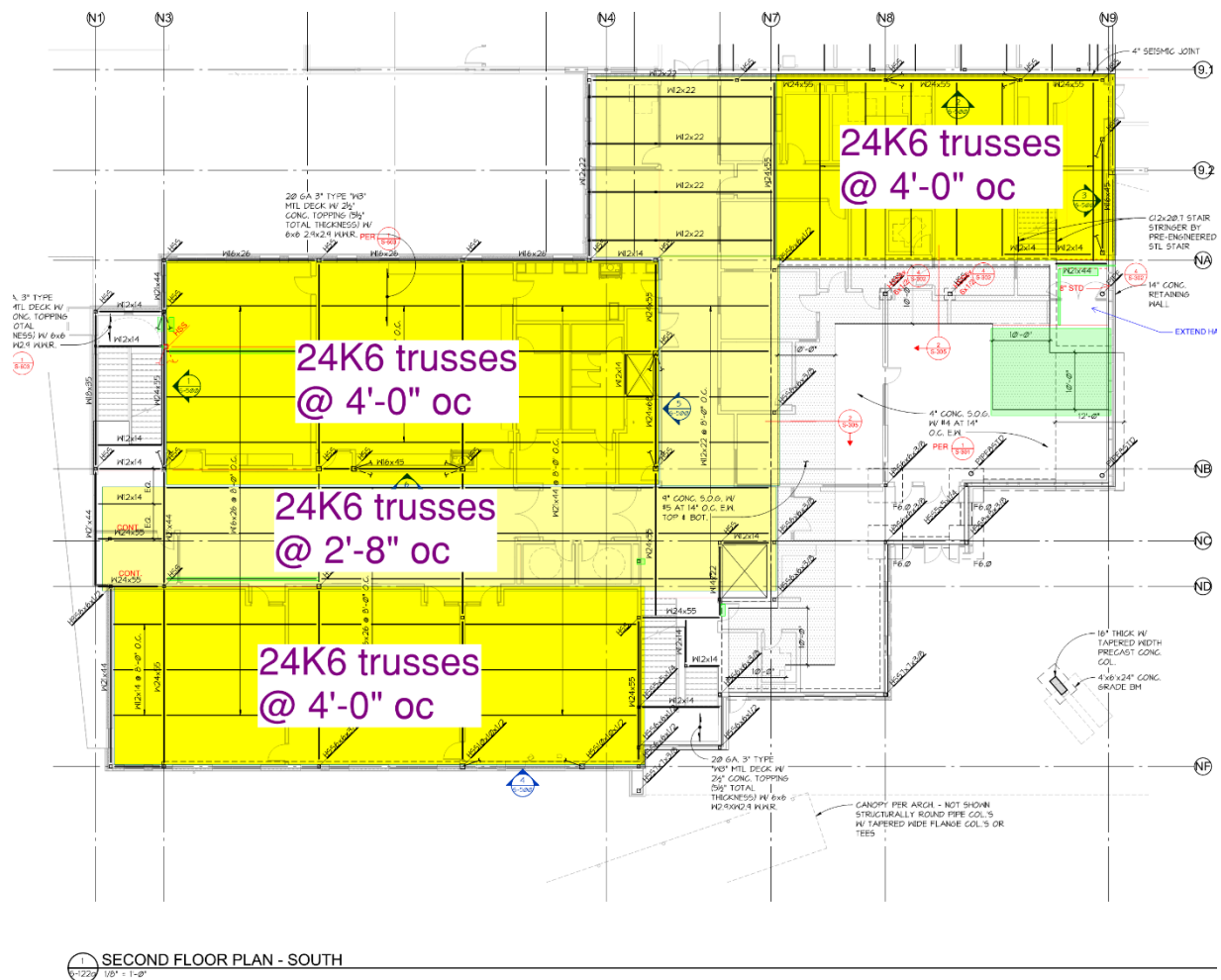
SKETCHES/DIAGRAMS:

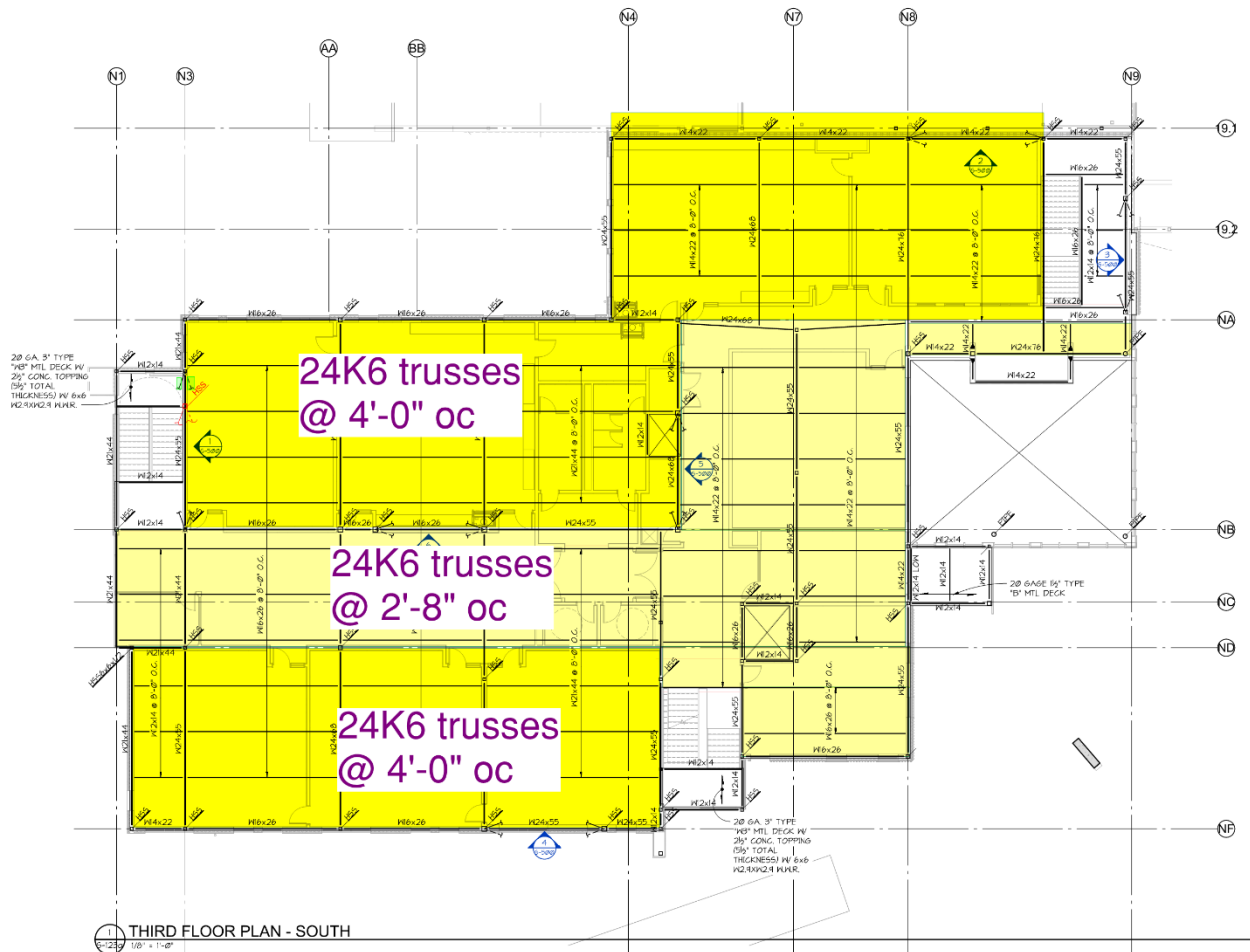
BASELINE CONCEPT



VALUE PROPOSAL CONCEPT







ASSUMPTIONS AND CALCULATIONS:

Gravity floor loads were based on loads provided in the structural drawing set. Floor dead loads are 65psf. Floor live loads vary from 50psf in classrooms to 100psf in the first-floor corridors. Most floor spans are approximately 20 to 24 feet. Generally, the trusses can support 520 pounds per foot, which results in 4.5' joist spacing at the classrooms and 3' joist spacing at the first-floor corridors.

COST ESTIMATE FORM

COMPONENT:

Floor Framing

S1

[illegible]

MENG Analysis

Proposal

S1

	PROPOSAL	S2
COMPONENT: Lateral System	AUTHOR	MKW
BASELINE CONCEPT: Structural design lateral force resisting system consists of a rigid or semi-rigid diaphragm with steel buckling-restrained braced frames.		
VALUE PROPOSAL SYNOPSIS: Lateral force resisting system that consists of a rigid or semi-rigid diaphragm with steel stud shearwalls.		
SCHEDULE SAVINGS: Improved		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Degraded	Improved
FUNCTIONS:		
Transfer Load	Stiffen Assembly	Prohibit Collapse
ADVANTAGES: - Removes expensive components - Low-cost materials - Uses walls in current design - Improved redundancy		DISADVANTAGES: - Increased concrete footings - Makes walls part of primary structure - Restricts penetration opportunities

DISCUSSION AND JUSTIFICATION

This proposal is about using the designed steel stud walls as shearwalls to resist the lateral forces instead of using buckling-resistant braced frames. This increases structural efficiency as the steel studs are already part of the project, and this proposal suggests adding detailing for connections and sheathing to make the wall framing into shearwalls. Light gauge steel stud shearwalls are redundant and efficient, as opposed to braced frames which require intense inspection, specialty welding, installation requirements, and heavier materials. However, a much larger area of wall is required than the areas required for the braced frames. Virtually all of the walls in the addition building longer than six feet would need to be part of the lateral force resisting system. Shearwalls must be continuous floor to floor, with minimal penetrations, which can complicate mechanical, electrical, and duct systems.

SCHEDULE SAVINGS:

Improved: The proposed lateral solution using shearwalls would eliminate the complicated construction and inspection involved with the buckling restrained braced frames. The steel walls are already part of the design schedule, and the time required to provide shear connections and sheathing is small compared to the braced frame construction.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained. Both systems are commonly used and accepted structural systems for lateral force resistance that conform to building code requirements. Both systems should protect life safety in a seismic or lateral force event. Depending upon the design considerations, both systems may experience damage during a strong seismic event that may require repair or replacement.

Maintenance & Operations:

Degraded. Maintenance is not an anticipated need for either system. However, the braced frame system allows for more space and less interference for other building systems. Shearwalls are continuous solid walls with limited openings which would hinder changes to the wall layout, addition of openings, and alteration of other building systems.

Functionality:

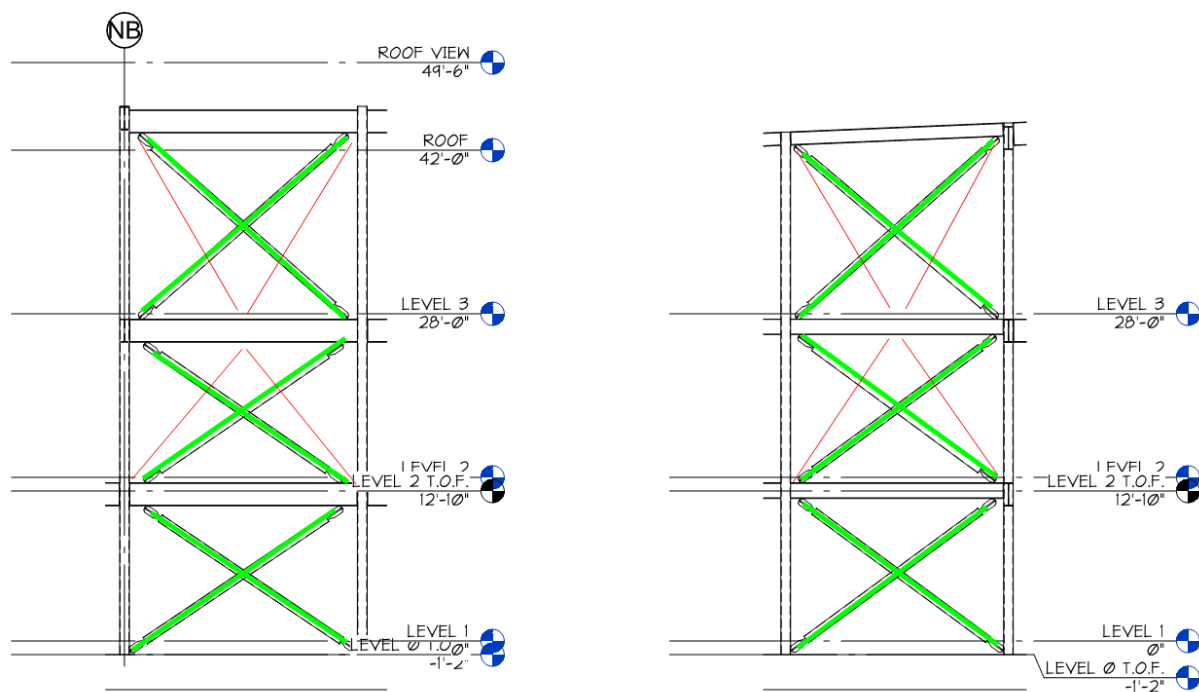
Improved: Shearwalls are distributed throughout the building, as opposed to the braced frames which are much more limited. The redundancy provided by shearwalls allows for a wider distribution of load resistance.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

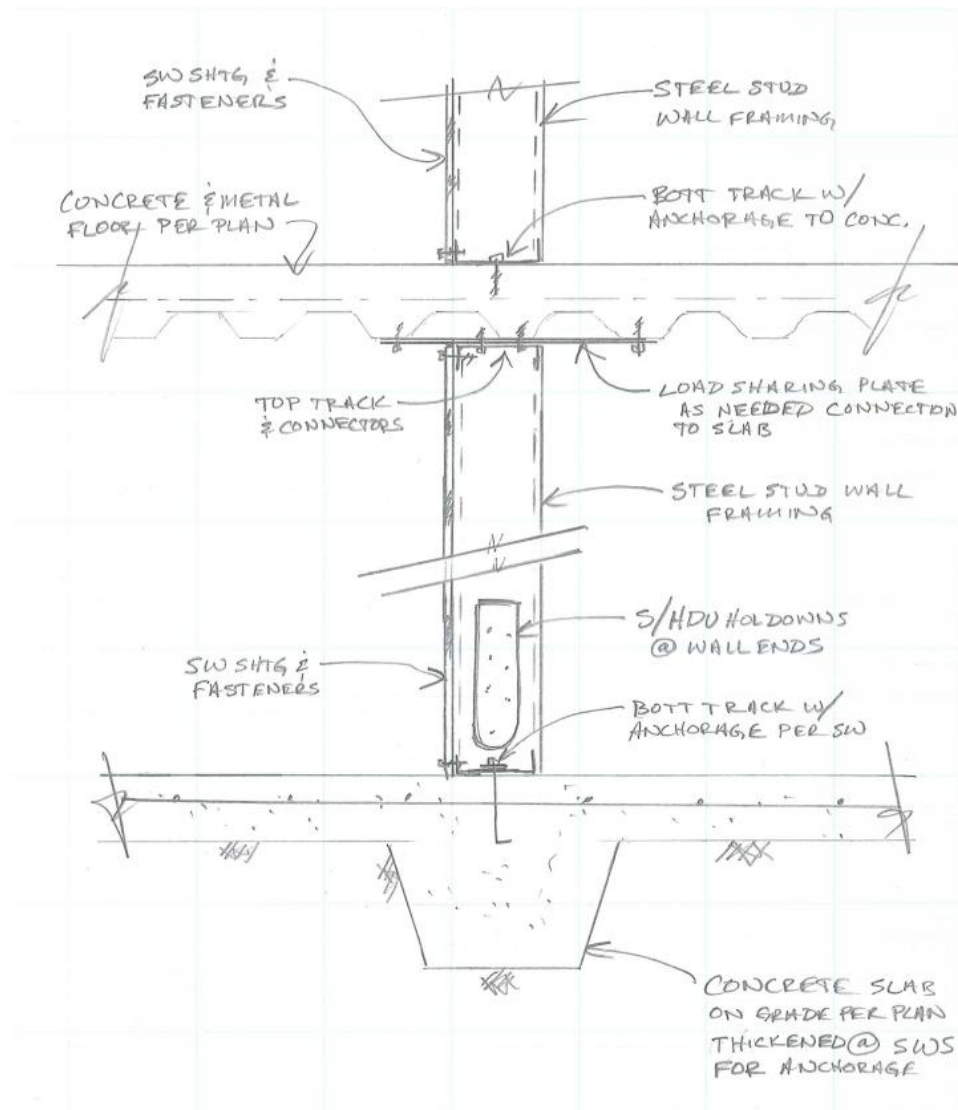
They would consider this option with additional information provided.

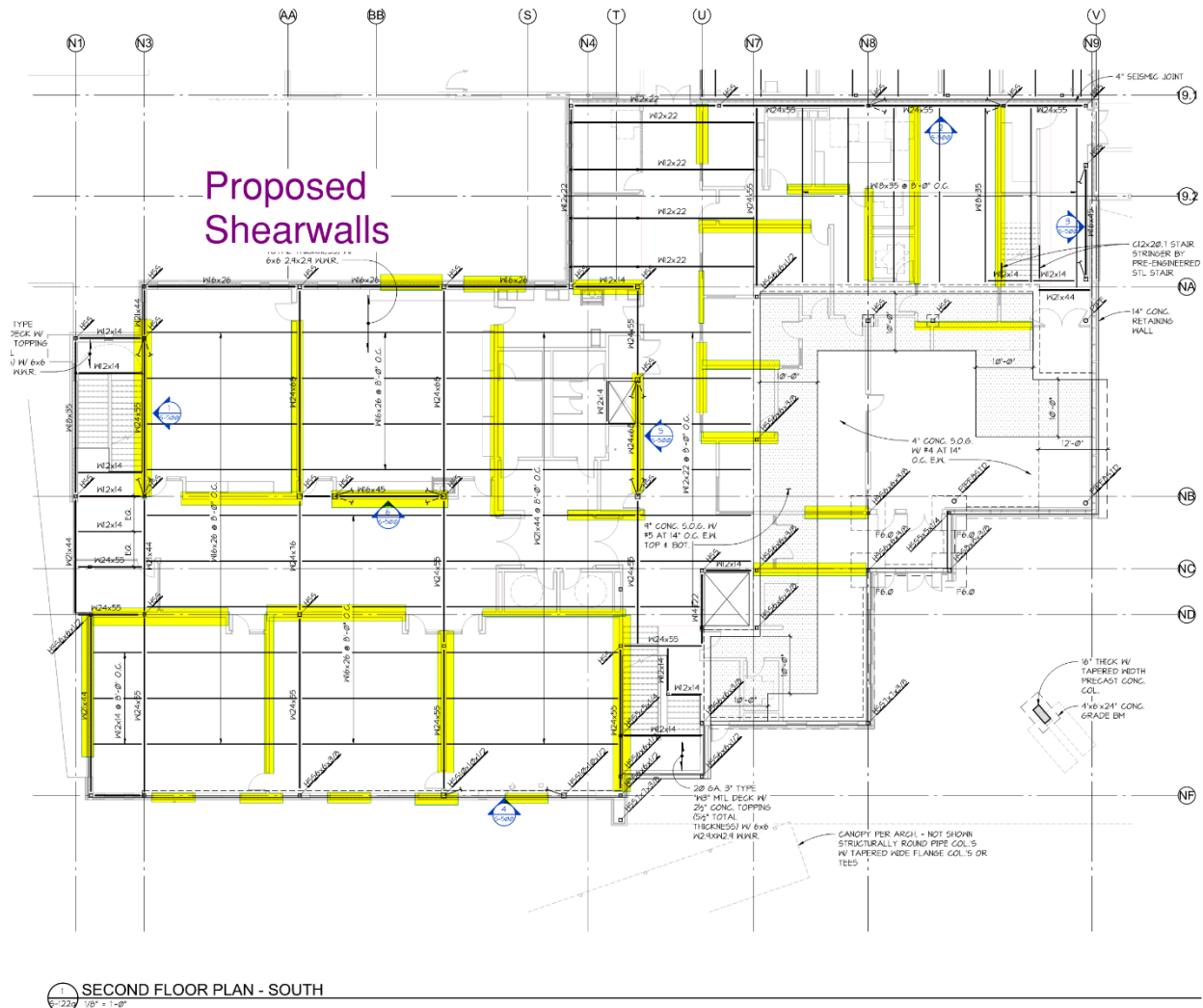
SKETCHES/DIAGRAMS:

BASELINE CONCEPT



VALUE PROPOSAL CONCEPT





ASSUMPTIONS AND CALCULATIONS:

A seismic base shear was not provided by the EOR. Our calculation of the seismic base shear is based on the provided roof and floor weights, and estimated weights for the interior and exterior walls. The proposed seismic system is B23: Building Frame Systems with Light-frame (cold-formed steel) walls, per ASCE 7-16, Table 12.2. We calculated a base shear of 440k, or 308k (ASD). To accommodate this shear force, we anticipate needing approximately 203 feet of single-sided shearwall or 102 feet of double-sided shearwall in each primary direction on the upper floor. We need approximately 370 feet of single-sided shearwall or 185 feet of double-sided shearwall on the main floor. We need approximately 352 feet of single-sided shearwall or 175 feet of double-sided shearwall on the basement level.

S2

Lateral System

Proposal

S2

	PROPOSAL	A1
COMPONENT: Stair Tower	AUTHOR	BC
BASELINE CONCEPT: The current proposal includes a stair tower extending to the roof. This creates a very prominent architectural feature (beacon) that helps identify the main entry area.		
VALUE PROPOSAL SYNOPSIS: The revised proposal includes the same functions while utilizing less curtain wall glazing and still maintaining the goal of helping to draw attention to the building entrance.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Improved	Maintained
FUNCTIONS:		
Endure Time	Clarify Entry	Define Aesthetic
ADVANTAGES: • Reduced labor • Improved maintenance • Life cycle cost savings • Improved energy efficiency • Reduced material costs		DISADVANTAGES: • Re-design required • Less prominent “beacon” • Changes aesthetic

DISCUSSION AND JUSTIFICATION

The stair tower design is very striking and draws attention to the main entry. Focus on the main entry is one of the major design goals of this project. A secondary design function of the stair tower is to provide direct access to the roof for maintenance. This revision proposes to accomplish both goals while reducing the cost of materials and the amount of energy loss through the curtain wall system. Other opportunities may include adding LED lighting with color controls. This could be used for special events to draw more attention and excitement to the school when utilized.

SCHEDULE SAVINGS:

Maintained.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained.

Maintenance & Operations:

Improved.

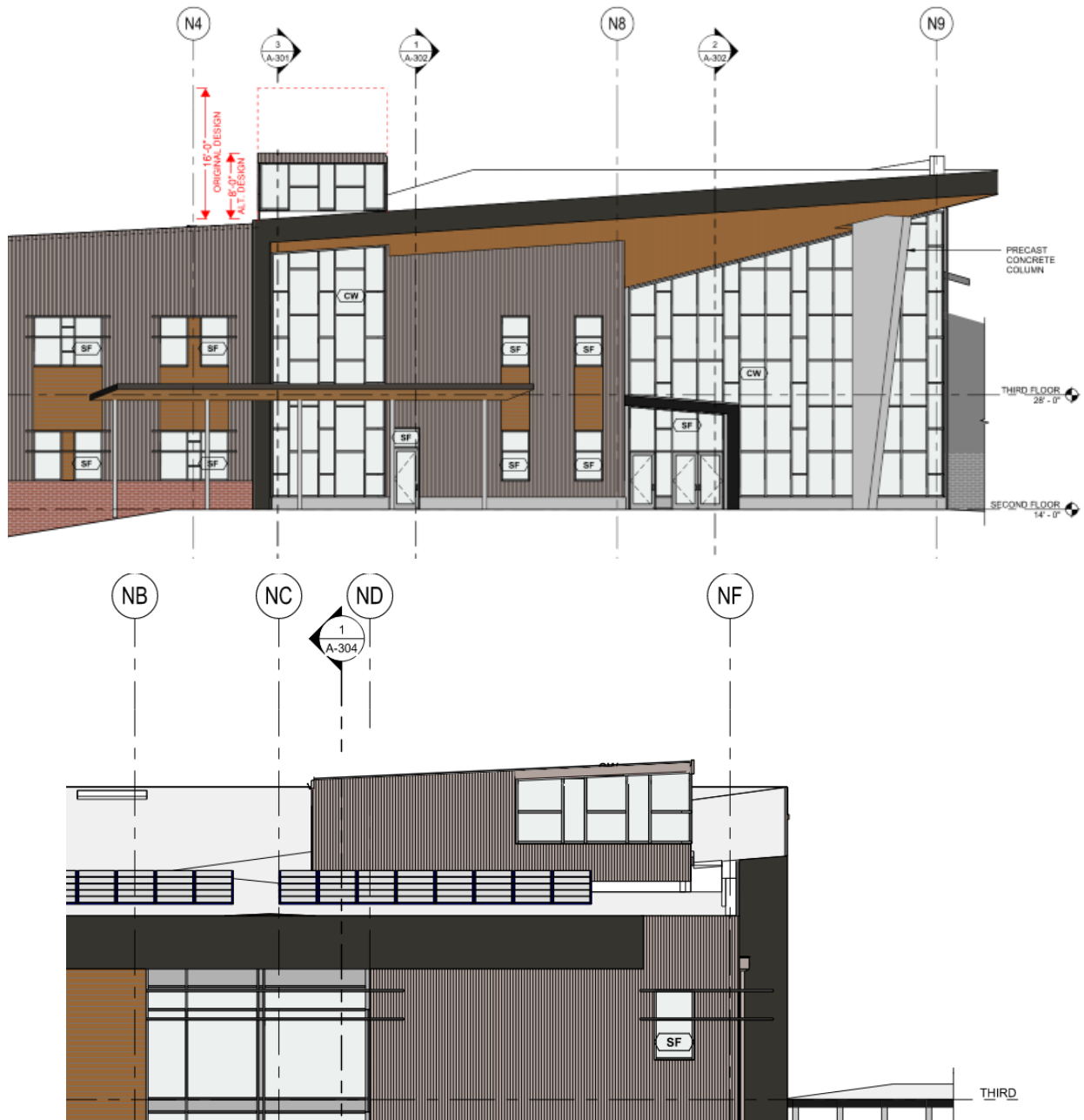
Functionality:

Maintained.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

It was noted during a meeting with the owners and design team that the original scheme was presented to the public and board, and it was very well received. The goal of the alternate proposal is to keep the main concept of the stair tower, however somewhat smaller.

VALUE PROPOSAL CONCEPT



ASSUMPTIONS AND CALCULATIONS:

The cost comparison includes the initial reduced cost for materials and installation. Life cycle costs are not included but will add to the value over time due to energy savings.

Bonney Lake High School Expansion & Intersection Improvements

COST ESTIMATE FORM

Stair Tower

A1

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Curtain wall window system	600	SF	130	\$78,000	Curtain wall window system	250	SF	130	\$32,500
Corrugated metal panel - cool parchement	760	SF	30	\$22,420	Corrugated metal panel - cool parchement	530	SF	30	\$15,640
HSS 10x10x1/2	3,123	LB	4	\$10,930	9-1/2"x9-1/2" square tube columns	1,499	LB	4	\$5,250
b.8L - 2-1/2" rigid insulation, WRB, 5/8" sheathing, 8" metal stud framing, batt insulation, vapor barrier, 5/8" gypsum board	760	SF	36	\$27,040	b.8L - 2-1/2" rigid insulation, WRB, 5/8" sheathing, 8" metal stud framing, batt insulation, vapor barrier, 5/8" gypsum board	530	SF	36	\$18,860
Subtotal				\$138,390	Subtotal				\$72,250
General Contractor Markup	26.41%			\$36,549	General Contractor Markup	26.41%			\$19,081
Total to nearest \$1000				\$175,000	Total to nearest \$1000				\$91,000
					Difference				\$84,000

Proposa

A1

	PROPOSAL	A2
COMPONENT: Study Nooks	AUTHOR	BC
BASELINE CONCEPT: The current floor layout includes a “Flex Space” on the 2 nd and 3 rd floors. This space is open and allows for several different flexible functions.		
VALUE PROPOSAL SYNOPSIS: This proposal includes adding small areas (study nooks) that fit into the “Flex Space”. This will provide a bit more program structure to the areas while still maintaining ongoing flexibility.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Maintained	Improved
FUNCTIONS:		
Educate Students	Improve Flexibility	Promote Timelessness
ADVANTAGES: - Improves useability - Enhances group dynamic - Provides additional study options - Allows for spontaneous work		DISADVANTAGES: - Slightly less flexibility - Increased furniture costs - Added structural elements

DISCUSSION AND JUSTIFICATION

Adding dedicated study nook furniture will allow students to have impromptu group and individual study sessions while maintaining a close connection to their classrooms. The study nooks overlooking the Welcome Center will have a small cantilever floor allowing an open view down into the spaces. The study nooks in the east "Flex Space" will have ample daylight from the adjoining windows.

SCHEDULE SAVINGS:

Maintained.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained.

Maintenance & Operations:

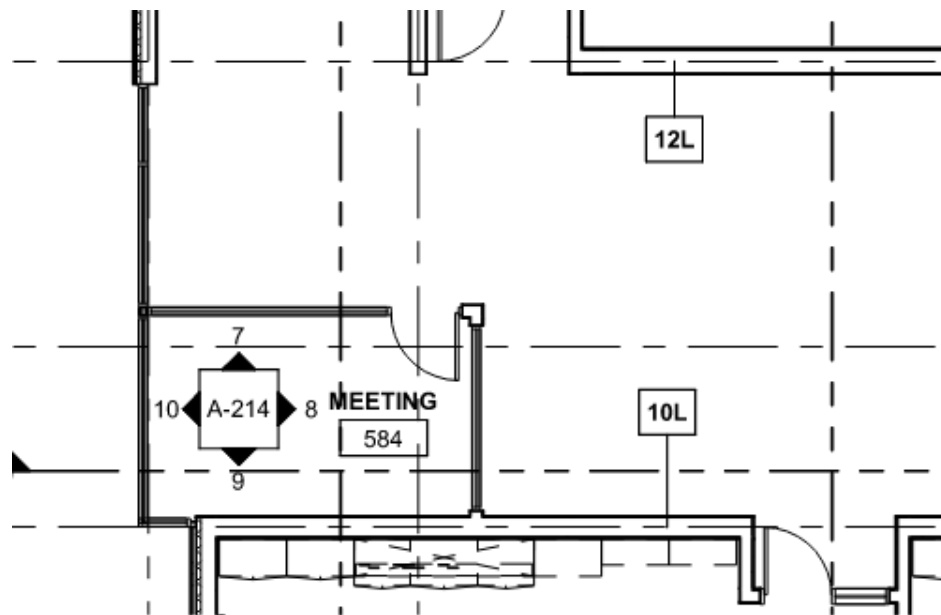
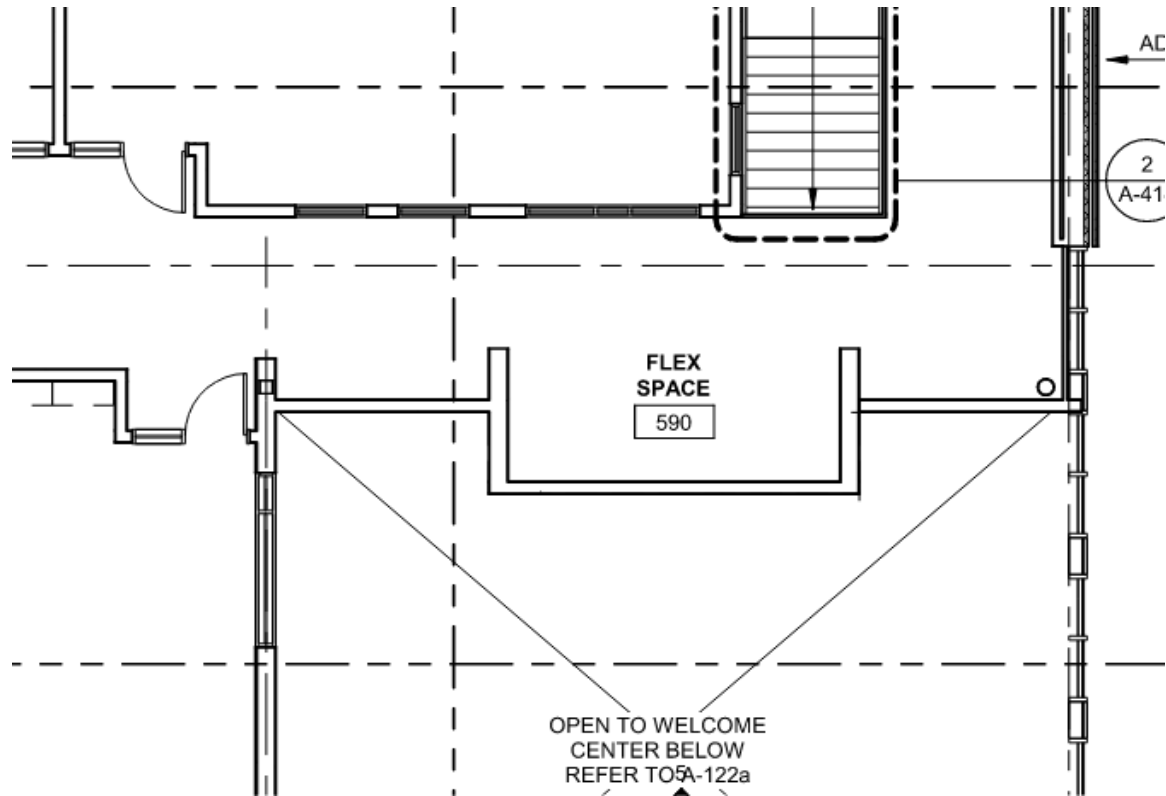
Maintained.

Functionality:

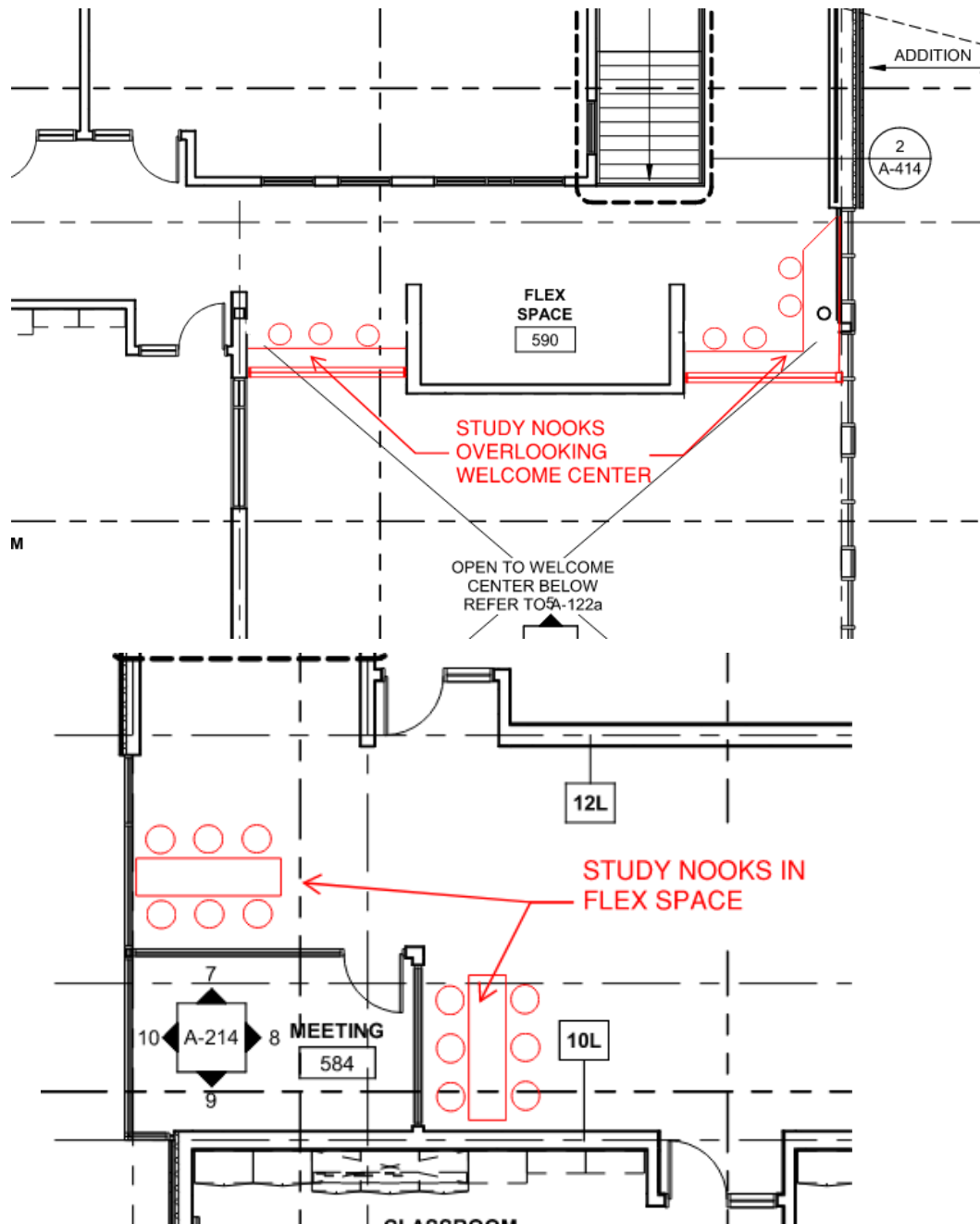
Improved.

SKETCHES/DIAGRAMS:

BASELINE CONCEPT



VALUE PROPOSAL CONCEPT



ASSUMPTIONS AND CALCULATIONS:

The Study Nook furniture can be removeable to maintain the function of the Flex Space.
The Study Nooks over the Welcome center will require a small structural cantilever.

A2

Study Nooks

Proposal

A2

	PROPOSAL	M1
COMPONENT: Integrated Design	AUTHOR	DCS
BASELINE CONCEPT: Addition is designed to the code-minimum thermal envelope with full-size HVAC system.		
VALUE PROPOSAL SYNOPSIS: Design the addition with a premium thermal envelope with down-sized HVAC system.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Maintained	Improved
FUNCTIONS:		
Conserve Energy	Insulate Space	Condition Space
ADVANTAGES: - Life cycle cost savings - Improved energy efficiency - Reduced carbon footprint - More clean buildings ready - Improved acoustic performance		DISADVANTAGES: - Requires integrated design - May require architectural assembly detail revision

DISCUSSION AND JUSTIFICATION

The project WSSP Checklist calls for energy efficiency performance beyond WA State Energy Code (WSEC) minimums. The current design includes code minimum thermal envelope (insulation), with high efficiency mechanical, plumbing, and electrical systems. This proposal suggests a more balanced approach with an upgraded thermal envelope and down-sized HVAC system, which together result in lower life cycle cost, improved occupant thermal comfort, and increased building acoustic performance.

SCHEDULE SAVINGS:

No impact on schedule.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained. No change in reliability.

Maintenance & Operations:

Improved. Slightly less maintenance as mechanical (HVAC) equipment will be downsized.

Functionality:

Improved. Functionality is increased as occupant thermal comfort and building acoustic performance are improved.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

Sounds good.

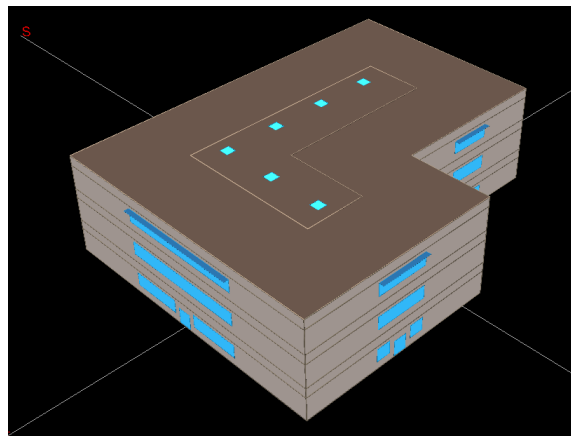
SKETCHES/DIAGRAMS:

BASELINE AND VALUE PROPOSAL CONCEPTS (CURRENT AND PROPOSED)

Component	Units	Code Minimum	Current Design	Proposed Design
Foundation	R-value	R-10 for 2 ft	R-10 for 2 ft	R-15 for 2 ft
Wall	R-value	R-13 +R-10 ci	R-21 batt	R-21 batt + R-10 ci
Door	U-value	U-0.60	U-0.60	R-0.30
Window	U-value	U-0.34	U-0.34	U-0.22
Roof	R-value	R-38 ci	R-38 ci	R-49 ci
Mechanical	n/a	n/a	Full-size	Down-sized

ASSUMPTIONS AND CALCULATIONS:

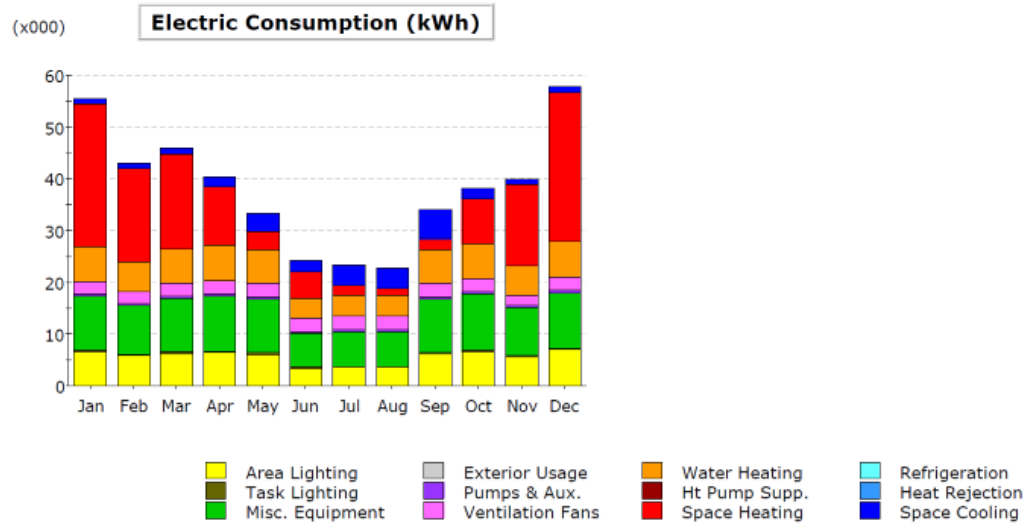
Energy life cycle costed proposals for this value engineering study are based on a shoebox energy model with the following simplified geometry, code-minimum thermal envelope, four-pipe HVAC system with air-source heat pumps, fan coil unit heating & cooling, and dedicated outside air system (DOAS) ventilation; heat pump domestic hot water heaters; LED lighting with controls per code; modest plug loads; and short-day summer school. Monthly and annual energy use and cost are estimated as shown in the following graphs.



Monthly Energy Use:

Project/Run: BLHS - Baseline Design

Run Date/Time: 11/16/25 @ 22:22



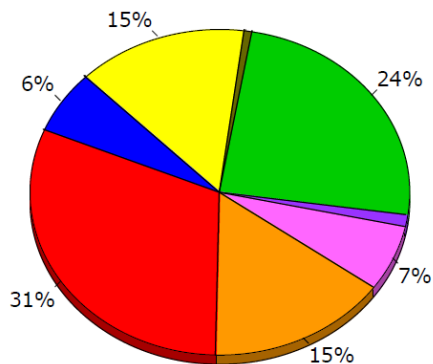
Electric Consumption (kWh x000)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Space Cool	1.06	1.01	1.24	1.72	3.31	2.23	4.11	4.15	5.54	1.97	0.99	1.11	28.43
Heat Reject.	-	-	-	-	-	-	-	-	-	-	-	-	-
Refrigeration	-	-	-	-	-	-	-	-	-	-	-	-	-
Space Heat	27.55	17.92	18.31	11.32	3.69	5.01	1.74	1.17	2.39	8.66	15.75	28.72	142.22
HP Supp.	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Water	6.62	6.00	6.63	6.90	6.54	3.85	3.95	3.94	6.42	6.76	5.61	6.87	70.08
Vent. Fans	2.45	2.22	2.45	2.56	2.45	2.62	2.71	2.71	2.45	2.56	2.10	2.56	29.85
Pumps & Aux.	0.44	0.38	0.42	0.43	0.41	0.43	0.45	0.44	0.42	0.43	0.36	0.46	5.06
Ext. Usage	-	-	-	-	-	-	-	-	-	-	-	-	-
Misc. Equip.	10.39	9.39	10.39	10.61	10.39	6.50	6.71	6.71	10.28	10.72	9.27	10.72	112.08
Task Lights	0.29	0.26	0.29	0.30	0.29	0.15	0.16	0.16	0.29	0.30	0.24	0.30	3.00
Area Lights	6.62	5.78	6.27	6.43	6.11	3.41	3.54	3.55	6.23	6.64	5.59	6.95	67.12
Total	55.40	42.96	45.98	40.27	33.18	24.20	23.36	22.83	34.00	38.04	39.92	57.69	457.83

Annual Energy Use:

Annual Energy Consumption by Enduse

	Electricity kWh (x000)	Natural Gas Btu	Steam Btu	Chilled Water Btu
Space Cool	28.43	-	-	-
Heat Reject.	-	-	-	-
Refrigeration	-	-	-	-
Space Heat	142.22	-	-	-
HP Supp.	-	-	-	-
Hot Water	70.08	-	-	-
Vent. Fans	29.85	-	-	-
Pumps & Aux.	5.06	-	-	-
Ext. Usage	-	-	-	-
Misc. Equip.	112.08	-	-	-
Task Lights	3.00	-	-	-
Area Lights	67.12	-	-	-
Total	457.83	-	-	-

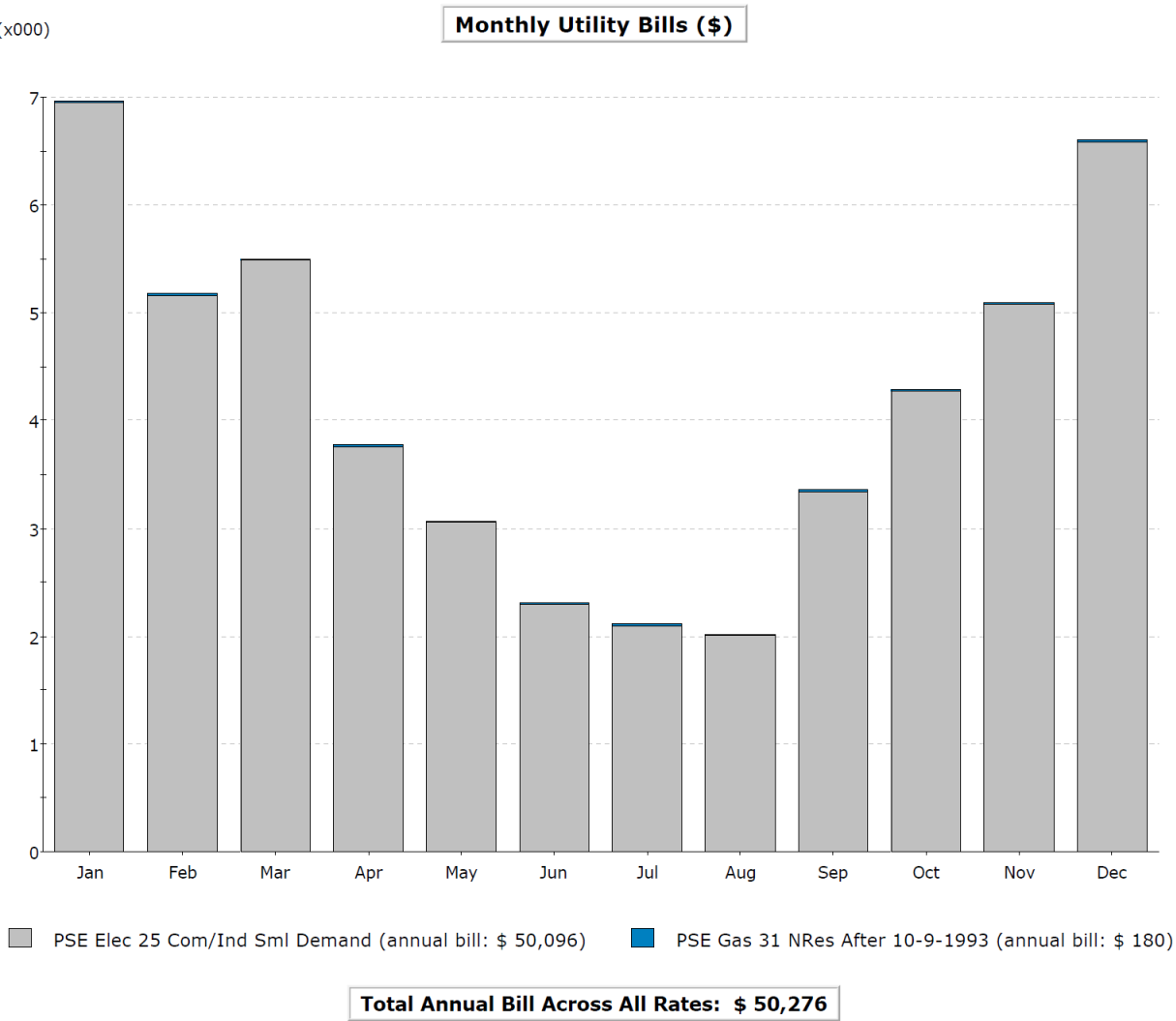


Electricity



Annual Energy Cost:

(x000)



COMPONENT LIFE CYCLE COST ANALYSIS (LCCA)

Client:	Sumner-Bonney Lake School District	M1
Project:	Bonney Lake High School Expansion & Intersection Improvements	proposal #
Date:	12-Sep-25	
By:	DCS	
COMPONENT	Integrated Design	
COMPONENT #	M1	
Escalation rate	0.03	
Discount rate	0.04	
Study Period	50 Yrs.	

Notes: Per the shoebox energy model annual energy cost is \$50K/yr, with 45% or \$22.5K/yr for HVAC; assume half this is envelope load or \$11.3K/yr. This proposal reduces envelope load by approximately 25%, saving \$3K/yr. Assume 1% of initial cost for annual maintenance cost, 10% for renewal year 25. Assume 10% salvage value.

Instructions: Enter escalation, discount, and study period above.
Enter annual costs, replacement costs (and appropriate replacement year), and salvage value.
Enter these costs in the shaded cells using today's (current) dollars. For annual costs, escalation rates will be automatically entered, but can be individually overwritten below for differential escalation.
All costs will automatically be escalated and discounted.

ALTERNATIVE A : Current				ALTERNATIVE B: Proposed				DIFFERENCE
INITIAL COSTS				INITIAL COST				DIFFERENCE
INITIAL COST				INITIAL COST				
8,224,000				\$ 8,015,000				\$209,000.00
O & M ANNUAL COSTS								
STAFFING OPERATIONS								
STAFFING MAINTENANCE								
SUPPLIES OPERATIONS								
SUPPLIES MAINTENANCE								
Cost in current				Cost in				
Subcomponents	\$	Esc.	Pres. Worth \$	Subcomponents	current \$	Esc.	Pres. Worth \$	
Annual maintenance cost	82,240	0.03	\$ 3,319,886	Annual maintenance cost	80,150	0.03	\$ 3,235,516	\$84,369.67
Annual energy cost	50,276	0.03	\$ 2,029,555	Annual energy cost	47,276	0.03	\$ 1,908,450	\$121,104.80
		0.03	\$ -			0.03	\$ -	\$0.00

	PROPOSAL	M2
COMPONENT: Thermal Generation	AUTHOR	DCS
BASELINE CONCEPT: Campus central plant capacity is increased, and most equipment is replaced. This includes two aging air-cooled chillers, most pumping, and significant piping and controls. The work excludes three newer heating hot water boilers installed in 2010.		
VALUE PROPOSAL SYNOPSIS: The campus central plant chillers are renewed. A new, small thermal generation plant is provided for the addition, including one air-source heat pump with electric boiler back-up to produce heating, hot and chilled water, pumps, piping, and controls.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Maintained	Maintained
FUNCTIONS:		
Heat Fluid	Cool Fluid	Pump Fluid
ADVANTAGES: • Reduced work at existing • Lower first cost • Simplified controls • More energy-efficient addition building • Makes use of extra space in the addition large mechanical room		DISADVANTAGES: • Air-to-water heat pumps at addition building • Renewed chillers instead of new heat pumps at existing plant • Requires an outside air-cooled heat pump adjacent to, or on the roof of the addition

DISCUSSION AND JUSTIFICATION

The current design both increases existing central plant capacity and replaces most of the existing plant equipment, except the three newer (2010) gas-fired boilers. While the existing two air-cooled Petra chillers are aging, they are not at end-of-life and can be renewed for a fraction of the cost of replacing them with new air-to-water heat pumps. The new addition building is remote from the central plant, hence there is pumping energy loss from the central plant to the addition. The new addition hydronic heating hot water system operates at lower temperature (140 deg F) than the existing building (180 deg F) during cold weather; hence the current design has two different central plant heating hot water temperature loops.

The existing school buildings are capable of full heating and cooling from the existing four-pipe boiler/chiller central plant; with new (2010) boilers and aging (2004) chillers. While the chillers are aging, they are high-quality Petra chillers, and can be renewed (re-conditioned) for 10 to 15-years additional life, at which time the entire HVAC system can be fully modernized. The existing four-pipe hydronic system operates at 180 deg F heating and 45 deg F chilled water temperature.

The new addition is planned to operate on a four-pipe system, but with lower heating water temperature at 140 instead of 180 deg F.

Since the existing campus buildings and the new addition operate at different heating hot water temperatures, additional heat exchangers, pumps, piping, and controls are needed. This increases project cost, complicates maintenance, and reduces energy efficiency. Additionally, the new addition is remote from the existing plant, which requires additional pump energy, further reduces energy efficiency, and complicates ceiling space in the existing school building.

This proposal suggests a simpler approach with a new, small four-pipe hydronic plant with one air-source heat pump and electric boiler back-up; redundancy is provided for heating, but not for cooling. The substantial cost savings will more than pay for the renewal of the existing two air-cooled chillers. Plus, there is no need to upgrade existing building controls or fan coils.

SCHEDULE SAVINGS:

The smaller, simpler plant for the addition, and renewal instead of replacement of the existing chillers, shortens project schedule and simplifies phasing.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Improved. Reliability is increased due to simpler systems, but also reduced with only double, instead of quadruple, redundancy at the heating & cooling water plants.

Maintenance & Operations:

Improved. Maintenance & Operations is simplified due to a simpler HVAC system, noting there will be one larger heating and cooling plant for the existing buildings, and one smaller plant for the addition.

Functionality:

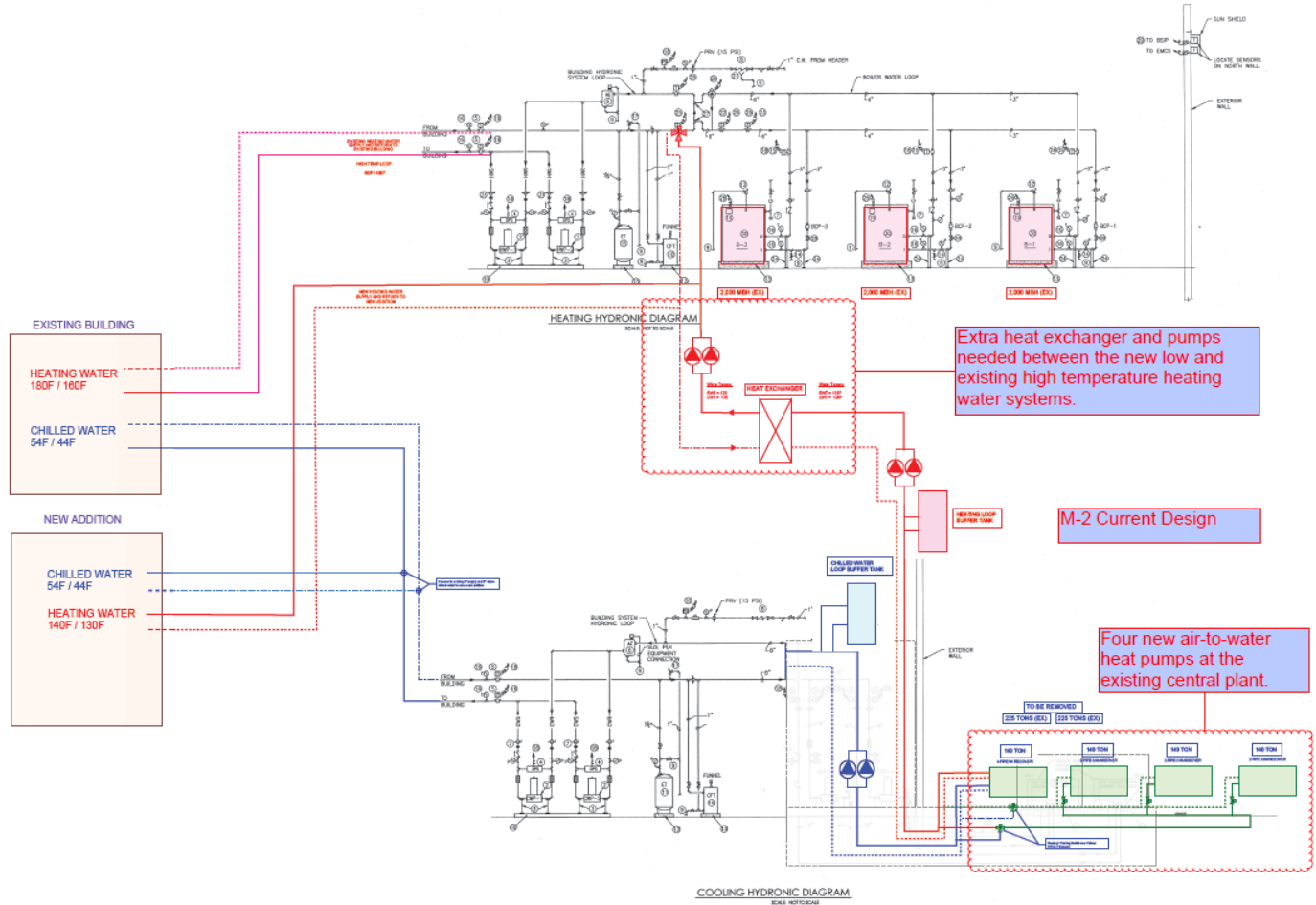
Maintained. Functionality remains substantially the same with full heating and cooling throughout the existing and new buildings.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

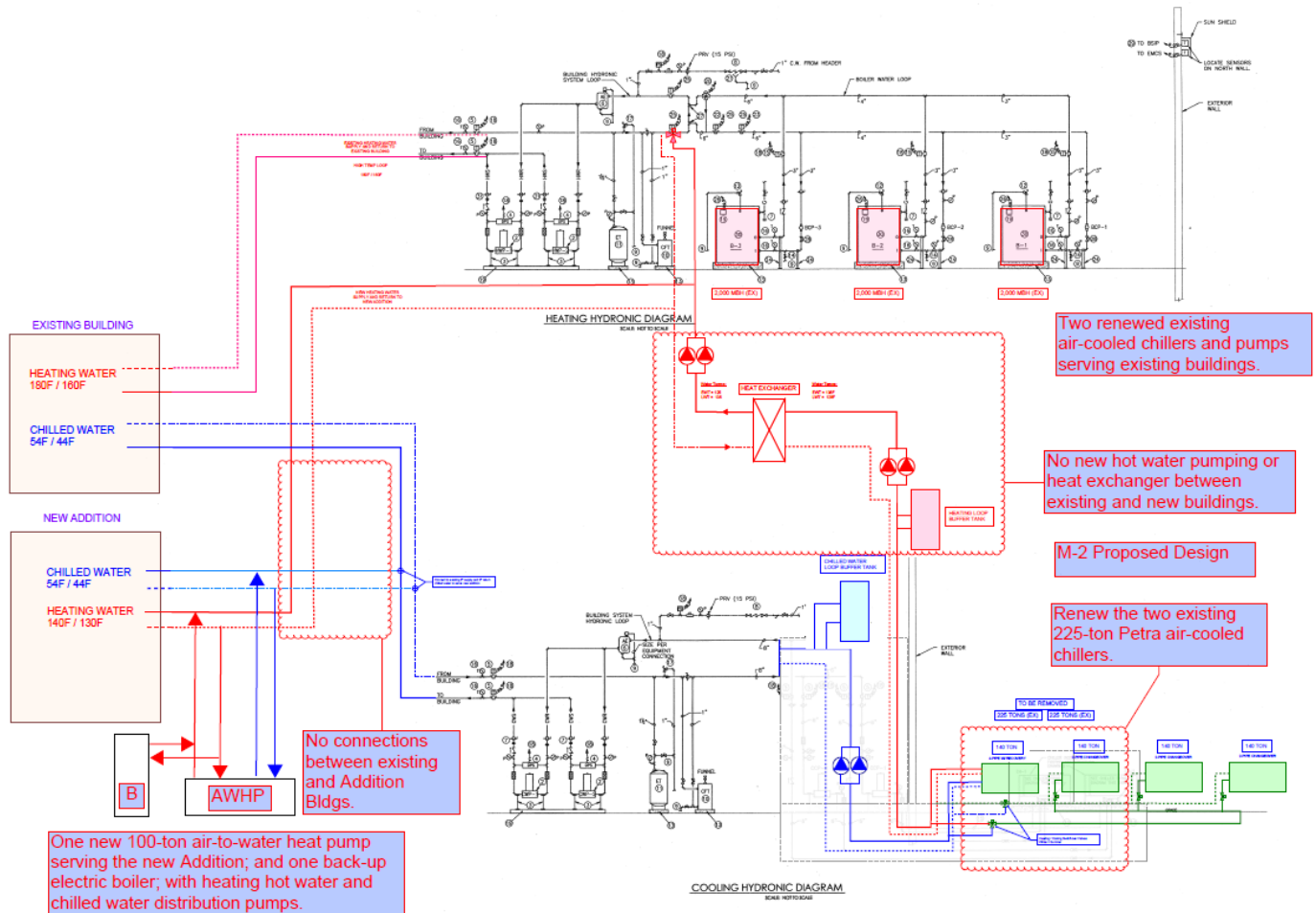
Interested

SKETCHES/DIAGRAMS:

BASELINE CONCEPT



VALUE PROPOSAL CONCEPT



ASSUMPTIONS AND CALCULATIONS:

See above narrative, diagrams, and related cost estimate.

COST ESTIMATE FORM

COMPONENT:

Thermal Generation

M2

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Central Plant with four heat pumps	4	EA	679,500	\$2,718,000	Renew existing Petra chillers	2	EA	50,000	\$100,000
Central Plant electrical work	1	EA	150,000	\$150,000	New air-source heat pump at Addition; 100-ton capacity	100	TON	4,200	\$420,000
					Back-up electric boiler 340-kW	1	LS	100,000	\$100,000
					Addition plant, electrical work	1	LS	100,000	\$100,000
Subtotal				\$2,868,000	Subtotal				\$720,000
General Contractor Markup	26.41%			\$757,439	General Contractor Markup	26.41%			\$190,152
Total to nearest \$1000				\$3,625,000	Total to nearest \$1000				\$910,000
					Difference				\$2,715,000

MENG Analysis

Proposal

M2

	PROPOSAL	M3
COMPONENT: Energy Source	AUTHOR	DCS
BASELINE CONCEPT: Central plant energy source from atmospheric air via four new air-to-water heat pumps, plus three existing gas-fired boilers used for heating during cold weather.		
VALUE PROPOSAL SYNOPSIS: Central plant energy source from geothermal ground exchanger via new water-to-water heat pumps, plus three existing gas-fired boilers used during freezing cold weather.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Improved	Improved	Improved
FUNCTIONS:		
Conserve Energy	Heat Fluid	Cool Fluid
ADVANTAGES: • More energy efficient • Lower life cycle cost • Reduced carbon footprint • More CBPS ready • Reduced service yard noise • Operates during extreme cold		DISADVANTAGES: • Requires coordination between mechanical and site work • Distance between plant and geo-exchanger is a concern

DISCUSSION AND JUSTIFICATION

The current design heating and cooling energy source is air; with low-efficiency air heat pumps generating hydronic heating and chilled water cooling water with coefficients of performance (COP) in the 2 to 3 COP-range. Additionally, the air-source heat pumps occupy significant space in the service yard and generate unwanted noise, especially during heavy mid-winter and mid-summer loads. This proposal suggests a significantly more efficient ground-source heat pump system with COP in the 4 to 5-range, nearly twice as efficient at the current design, plus out-of-sight, out-of-mind heat pumps, which can be located in the existing reportedly over-sized boiler room. The complete reconstruction of two of the existing sports fields presents a unique cost-saving opportunity to install a horizontal ground-exchanger for the proposed ground-source heat pump system.

SCHEDULE SAVINGS:

May add to schedule to construct the geothermal ground exchanger as part of the site redevelopment work. Assuming the ground exchanger is under the sports fields, the ground exchanger must be installed before the sport fields can be completed.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Improved. Reliability is improved as geo-exchangers are passive under-ground devices with no moving parts and no direct exposure to atmospheric weather. Plant equipment reliability is improved with more stable condenser water temperatures.

Maintenance & Operations:

Maintained. Maintenance & Operations on one hand is simpler as geothermal systems are not directly subject to weather; on the other hand, they are less familiar to some, and the geo-exchanger condenser water system adds an element of complexity compared to air-cooled heat pumps.

Functionality:

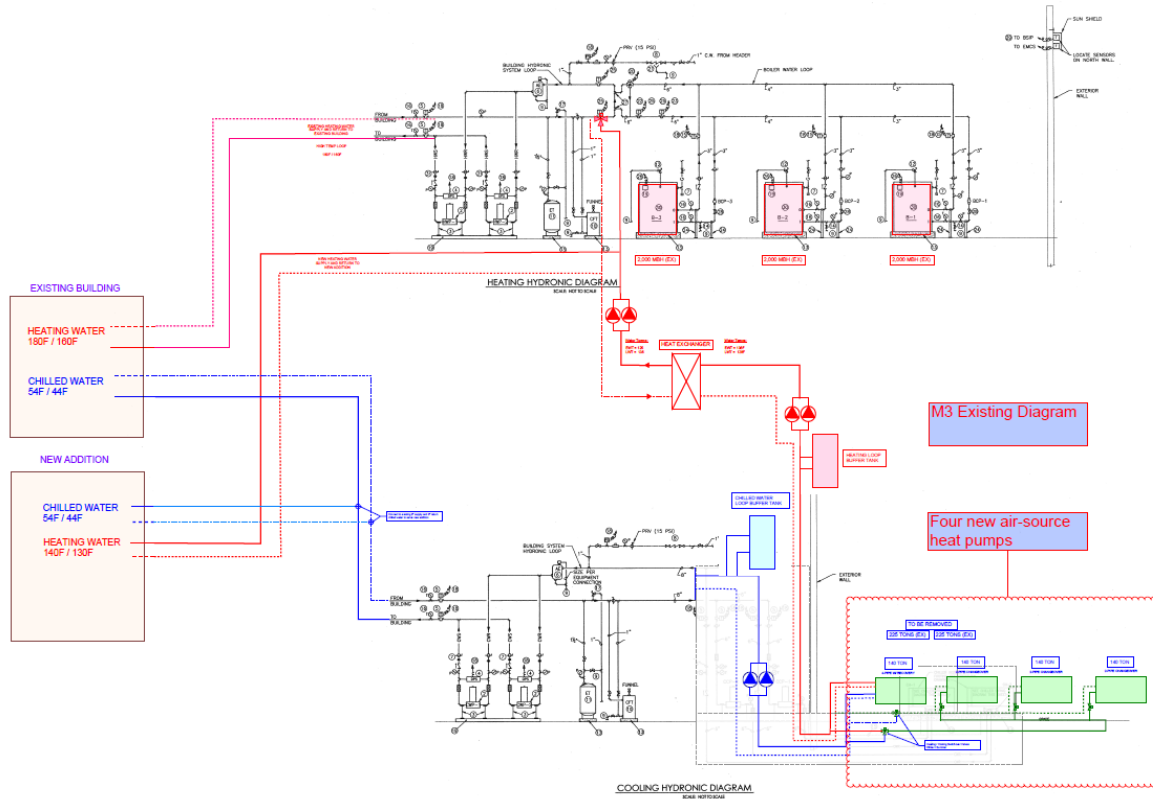
Improved. Functionality is improved as geothermal systems are more reliable during extreme weather.

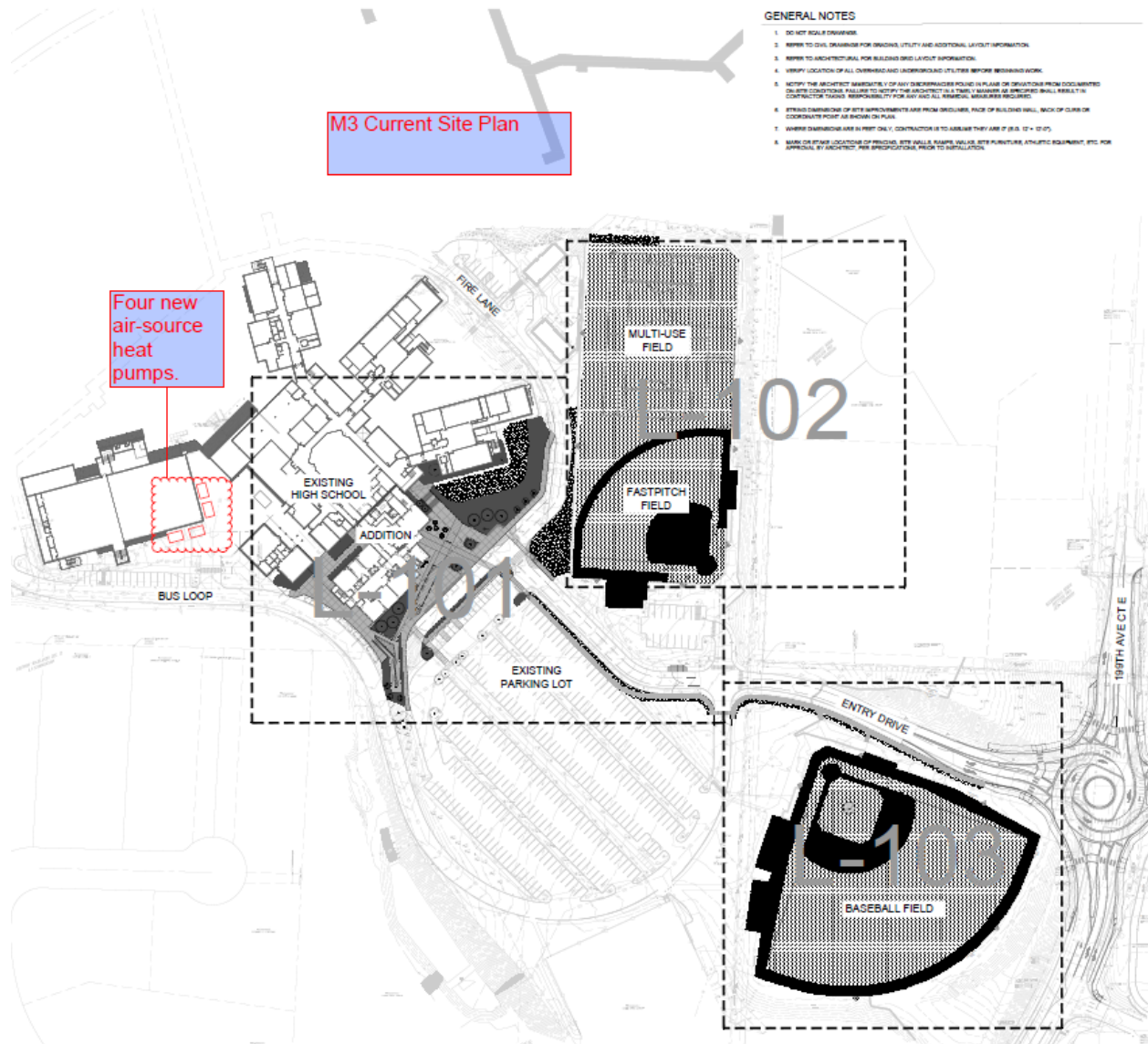
REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

Interested.

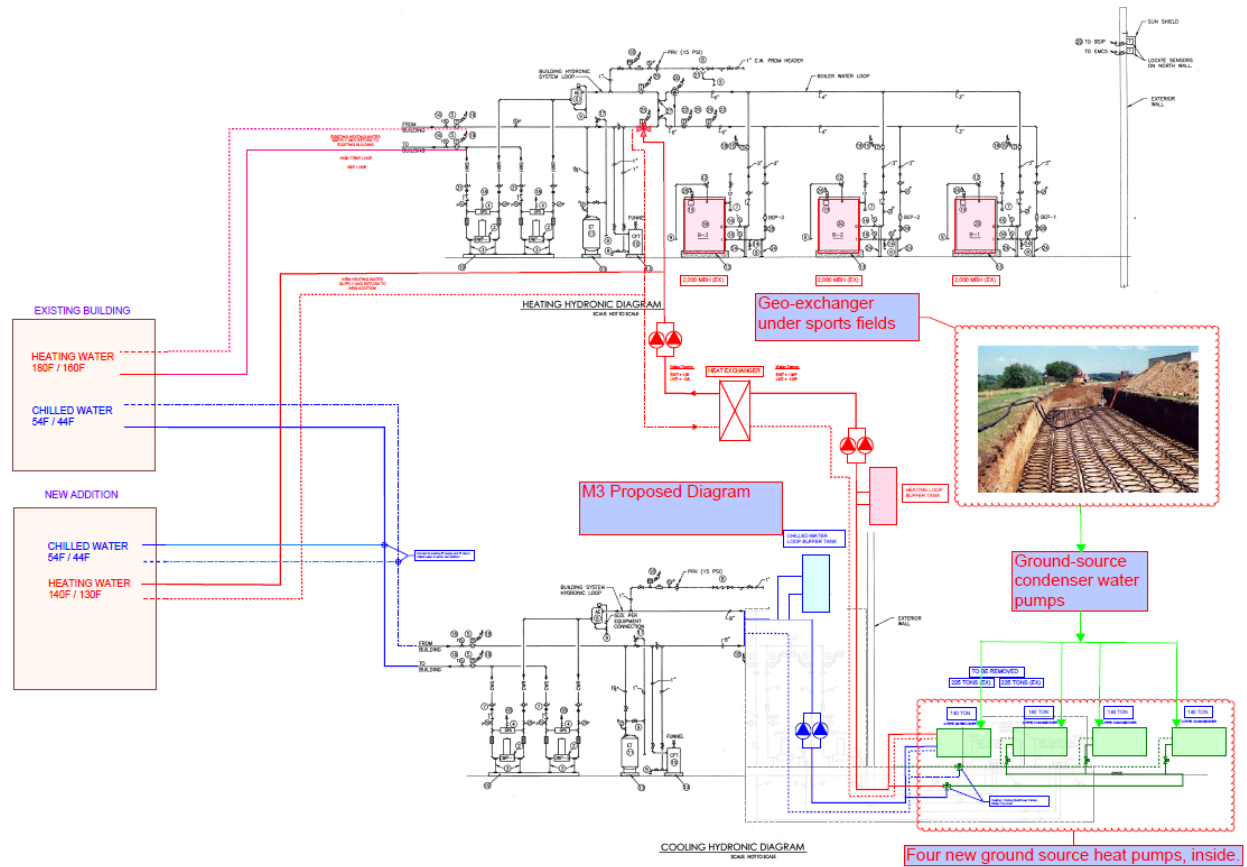
SKETCHES/DIAGRAMS:

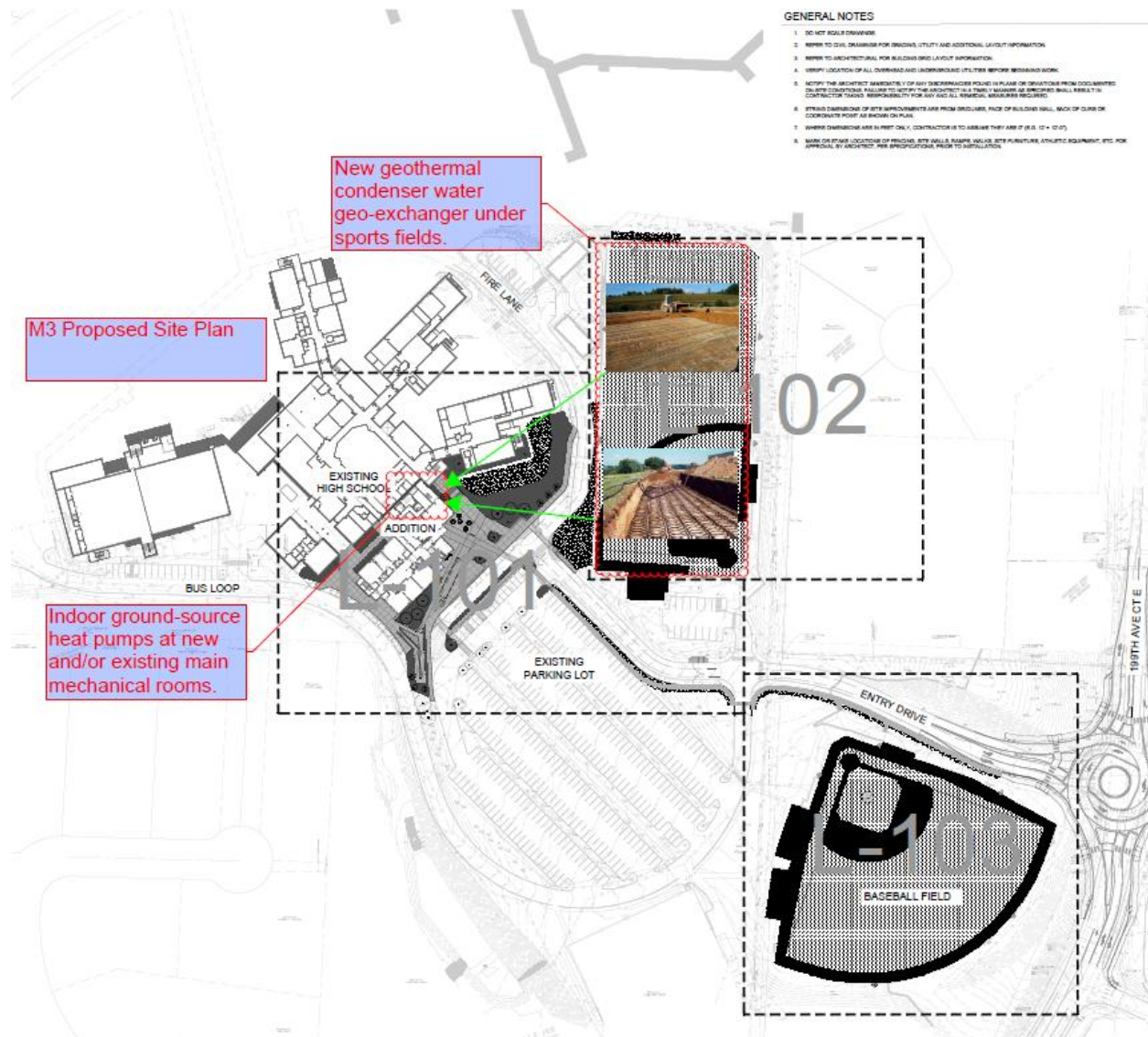
BASELINE CONCEPT





VALUE PROPOSAL CONCEPT





ASSUMPTIONS AND CALCULATIONS:

See above narrative and diagrams, plus accompanying cost estimate.

COST ESTIMATE FORM

COMPONENT:

Energy Source

M3

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Central Plant with four air-source heat pumps	4	EA	679,500	\$2,718,000	Central plant geothermal heat pumps	4	EA	450,000	\$1,800,000
Central Plant electrical work	1	LS	150,000	\$150,000	Geo-exchanger, horizontal	560	TON	1,500	\$840,000
					Central plant electrical work	1	LS	135,000	\$135,000
Subtotal				\$2,868,000	Subtotal				\$2,775,000
General Contractor Markup	26.41%			\$757,439	General Contractor Markup	26.41%			\$732,878
Total to nearest \$1000				\$3,625,000	Total to nearest \$1000				\$3,508,000
					Difference				\$117,000

MENG Analysis

Proposal

M3

COMPONENT LIFE CYCLE COST ANALYSIS (LCCA)

Client:	Sumner-Bonney Lake School District	M3
Project:	Bonney Lake High School Expansion & Intersection Improvements	proposal #
Date:	12-Sep-25	
By:	DCS	
COMPONENT	Energy Source	
COMPONENT #	M3	
Escalation rate	0.03	
Discount rate	0.04	
Study Period	50 Yrs.	

Notes: Per the shoebox energy model annual energy cost is \$50K/yr, with 45% or \$22.5K/yr for HVAC. This increases central plant energy efficiency by an estimated 25%, saving \$6K/yr. Assume 1% of initial cost for annual maintenance cost, 10% for renewal year 25. Assume 10% salvage value.

Instructions: Enter escalation, discount, and study period above.
Enter annual costs, replacement costs (and appropriate replacement year), and salvage value.
Enter these costs in the shaded cells using today's (current) dollars. For annual costs, escalation rates will be automatically entered, but can be individually overwritten below for differential escalation.
All costs will automatically be escalated and discounted.

ALTERNATIVE A : Current				ALTERNATIVE B: Proposed				DIFFERENCE
INITIAL COSTS				INITIAL COST				DIFFERENCE
3,625,000				\$ 3,508,000				\$117,000.00
O & M ANNUAL COSTS								
STAFFING OPERATIONS								
STAFFING MAINTENANCE								
SUPPLIES OPERATIONS								
SUPPLIES MAINTENANCE								
Subcomponents	Cost in current \$	Esc.	Pres. Worth \$	Subcomponents	Cost in current \$	Esc.	Pres. Worth \$	
Annual maintenance cost	36,250	0.03	\$ 1,463,350	Annual maintenance cost	35,080	0.03	\$ 1,416,119	\$47,230.87
Annual energy cost	50,000	0.03	\$ 2,018,413	Annual energy cost	44,000	0.03	\$ 1,776,204	\$242,209.59
		0.03	\$ -			0.03	\$ -	\$0.00
SUBT. O & M OVER LIFE CYCL \$ 86,250 \$ 3,481,763				79,080 \$ 3,192,322				\$289,440.46
MAJOR REHAB REPLACEMENT COSTS								
Subcomponents	Cost in current \$	Yr.	Pres. Worth \$	Subcomponents	Cost in current \$	Yr.	Pres. Worth \$	
			\$ -				\$ -	\$0.00
Maintenance rehab	362,500	25	\$ 291,642	Maintenance rehab	350,800	25	\$ 282,229	\$9,412.99
			\$ -				\$ -	\$0.00
			\$ -				\$ -	\$0.00
			\$ -				\$ -	\$0.00
			\$ -				\$ -	\$0.00
			\$ -				\$ -	\$0.00
SUBT. REPLACEMENT			\$ 292,000				\$ 282,000	\$10,000.00
TOT. O & M & REPL. (Pres. Worth)				3,474,000				\$300,000.00
TOT. INITIAL, O&M, & REPL. (Pres. Worth)				6,982,000				\$417,000.00
	Cost in current \$				Cost in current \$			
SALVAGE VALUE	362,500	50	\$ 235,000		350,800	50	\$ 227,000	\$8,000.00
TOT. INITIAL, O&M, REPL. MINUS SALVAGE			7,164,000				6,755,000	\$409,000.00

	PROPOSAL	E1
COMPONENT: Transfer Power	AUTHOR	CT
BASELINE CONCEPT: Addition of an automatic transfer switch to facilitate the new classroom addition emergency power.		
VALUE PROPOSAL SYNOPSIS: Upsize existing automatic transfer switch to an increased ampacity automatic transfer switch to include the new classroom addition emergency power.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Improved	Improved
FUNCTIONS:		
Transfer Energy	Improve Function	Connect Existing
ADVANTAGES: - Improved reliability - Reduce maintenance - Less cost		DISADVANTAGES: - Increase breaker size at generator - Reduces redundancy

DISCUSSION AND JUSTIFICATION

The proposed system is introducing a Life Safety transfer switch for just the new addition. The existing building currently has a stand-by automatic transfer switch with battery back up at the egress lights to meet code. There is no good way to segregate the existing branch circuits that are on the generator from Life Safety and Stand-by without a lot of rewiring.

This study evaluated upgrading the existing transfer switch to a larger transfer switch, upgrading the existing emergency distribution panel, and sub-feeding a stand-by panel to support the loads that the district desires to be supported by an emergency generator. This eliminates the need for an additional automatic transfer switch to be added.

SCHEDULE SAVINGS:

Maintained.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Improved. The transfer switches would need to be sequenced in order for the life safety transfer to occur first.

Maintenance & Operations:

Improved. By enlarging the existing transfer switch and emergency distribution panel in the existing school, the district can add additional loads that they desire to be on the generator as well as facilitating the egress lighting in the new addition.

This will include providing battery back-up or a centralized inverter for the egress lighting in the corridors.

Functionality:

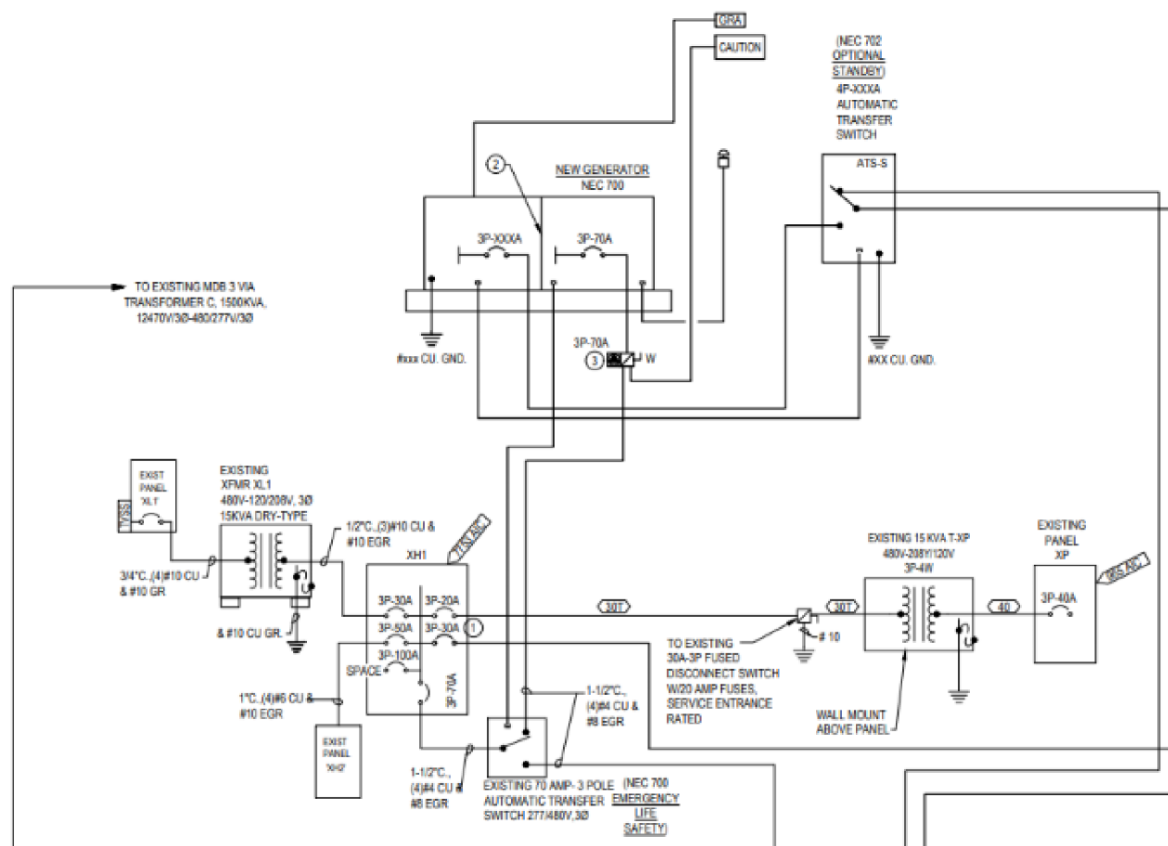
Improved. It will allow the IDF room to be on the generator as well as the corridor lighting without routing multiple feeders from the existing building electrical stand-by distribution panel.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

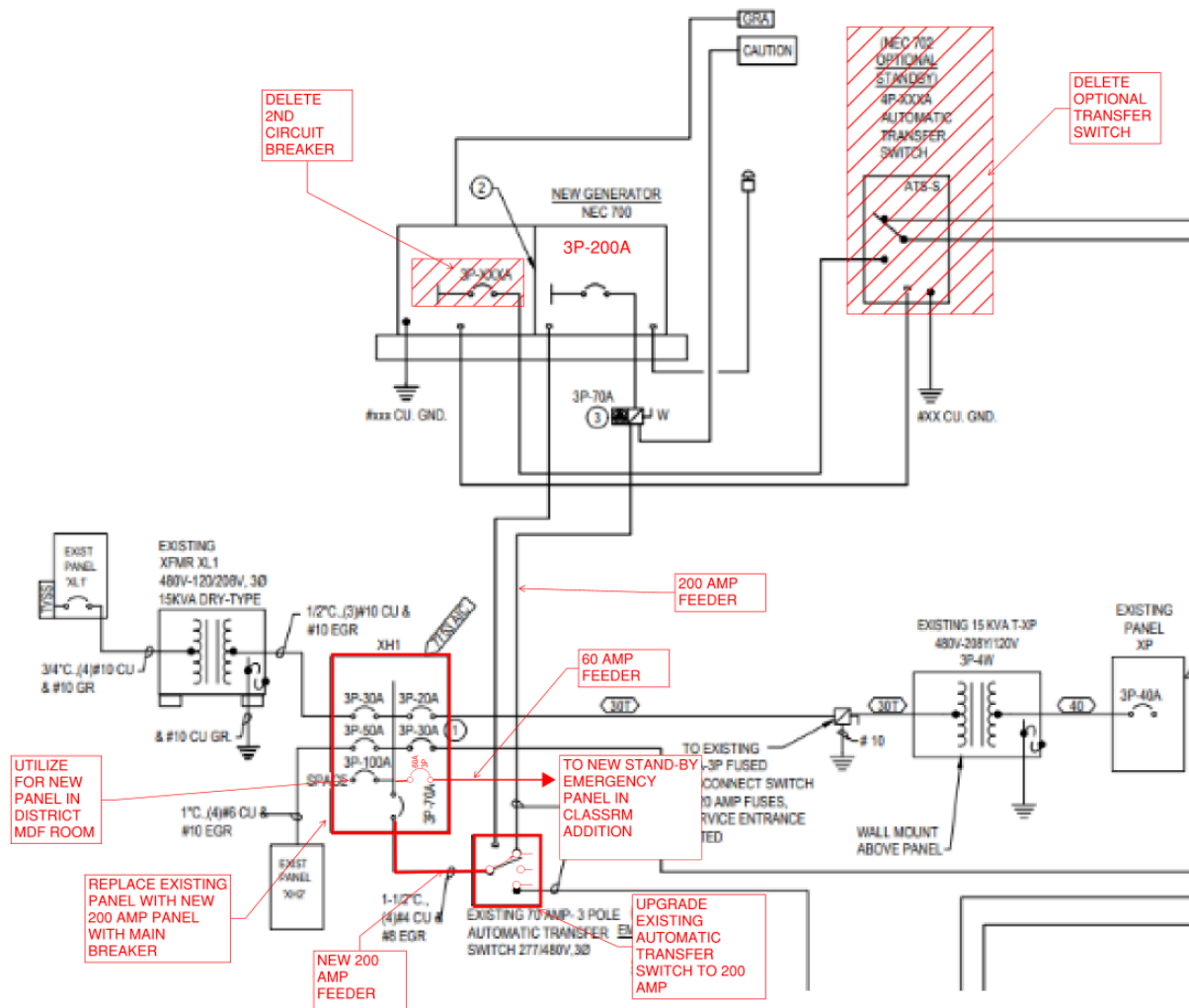
1. The district expressed that there is a need to add additional items to the generator. This will help facilitate that without segregating the existing building lighting and the other loads that are currently on the existing transfer switch.
2. Since the generator is slated to be replaced due to age, upsizing the existing emergency equipment panel and automatic transfer switch allows for the addition of new loads.

SKETCHES/DIAGRAMS:

BASELINE CONCEPT



VALUE PROPOSAL CONCEPT



ASSUMPTIONS AND CALCULATIONS:

Due to a quick estimated load calculation for the new generator; upgrading of the existing transfer switch and existing Emergency Distribution Panel to a 200 Amp will facilitate the new addition and estimated new loads that the district would like on the generator.

STAND-BY GENERATOR ESTIMATED LOAD					
CLASSROOM ADDITION					
	Qty	Watts	Total W	Total kW	Total Amps
Egress Lighting	40	20	800		
IDF	1	5000	5000		
			5800	5.8	7.0
EXISTING SCHOOL					
Based off PAC -					
metered and added				21.25	
load					
Add MDF				30	
Add Cooler				10.21	
Add Freezer				5.21	
Total Est. Load				72.47	87.0
Estimated Gen Size				150	180.0

COST ESTIMATE FORM

COMPONENT:

Transfer Power

E1

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
60 Amp Automatic Transfer Switch	1	EA	20000	\$20,000	200 Amp Automatic Transfer Switch	1	EA	35000	\$35,000
60 Amp panel	2	EA	2500	\$5,000	200 Amp feeder from gen set to new panel	60	LF	80	\$4,800
60 Amp feeder	300	LF	15	\$4,500	New 200 Amp panel	1	EA	10000	\$10,000
100 Amp circuit breaker	1	EA	500	\$500	Demo existing panel	1	LS	2500	\$2,500
					Reconnect existing branch circuits	1	LS	8000	\$8,000
					60 Amp panel	1	EA	2500	\$2,500
					60 Amp feeder	300	LF	15	\$4,500
					200 amp circuit breaker	1	EA	1200	\$1,200
Subtotal				\$30,000	Subtotal				\$68,500
General Contractor Markup	26.41%			\$7,923	General Contractor Markup	26.41%			\$18,091
Total to nearest \$1000				\$38,000	Total to nearest \$1000				\$87,000
					Difference				(\$49,000)

MENG Analysis

Proposal

E1

	PROPOSAL	E2
COMPONENT: Store Power	AUTHOR	CT
BASELINE CONCEPT: Emergency power from a new upgraded emergency diesel generator serves the existing building and addition.		
VALUE PROPOSAL SYNOPSIS: Retain the existing generator to continue to serve the existing building and utilize a battery energy storage system (BESS) for all emergency and stand-by power for the addition.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Improved	Degraded	Improved
FUNCTIONS:		
Conduct Energy	Power Devices	Transmit Energy
ADVANTAGES: • Maintain existing emergency distribution system in existing school • Existing generator can be maintained or integrated with a larger generator • Potential grant from utility for implementing • Utilization of power produced by PV system ILO putting it back to the grid		DISADVANTAGES: • Increased costs • Increased maintenance • Newer technology • Allocation of space for the unit on site • Replacement battery cost

DISCUSSION AND JUSTIFICATION

A Battery Energy Storage System (BESS) is designed to store electrical energy for later use. A BESS can respond quickly to fluctuations in energy generation and consumption, making them ideal for integrating renewable energy sources. They can deliver full power for durations ranging from 1 to 4 hours or longer depending on the battery bank sizing.

Key Functions and Applications:

- **Backup Power:** Provides backup power during outages and optimizes energy use from renewable sources.
- **Grid Stabilization:** Helps provide a steady power flow, balancing supply and demand on the grid during peak usage times.
- **Peak Shaving:** The systems stored energy can be utilized during peak demand periods to reduce the need for additional power generation, lowering energy costs.

SCHEDULE SAVINGS:

Maintained. The unit is self-contained so it is delivered to the site and set in place by the manufacturer.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Improved. It provides power in case there is an issue with the generator starting.

Maintenance & Operations:

Degraded. While the unit comes self-contained with a sprinkler system, the batteries have a life expectancy of approximately 10-years with lithium-ion batteries or 3-5 years with lead acid batteries. The system will require regular maintenance that can be provided by the manufacturer of the system.

Functionality:

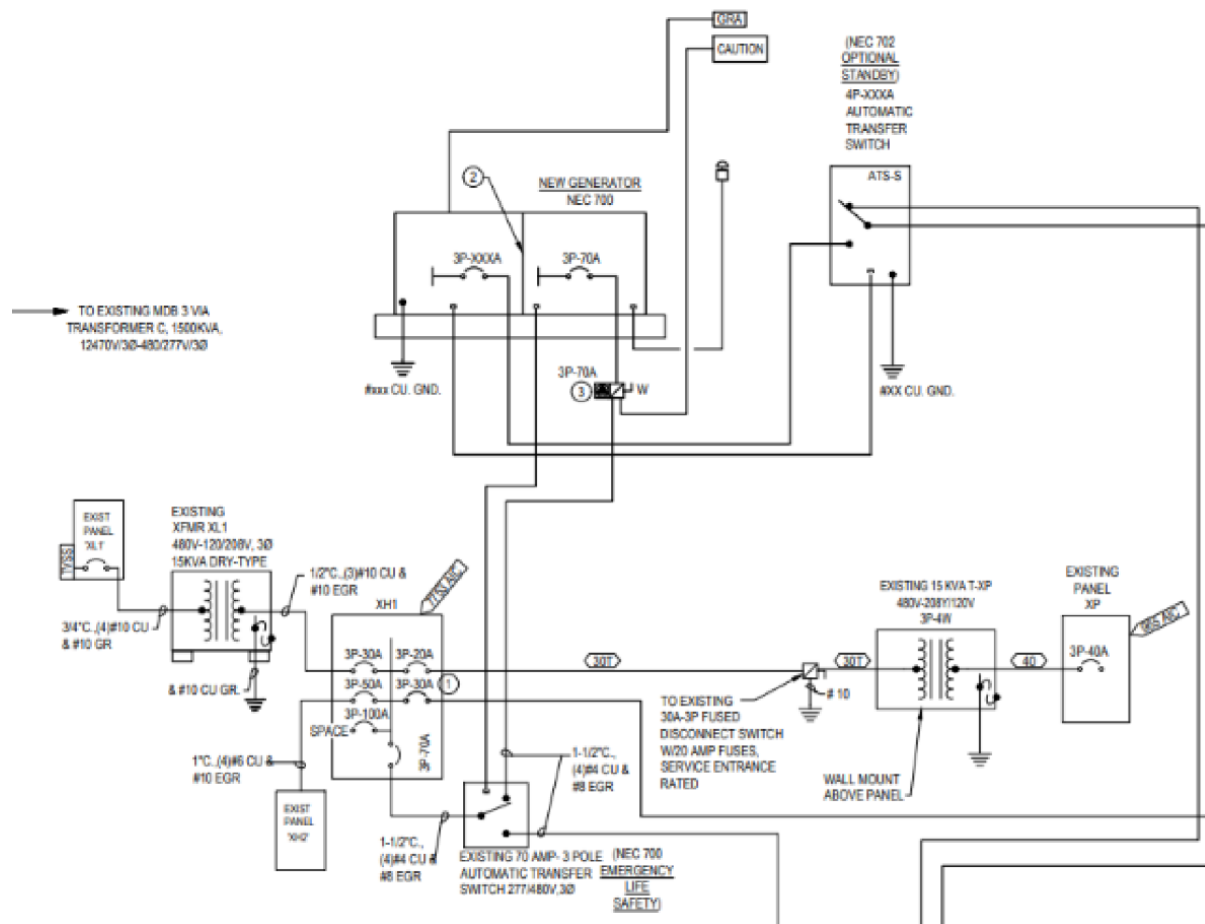
Improved. Since a BESS system takes advantage of the power produced by the solar array on the roof, the district is able to utilize the energy for the facility during peak demands and power outage with the diesel generator being a back-up during prolonged outages. The BESS system has zero emissions and noise attributing to the environment.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

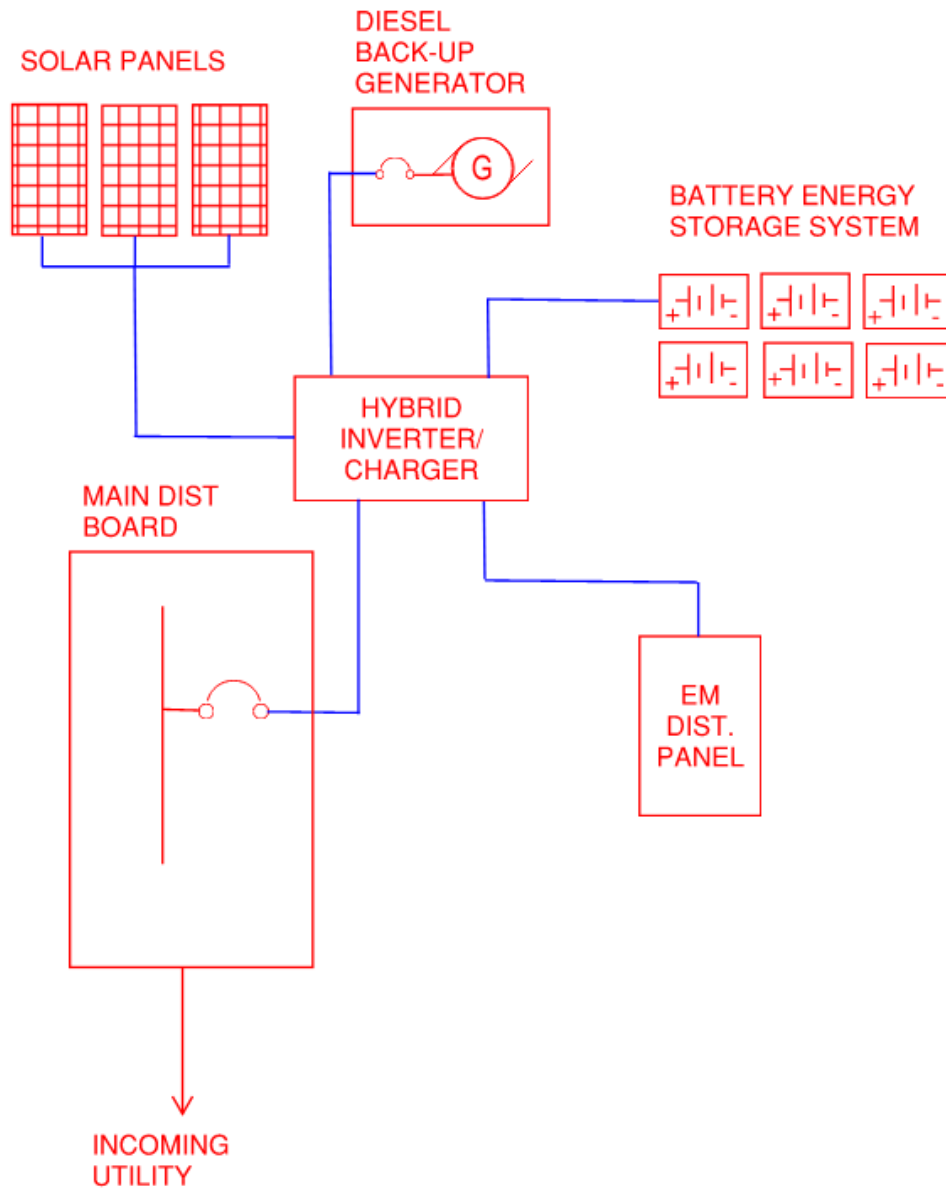
1. The district expressed an interest in what a battery energy storage system could do to support the facility.
2. The district is working on demand response with the utility company for mitigation of their peak power needs.
3. The district will be able to harvest and store energy that the photovoltaic system produces.
4. It would be a great student learning opportunity.

SKETCHES/DIAGRAMS:

BASE LINE CONCEPT



VALUE PROPOSAL CONCEPT





BESS units come self-contained with batteries, power management system, inverter/chargers, distribution panel and fire suppression system.

ASSUMPTIONS AND CALCULATIONS:

BESS systems are rated based off kilo-watt of load and duration that the load is to be sustained – kilo-watt hour (KwH). The following estimated calculation assumes 3 hours of run time.

BESS					
CLASSROOM ADDITION					
	Qty	Watts	Total W	Qty HRS	Total KwH
Egress Lighting	40	20	800		
IDF	1	5000	5000		
			5800	3	17.4
EXISTING SCHOOL					
Based off PAC - metered and added load			21.25	3	63.75
Add MDF			30	3	90
Add Cooler			10.21	3	30.63
Add Freezer			5.21	3	15.63
Total Est. Size					217.41

COST ESTIMATE FORM

COMPONENT:

Store Power

E2

CURRENT DESIGN					VE PROPOSAL				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Upgrade Existing Generator	1	EA	100000	\$100,000	BESS System	211	kWH	600	\$126,600
200 amp Automatic Transfer Switch (based off GFT estimate)	1	EA	20000	\$20,000	200 amp feeder	112	LF	80	\$8,960
					Integration with PV and generator	1	LS	20000	\$20,000
Subtotal				\$120,000	Subtotal				\$155,560
General Contractor Markup	26.41%			\$31,692	General Contractor Markup	26.41%			\$41,083
Total to nearest \$1000				\$152,000	Total to nearest \$1000				\$197,000
					Difference				(\$45,000)

MENG Analysis

Proposal

E2

	PROPOSAL	E3
COMPONENT: Electrical Service	AUTHOR	CT
BASELINE CONCEPT: 480 –volt power from existing school to the new addition with transforming voltage to 208-volt for distribution.		
VALUE PROPOSAL SYNOPSIS: Do not install a 480-volt feeder and transformer, provide 208/120 volt distribution panels only for the addition, especially as large mechanical systems are not needed in the Addition.		
SCHEDULE SAVINGS: Maintained		
QUALITATIVE BENEFITS:		
RELIABILITY	M&O	FUNCTIONALITY
Maintained	Improved	Improved
FUNCTIONS:		
Deliver Energy	Transform Energy	Power Devices
ADVANTAGES: • Less equipment • Less ventilation required • Smaller breaker at MDB-1 which does not take up as much space		DISADVANTAGES: • A smaller transformer will be required for the emergency lighting since the voltage will be 120-volt • Outside location may impact the visual appeal of the plaza

DISCUSSION AND JUSTIFICATION

Since all of the loads indicated for the new classroom building are 120/208 volt, and the large mechanical equipment that requires 480-volt is located in the existing building, there does not appear to be a need for a large 480-volt distribution board at the new classroom addition.

By placing the 480-208/120-volt; 300 kVA transformer outside, it eliminates the heat gain associated with it. This allows space within the electrical room for the PV inverters from the roof to be located.

SCHEDULE SAVINGS:

This would allow for the early procurement of the transformer and placement of it while the construction of the addition is going on.

QUALITATIVE BENEFITS – This proposal impacts the following:

Reliability:

Maintained. There is no additional impact.

Maintenance & Operations:

Improved. The transformer is located outside. There is an increased heat load from the transformer in the room.

Functionality:

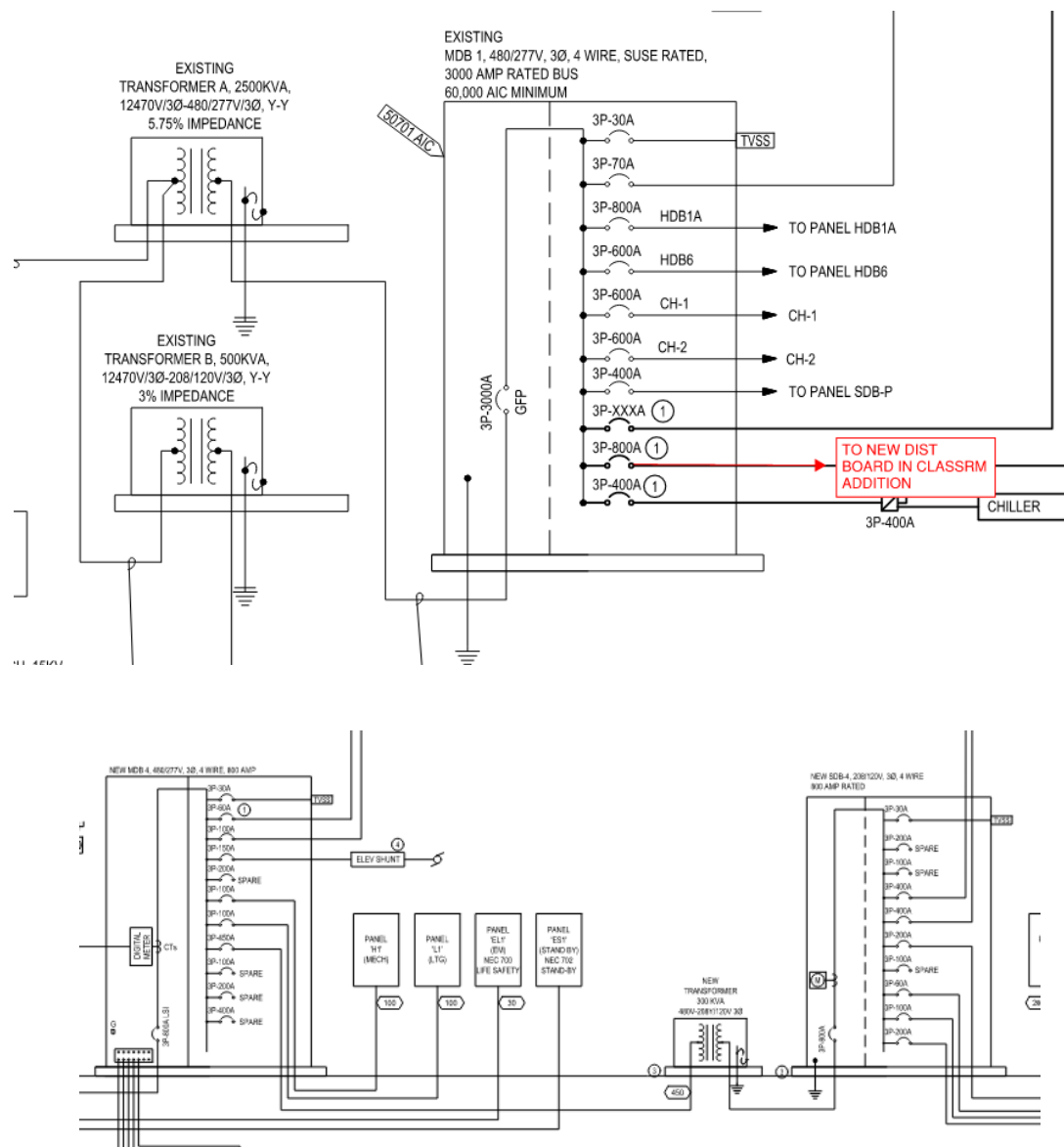
Maintained.

REVIEW COMMENTS: During the evaluation phase of the workshop, stakeholders participated in the mid-point evaluation phase review and had the following comments about this idea:

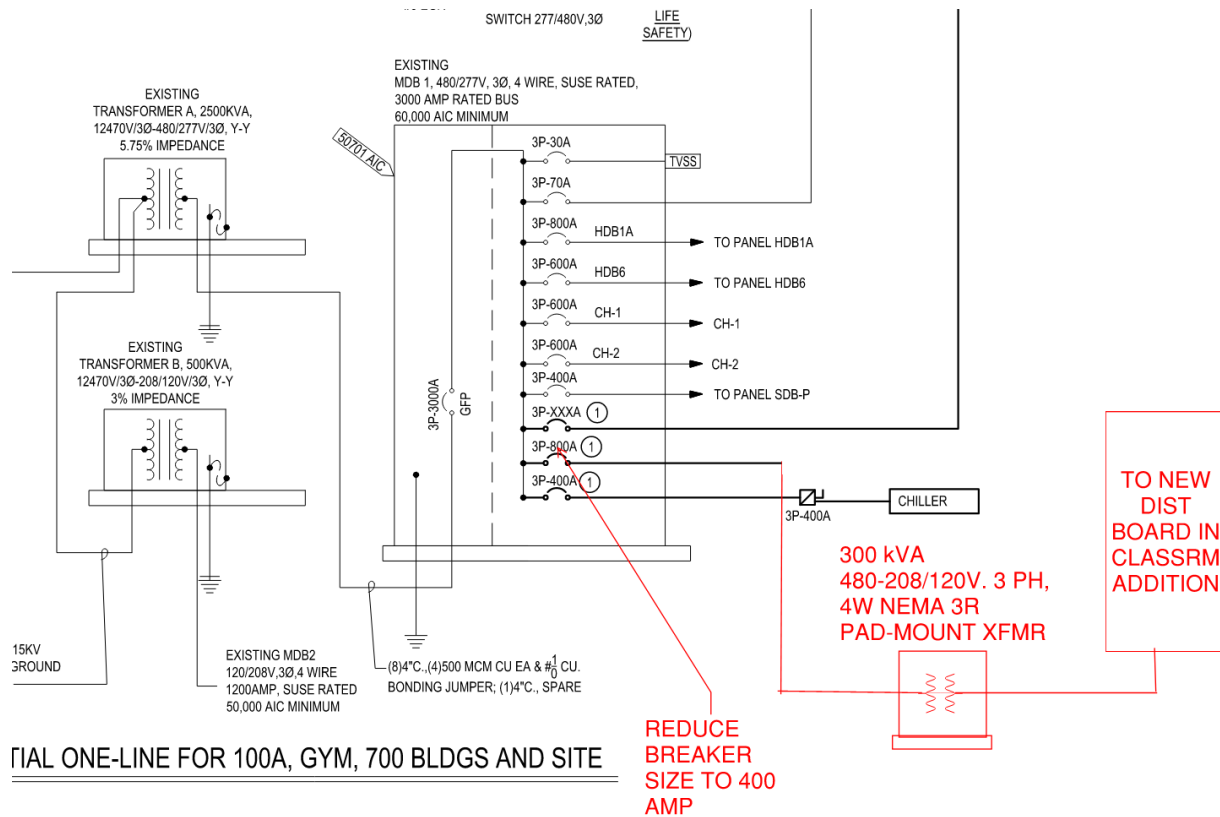
1. Less equipment.
2. Large mechanical equipment that requires the 480-volt is located in the main mechanical room of the existing school.

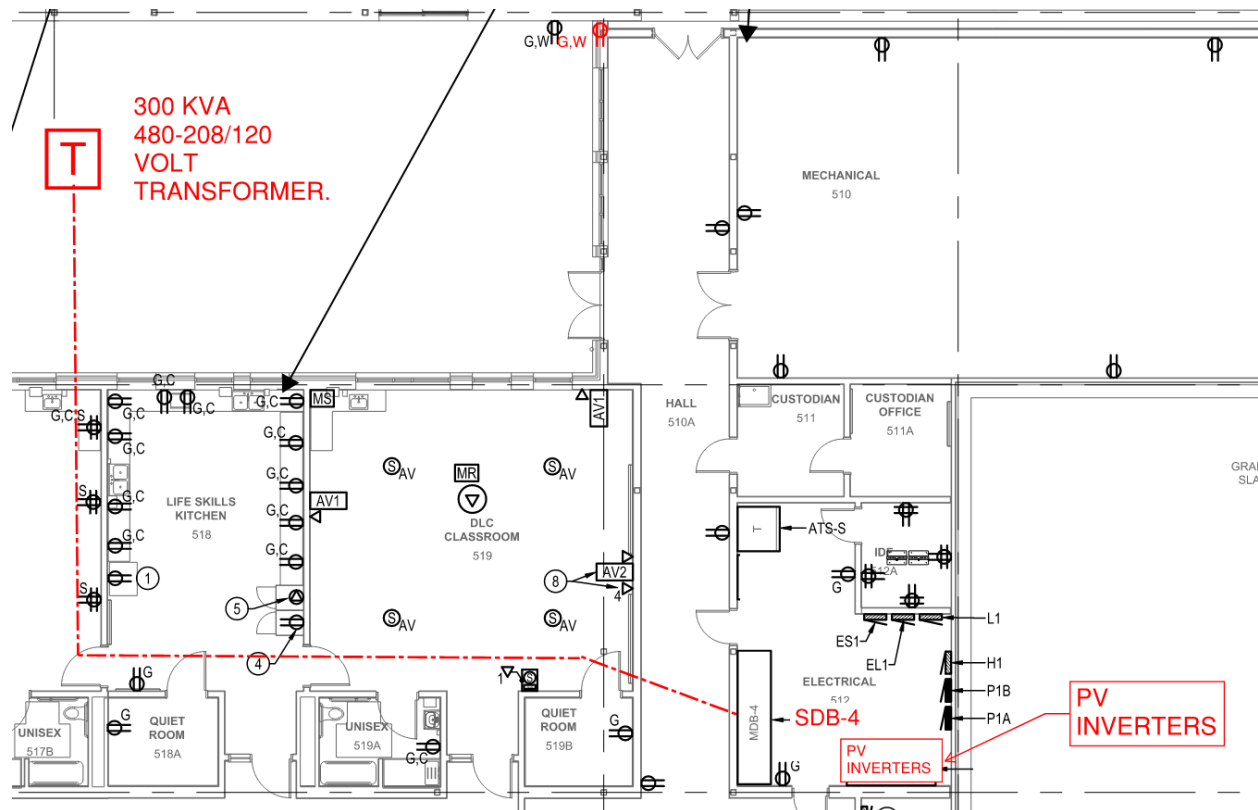
SKETCHES/DIAGRAMS:

BASELINE CONCEPT



VALUE PROPOSAL CONCEPT





ASSUMPTIONS AND CALCULATIONS:

1. The large mechanical loads requiring 480-volt power are located in the existing mechanical room and the existing service yard.
2. Since the lighting would now be fed with 120-volt; the emergency lighting panel will need to be 120/208V and a 30 kVA dry-type transformer to serve it.

COST ESTIMATE FORM

COMPONENT:

Electrical Service

E3

[illegible]

MENG Analysis

Proposal

E3

IV. TECHNICAL REPORTS

ID	Name
T1	Energy Grants
T2	Clean Buildings
T3	Construction Considerations

TECHNICAL REPORT	PROPOSAL	T1
COMPONENT: Energy Grants	AUTHOR	DCS
CURRENT CONCEPT While the Basis of Design (BOD) mentions Puget Sound Energy (PSE) energy efficiency grants, specifics are not provided. Some new systems, such as the Addition building envelope are code minimum. CONSIDERATIONS Apply for PSE energy efficiency grants for both existing space modernization, and for new Addition energy performance above code minimum. Puget Sound Energy (PSE) provides both power and natural gas to the BLHS campus and is currently offering the most generous grants ever to reduce owner direct cost for energy efficiency measures (ECMs) above code minimum. For example, the premium thermal envelope upgrades suggested in proposal M1 – Performance Engineering, may be heavily subsidized by PSE. Of specific interest is PSE’s new energy use intensity (EUI) grant program, which pays a generous incentive immediately upon project completion, with a one-year performance bonus after the first year of full occupancy. In general, the PSE grants pay \$0.45/kW-hr saved during the first year of operation. Under the EUI grant, the baseline for new buildings is set at 85% of the WA State Clean Buildings EUI target for schools; for high schools this is $EUI\ 48 \times 0.85 = EUI\ 41$. For further information see the PSE business incentive webpage: https://www.pse.com/en/business-incentives 		

TECHNICAL REPORT

PROPOSAL

T2

COMPONENT: Clean Buildings

AUTHOR

DCS

CURRENT CONCEPT:

No WA State Clean Buildings Performance Standard rule compliance plan. However, an Energy Audit has been performed on the existing facility.

CONSIDERATIONS

Develop a roadmap to Net Zero Energy for long-term compliance with the WA State CBPS law.
BLHS existing and addition buildings are subject to the new WA State CBPS program, requiring buildings to meet energy performance targets which are reduced every five years, leading to net-zero by 2050, if not 2045. This technical report suggests the District develop a CBPS compliance roadmap for both the existing building and especially the new building. The following is a potential roadmap for the new addition.

Roadmap to Net Zero for BLHS Addition Building:

Compliance Cycle	Measure	EUI net	Comments
2030	Current design	25	Includes baseline PV
2035	Optimize controls, expanded thermal comfort envelope, reduce plug loads	20	Expand comfort envelope to 65 deg F heating and 78 deg F cooling
2040	Upgrade to ground-source heat pumps, add more rooftop PV	10	GSHP and more PV
2045	Add more PV, including at parking lot	0	Add PV as needed for Net Zero

Notes:

1. EUI net is gross annual energy use intensity, less on-site generated PV power.

TECHNICAL REPORT	PROPOSAL	T3
COMPONENT: Construction Considerations	AUTHOR	TB
CURRENT CONCEPT: Now that the GCCM has been selected, there are several items that should be considered for planning discussions to help leverage their involvement and maximize the success for the project, items that should be included in the document set, and ideas for consideration to assist during construction.		
CONSIDERATIONS Collaborative planning with the GCCM and District • Develop logistics plan. • Generate phasing & relocation plan for staff and students during construction. • Develop a schedule for contractor/worker access and delivery time restrictions to not coincide with bus, and student drop-off or pick-up times. • Identify truck route, loading and unloading zone, and traffic control strategy. • School district to identify all planned schedule events that must not be disrupted (testing, special facility use events, athletic schedules, family information or parent-teacher nights, etc.). • Schedule and coordinate major construction activities during summer break to utilize that time efficiently. • Develop a utility phasing plan for service continuity to existing school. • Plan for security clearance, surveillance, and badging requirements during construction for crew. • Establish a bidder outreach program to get the word out to contractors. • Provide owner-provided equipment schedule including all required utility and connection requirements (power, data, water, drain, etc). • GCCM expected to be a full-time participant integrated with independent constructability team collaborative review.		

TECHNICAL REPORT	PROPOSAL	T3
COMPONENT: Construction Considerations	AUTHOR	TB
Integrate GCCM plans in Document Sets • Develop phasing and safety plans with GCCM and in the drawing set and show detailed step by step, phase by phase in DD set. ◦ Include designated site circulation identifying safe walking area from parking lot around construction area. • Include robust air barrier test plan document in set; including requirements to support the air barrier test by independent third party. • Conduct a commissioning planning meeting during the design phase to facilitate development of the commissioning plan. • Plan for temporary storage of reused structural steel beam at stage, prior to re-install. Considerations and Ideas for During Construction • Early site prep package (demo canopy and begin scope in footprint of new) • Allow contractor to use bonded and secured warehouse for equipment delivery (considering occupied tight site, and support early order of long-lead items). • Use just-in-time delivery for weather sensitive items to reduce storage needs and theft risk. • Use field export to backfill retaining walls. • Enhanced turnover plan including O&M manuals, warranties, and supplier contacts. • Implement envelope commissioning. • Conduct air barrier test as soon as air barrier is complete and HVAC, but before installing ceiling tiles (to allow better access for any adjustments).		

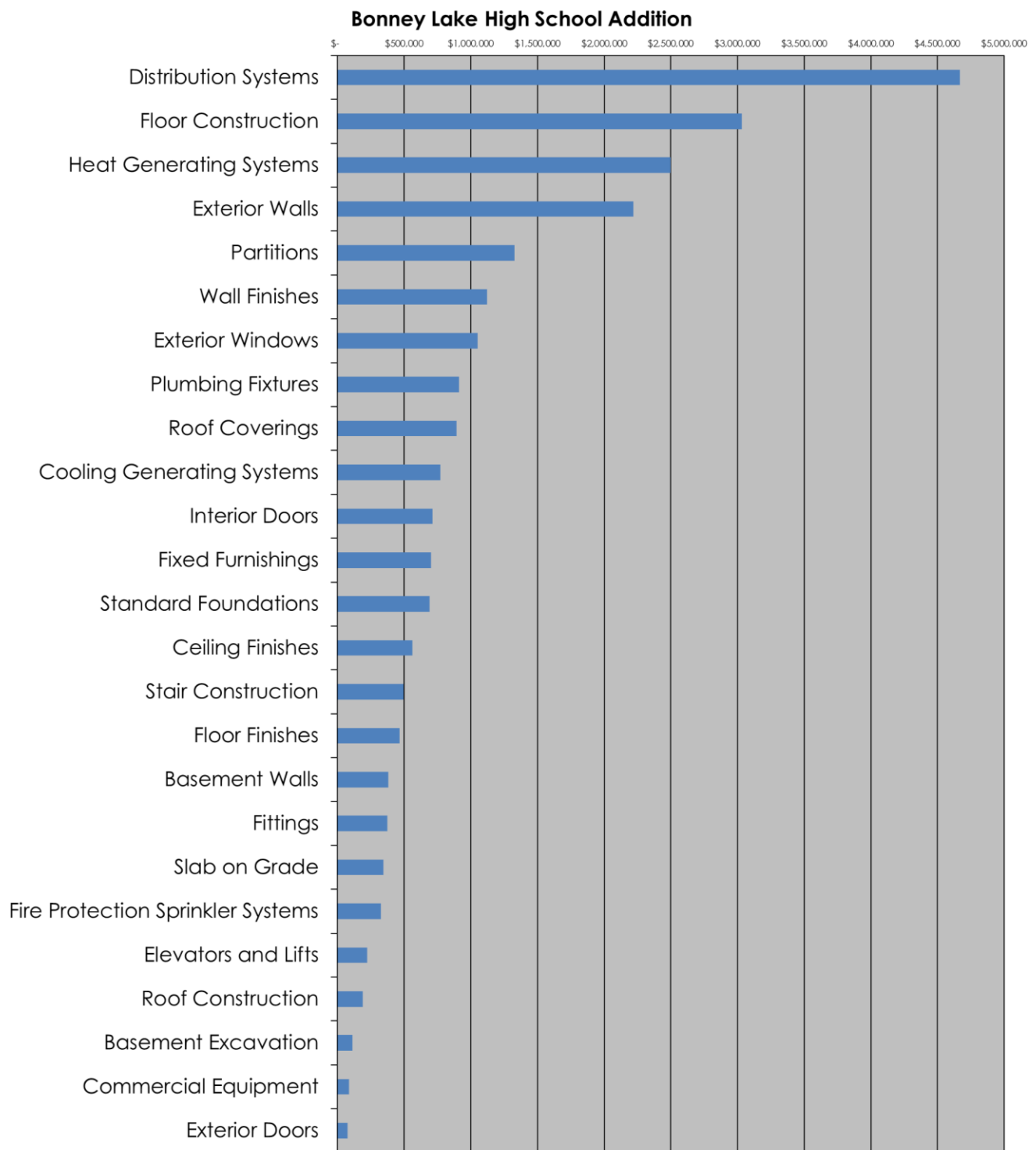
V. VALUE ENGINEERING

COST ANALYSIS

The VE team was not tasked to complete a detailed cost estimate review but as the various site and building systems were explored, the team did review and analyze certain cost categories and developed a cost model based upon the design team's cost estimates.

The cost model graph on the following page shows the cost estimate values broken down and grouped into function-based system costs (by Unifomat functions). This tool assists in evaluating where the largest percentage of the project's resources are being allocated towards different building functions.

For this project, distribution systems represent the highest first cost, followed by the floor structure and heat distribution system costs.



Cost Estimate Comments

The cost estimate is generally well formatted and appropriate for this level of design. As noted previously, the VE team was not tasked to complete a detailed cost estimate review, but as the various site and building systems were explored, the team did review certain cost items. Some potential cost estimate issues were identified and listed in the summary below. The adjusted costs are used in specific VE proposals in order to make apples-to-apples comparisons.

ID	Building Component	Current Cost	VE Proposal Cost	Difference	Comment
CE1	Netting price adjustment	\$384,000	\$111,000	\$273,000	Cost Adjustment
CE2	Existing stormwater replacement	\$0	\$190,000	(\$190,000)	Missing Scope
CE3	Interior doors price adjustment	\$600,000	\$234,000	\$366,000	Cost Adjustment
This is not a comment on the bottom overall estimate. The VE team highlighted these items for further study based on differing cost opinion discussions.					

COST ESTIMATE FORM

COMPONENT:

Netting price adjustment

CE1

CURRENT ESTIMATE					ADJUSTED ESTIMATE				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
Field Netting	675	LF	450	\$303,750	Field Netting (30ft high)	675	LF		
					Netting (Nylon or HDPE)	20,250	SF	1	\$20,250
					Steel Poles + Footings	675	LF	70	\$47,250
					Cabling & Hardware	675	LF	10	\$6,750
					Installation Labor	675	LF	20	\$13,500
Subtotal				\$303,750	Subtotal				\$87,750
General Contractor Markup	26.41%			\$80,220	General Contractor Markup	26.41%			\$23,175
Total to nearest \$1000				\$384,000	Total to nearest \$1000				\$111,000
					Difference				\$273,000

MENG Analysis

Proposal

CE1

CE2

COMPONENT: Existing stormwater replacement

MENG Analysis

Proposal

CE2

COST ESTIMATE FORM

COMPONENT:

Interior doors price adjustment

CE3

CURRENT ESTIMATE					ADJUSTED ESTIMATE				
ITEM	QTY	UNIT	UNIT COST	TOTAL COST	ITEM	QTY	UNIT	UNIT COST	TOTAL COST
HM door and frame					HM door and frame				
Flush					Flush				
Single, 3'-0" x 7'-0"	10	EA	3150	\$31,500	Single, 3'-0" x 7'-0"	10	EA	1200	\$12,000
Double, 6'-0" x 7'-0"	4	EA	5850	\$23,400	Double, 6'-0" x 7'-0"	4	EA	2200	\$8,800
90-minute fire rated rated, double, 6'-0" x 7'-0"	3	EA	10290	\$30,870	90-minute fire rated rated, double, 6'-0" x 7'-0"	3	EA	2800	\$8,400
Wood door/HM frame					Wood door/HM frame				
Flush					Flush				
Single, 3'-0" x 7'-0"	45	EA	3780	\$170,100	Single, 3'-0" x 7'-0"	45	EA	1500	\$67,500
Three-quarter lite					Three-quarter lite				
Single, 3'-0" x 7'-0"	22	EA	4095	\$90,090	Single, 3'-0" x 7'-0"	22	EA	1800	\$39,600
Double, 6'-0" x 7'-0"	5	EA	7740	\$38,700	Double, 6'-0" x 7'-0"	5	EA	3000	\$15,000
Aluminum door and frame					Aluminum door and frame				
Fully glazed, 1/4" tempered glass					Fully glazed, 1/4" tempered glass				
Single, 3'-0" x 7'-0"	7	EA	5220	\$36,540	Single, 3'-0" x 7'-0"	7	EA	2500	\$17,500
Double, 6'-0" x 7'-10"	3	EA	12239.46	\$36,718	Double, 6'-0" x 7'-10"	3	EA	4000	\$12,000
Double, 8'-0" x 7'-10"	1	EA	16469.28	\$16,469	Double, 8'-0" x 7'-10"	1	EA	4500	\$4,500
Subtotal				\$474,388	Subtotal				\$185,300
General Contractor Markup	26.41%			\$125,286	General Contractor Markup	26.41%			\$48,938
Total to nearest \$1000				\$600,000	Total to nearest \$1000				\$234,000
					Difference				\$366,000

MENG Analysis

Proposal

CE3

RISK ANALYSIS

While risk management should be conducted throughout the life cycle of a project, risk analysis integrated into the review study utilizes the skills and experience of the independent subject matter expert design professionals on the VE team. The process is used in identifying, evaluating, and prioritizing potential project risks to assist the owner and design team with risk management plan activities.

During the Evaluation Phase of the VE workshop, the team identified potential risks for the project quality, scope, cost, and schedule based on the current status of documents provided for the study. The team then conducted qualitative risk analysis with the nominal group technique collectively assessing risk probability, resultant potential cost, and project schedule impacts for each risk item on a scale of 1 to 5 (with 5 being high probability or impact). To prioritize the risk register, these impact scores are entered into a simple weighting formula:

$$\text{Risk Priority} = (\text{Probability} \times \text{Cost Impact}) + (\text{Probability} \times \text{Schedule Impact})$$

The resultant weighted scores are then sorted and graphed. The project risk priority graph appears on the following page.

The intent of the risk analysis exercise is to identify major risk elements unique to each project for the benefit of the owner and design team and identify potential mitigation strategies where risk exposure can be controlled or reduced. The owner should collaborate with the design team to mitigate these risks.

Generally, with the selection of the GCCM for this project, many of the higher level risks that would ordinarily be present have been significantly reduced, mitigated, or transferred (example: Occupied Site), and so overall the review team feels this project is a moderate level risk.

Notes on the top two identified risk drivers:

Tariffs:

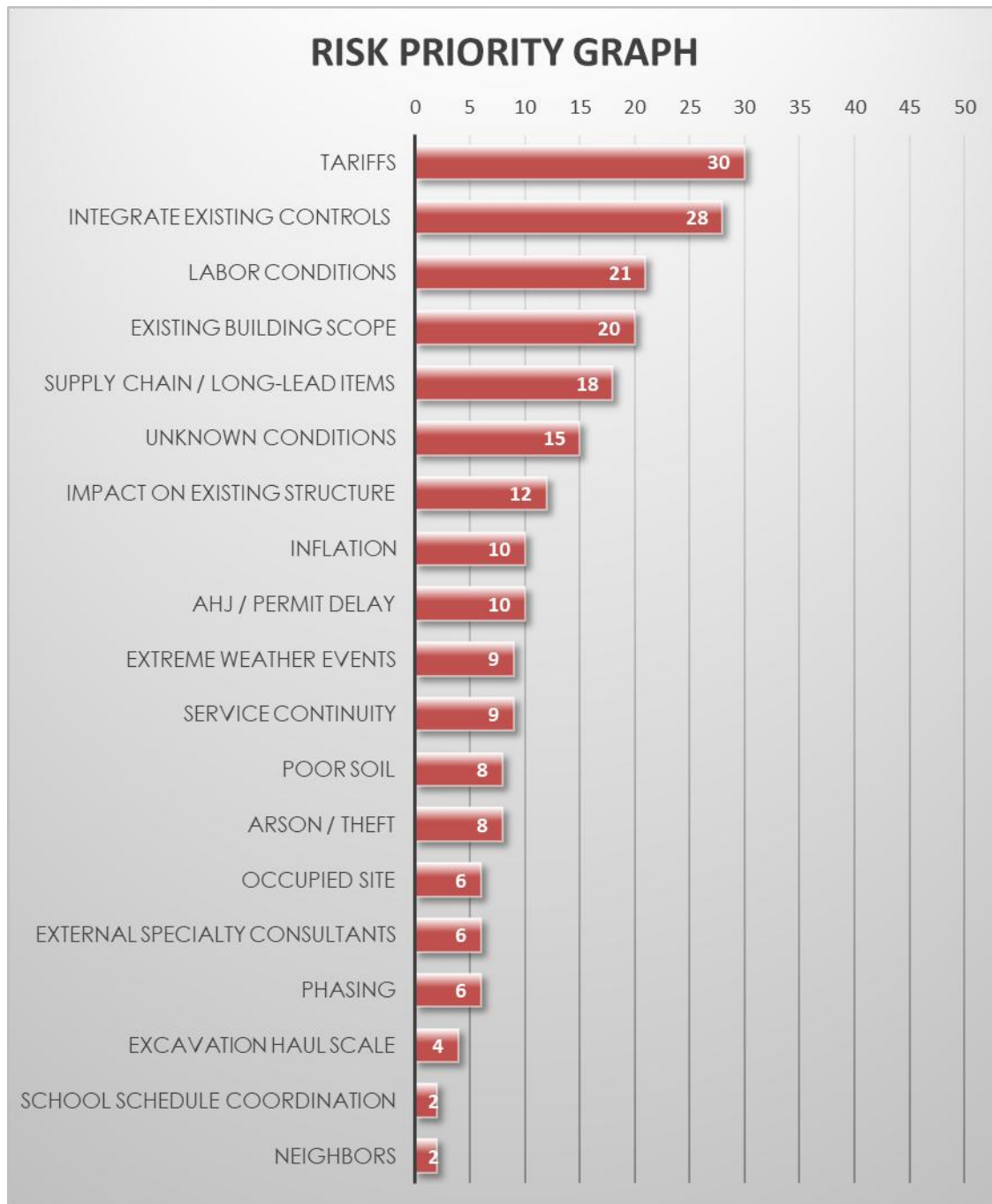
Market volatility is still present driven by the shifting uncertainty and price escalation related to tariffs on imported materials and components. With this project's reliance on building components like steel, concrete, mechanical units, transformers, panels, lighting, etc., these elements impart additional pricing risks due to price uncertainty. Rather than paying excessive costs to sub-contractors who may inflate bids to cover cost uncertainties, consider requiring subcontractors to provide current unit cost pricing lists with their bids for comparison with any claims for actual implemented new tariffs. Additionally, consider having the GCCM carry a tariff contingency line item to cover potential tariff costs increases.

Integrate Existing Controls:

The current concept and cost estimate also calls for retaining the existing buildings Honeywell based HVAC controls and extending the existing control system to the addition, and while it is more open source than some for extending, the existing system has known issues, and is nearing functional obsolescence. Consider adding a wholesale control system replacement to avoid integration risks and support improvements for the entire school functionality, plus CBPS compliance.

Washington State requires an independent constructability review as an additional risk mitigation technique to evaluate the construction documents for errors, omissions, and inconsistencies that could lead to RFI's and change orders. With the selection of GCCM delivery, it will be especially beneficial for the GC/CM team to join and fully integrate into the independent constructability review team.

As soon as the bid release date is set, schedule this project's constructability review to occur three weeks prior to releasing the documents for bid. This ensures there is sufficient development and detail provided for review, but still allows time to incorporate recommendations prior to releasing the bid documents).



Function Analysis

The process of Functional Analysis is unique to Value Engineering compared to other quality and cost control systems. The process frames the project's core needs, identifies the greatest opportunities for value improvement, and helps the team focus on what really matters. Additionally, the way the brain organizes information helps prepare the team for the Creativity Phase by prompting the divergent thinking needed to generate more ideas. You will see functions identified from this workshop in each proposal, and on the creativity alternative sheets in the appendix of this report.

Random Function Identification Technique

During the Function Analysis Phase, the team collaboratively evaluated project needs and conducted synergistic brainstorming to define the functions of the design.

- The Appendix section includes all the functions identified by the team in this collaborative exercise.

The identified functions were then prioritized to determine the top eight functions with the greatest opportunity for improved value based on the owners' success criteria and risk.

- The resulting prioritized functions can be seen on the creativity sheets in the report Appendix.

VI. VE METHODOLOGY

Value engineering provides an independent, impartial project review by a team assembled specifically for this study. Value Engineering itself is an organized creative process, which examines the proposed project and identifies alternatives to optimize cost and performance and assures compliance with project requirements. Through a structured system of investigation, idea generation, and analysis, the independent multidisciplinary team is able to consider and identify alternatives for site design, budget, schedule, and construction methods concurrently in a concentrated study.

After the initial presentation by Sumner-Bonney Lake School District and the design team, the VE team analyzed the budget and defined the basic functions of each project component. The VE team looked for ways to eliminate or modify design elements that add either first cost or life cycle cost without contributing to its required function. Specific proposals and reports were prepared and analyzed by the group for conformance to the project and VE study goals, prior to final prioritization.

Prioritization and brainstorming activities were conducted in group sessions alternating with additional small group and individual study sessions. All members supported an open-minded attitude to new suggestions, and all alternatives were considered valid until rejected by the entire team. Although the earlier sections of this report only elaborate or include the preferred alternatives, the appendix of this report includes all of the brainstorming notes from the workshop.



VII. APPENDIX

VALUE ENGINEERING TEAM

**Timothy Buckley, AIA, CVS, LEED AP
BD+C**

Team Facilitator

MENG Analysis

2001 Western Ave, Suite 200
Seattle, WA, 98121
(206) 838-9797 Office
(360) 608-2009 Cell
timothy@menganalysis.com

Preeti Pustake

**Construction Management
DCW Cost Management, LLC.**

415 1st Ave. N #9671
Seattle, WA
(206) 259-2990 Office
preeti@DCWCost.com

**Mark Van Wormer, PE
Civil**

Van Wormer Consulting, PLLC

14622 NE 174th Street
Woodinville, WA 98072
(206) 963-1694 Cell
marksvanwormer@outlook.com

**Melissa White, PE,
Structural**

Dibble Engineers

1029 Market Street
Kirkland, WA 98033
(425) 553-1843 Direct
Melissa@DibbleEngineers.com

**Doug Smith, PE, MBA, VMA, CCP, LEED
AP, cSBA**

Mechanical

MENG Analysis

2001 Western Avenue, Suite 200
Seattle, WA 98121
(206) 838-9797 Office
(425) 246-0949 Cell
doug@menganalysis.com

Brett Cato, LEED AP

Architectural

Independent Contractor

11275 Marine View Drive SW
Seattle, WA 98146
(206) 963-9619 Cell
bcato@icloud.com

Carrie Taylor

Electrical

Spees Design Build

23830 Pacific Hwy S Kent, WA 98032
(360) 490-6900 Cell
carriet@speesdb.com

Sarah Partap, MBA, VMA

Project Manager

MENG Analysis

2001 Western Avenue, Suite 200
Seattle, WA 98121
(206) 838-9797 Office
(206) 451-3462 Cell
sarah@menganalysis.com

Kara White, VMA

Project Coordinator

MENG Analysis

2001 Western Avenue, Suite 200
Seattle, WA 98121
(206) 838-9797 Office

OWNER AND DESIGN TEAM

Marina Tanay
Director of Capital Projects
Sumner-Bonney Lake SD
1202 Wood Ave.
Sumner, WA, 98390
(253) 891-6300 Office
(253) 470-8378 Cell
marina_tanay@sumnersd.org

John Boatman
Facilities Project Manager
Sumner-Bonney Lake SD
1202 Wood Ave.
Sumner, WA, 98390
(253) 891-6300 Office
(253) 247-2080 Cell
john_boatman@sumnersd.org

Matt Guilanians
Owner's Representative
Falcon and Fern
14033 165th Ave NE
Woodinville, WA, 98072
(206) 880-8505 Direct
matt@falconandfern.com

Christopher Spadafore
Architect
BCRA
2106 Pacific Avenue, Suite 300
Tacoma, WA 98402
(253) 627-4367 Office
cspadafore@bcradesign.com

Wade Jensen
Architect
BCRA
2106 Pacific Avenue, Suite 300
Tacoma, WA 98402
(253) 627 4367 Office
wjensen@bcradesign.com

Lori Robbin
Interior Design
BCRA
2106 Pacific Avenue, Suite 300
Tacoma, WA 98402
(253) 627-4367 Office
lrobbin@bcradesign.com

Sue Roulinski
Interior Design
BCRA
2106 Pacific Avenue, Suite 300
Tacoma, WA 98402
(253) 627-4367 Office
sroulinski@bcradesign.com

Andy Rasmussen
Landscape
FORA Landscape Architects
1430 NE 65th Street
Seattle, WA 98115
(206) 322-1732 Office
andy@fora.land

Callie Roberts
Landscape
FORA Landscape Architects
1430 NE 65th Street
Seattle, WA 98115
(206) 322-1732 Office
callie@fora.land

OWNER AND DESIGN TEAM (CONTINUED)

Marty Merriott

Civil

Sitts & Hill Engineers

4815 Center Street
Tacoma, WA 98409
(253) 474-9449 Office
martym@sittshill.com

Kathy Hargrave

Civil

Sitts & Hill Engineers

4815 Center Street
Tacoma, WA 98409
(253) 474-9449 Office
KathyH@sittshill.com

Luke Heath

Structural

PCS Structural Solutions

1250 Pacific Ave. Ste 701
Tacoma, WA, 98402
(253) 383-2797 Office
lheath@pcs-structural.com

Leah Cate

Structural

PCS Structural Solutions

1250 Pacific Ave. Ste 701
Tacoma, WA, 98402
(253) 383-2797 Office
lcate@pcs-structural.com

Jeff Hardwick

Mechanical / Plumbing

GFT Infrastructure, Inc.

6021 12th Street East, Suite 200
Fife, WA 98424
(253) 922-0446 Office
jrhardwick@gftinc.com

Steven Fernandes

Fire Protection Specialist

GFT Infrastructure, Inc.

6021 12th Street East, Suite 200
Fife, WA 98424
(253) 922-0446 Office

Michael Cozart

Electrical

GFT Infrastructure, Inc.

6021 12th Street East, Suite 200
Fife, WA 98424
(800) 800-5261 Office
mlcozart@gftinc.com

Kyung Kim

Electrical

GFT Infrastructure, Inc.

6021 12th Street East, Suite 200
Fife, WA 98424
(800) 800-5261 Office
kykim@gftinc.com

Steve Robinson

Acoustical Consultant

The Greenbusch Group

1448 Elliott Avenue West
Seattle, WA 98119
(206) 378-0569 Office

Gordon Adams

Door Hardware

Adams Consulting & Estimating

2337 N 57th Street
Seattle, WA 98103
(206) 528-0244 Office
gadams@centurylink.net

OWNER AND DESIGN TEAM (CONTINUED)

Dani Iltner
Building Envelope Consultant
Nexus BEC
2552 Jefferson Avenue
Tacoma, WA 98402
(253) 495-8544 Office
diltner@nexusbec.com

James Brigham
Geotechnical Engineer
Migizi Group
201 160th Street S., Suite 401
Spanaway, WA 98387
(253) 537-9400 Office

Andy Cluness
Cost Estimator
RC Cost Group
1015 Pacific Ave,
Tacoma, WA 98402
(253) 830-0543 Office
andy@rccostgroup.com

Melia Blankenship
Cost Estimator
RC Cost Group
1015 Pacific Ave,
Tacoma, WA 98402
(253) 830-0543 Office
melia@rccostgroup.com



KICKOFF MEETING SIGN-IN SHEET

MEETING INFORMATION	
MEETING TITLE	VE Kickoff - Bonney Lake High School Expansion
ORGANIZER	Kara White - MENG Analysis
LOCATION	Zoom
DATE	11/12/2025
TIME	8:00 AM

ATTENDEE NAME	ORGANIZATION	PROJECT ROLE	E-MAIL ADDRESS
Timothy Buckley	MENG Analysis	VE Facilitator	kara@menganalysis.com
Kara White	MENG Analysis	Project Coordinator	timothy@menganalysis.com
Doug Smith	MENG Analysis	VE Team Mechanical	doug@menganalysis.com
Brett Cato	Independent Contractor	VE Team Architect	bcato@cloud.com
John Boatman	Sumner - Bonney Lake SD	Capital Projects/Facilities Manager	john.boatman@sumnersd.org
Mark Van Wormer	Independent Contractor	VE Team Civil/Landscape	marksvanwormer@outlook.com
Matt Guilanians	Falcon & Fern	Bond Program Manager	matt@falconandfern.com
Marina Tanay	Sumner - Bonney Lake SD	Director of Capital Projects	marina_tanay@sumnersd.org
Sarah Partap	MENG Analysis	Project Manager	sarah@menganalysis.com
Alan McDaniel	Korsmo Construction	GCCM	amcdaniel@korsmo.com



ATTENDEE NAME	ORGANIZATION	PROJECT ROLE	E-MAIL ADDRESS
Andy Rasmussen	Fora Landscape	Architect	andy@fora.land
Carrie Taylor	Spees Design Build	VE Team Electrical	carriet@speesdb.com
Christopher Spadafore	BCRA	Project Manager	cspadafore@bcradesign.com
David Fillmore	Sitts & Hill Engineers	Civil Engineer	davidf@sittshill.com
Jeremy Doty	BCRA	Construction Administration Architect	jdoty@bcradesign.com
Jeff Hardwick	GFT	Mechanical	jrhawdwick@gftinc.com
Kyunghee Kim	GFT	Electrical	kykim@gftinc.com
Luke Heath	PCS Structural Solutions	Structural	lheath@pcs-structural.com
Lori Robbin	BCRA	Principal-in-Charge	lrobbin@bcradesign.com
Melissa White	Dibble Engineers	VE Team Structural	melissa@dibbleengineers.com
Sarah Fischer-Hersh	Korsmo Construction	Sr. Preconstruction Manager	sfischer@korsmo.com
Skott Young	Korsmo Construction	Cost Estimator	syoung@korsmo.com
Tim Eastman	GFT	Inc. Fire Alarm Systems	tceastman@gftinc.com
Wade Jensen	BCRA	Architect	wjensen@bcradesign.com
Ryan Rood	GFT	Technology & Security Systems	rarood@gftinc.com
Brian Mugane	GFT	Technology & Security Systems	bcmugane@gftinc.com

VIII. VE IMPLEMENTATION FORM

The VE Implementation form is used to track the acceptance of the Value Engineering proposals.

Our studies comply with WAC 392-343-080, WAC 392-344-065, and OSPI requirements, and the VE implementation form is used to track the acceptance of the value engineering proposals (Form D-7 submittal).

The following VE Implementation Form was received from the District on 01/08/2026 and replaces the blank form included in the draft report. This completed form documents the District's acceptance, modification, or rejection of the value engineering proposals and will be used by the School Board as part of the D-Form submittal process. Our studies comply with WAC 392-343-080, WAC 392-344-065, and OSPI requirements. Receipt of the completed VE Implementation Form also allows MENG Analysis to track and analyze results from our studies in order to continually improve the quality and effectiveness of our value engineering services.

CLIENT: Sumner-Bonney Lake School District											
PROJECT: Bonney Lake High School Expansion											
DATE: 1.7.2026											
Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION		
C1	Storm Vault	\$ 93,000		X					The team is planning to provide storage instead of adding vaults. This concept doesn't work for the baseball field due to the drainage direction. A metered discharge will be provided if needed.		
C2	ADA Ramp	\$ (77,000)		X					The idea is good, but this concept requires the existing driveway to shift. Grading wise, this doesn't seem feasible. A new retaining wall would be needed on the opposite side of the new roadway that is not included in the estimate.		
S1	Floor Framing	\$ 220,000		X					The floor joists will be closer on spacing. The floor vibrations can be an issue as they don't preform as well as the wide flange beams. Steel beams allow more flexibility to the owner and keeps the floor structure less deep.		
S2	Lateral System	\$ 199,000		X					Adding all of the light gage shear walls will compromise future flexibility to the school district. The weight of the floor structure is really heavy and the amount of shear walls would be excessive. In our experience, for multiple story buildings, brace frames are more efficient and cost effective.		
A1	Stair Tower	\$ 84,000		X					This is iconic part of the project. The back of the roof is set to the height needed for a man door to access the roof. BCRA looked at reducing the slope of the roof. It was reviewed by the team and did not have the desired aesthetics.		
A2	Study Nooks	\$ (11,000)		X					This proposal adds space to the flex area at the welcome center. There is some concern that a surface directly adjacent to the overlook at the welcome center could encourage students to sit on the countertop and become a fall hazard. The flex area at the classroom wings are already designed to be used by students with flexible furniture.		
M1	Integrated Design	\$ 209,000	X				\$ 209,000		The HVAC equipment sizing being reduced by 10% and 5% as stated it the cost savings is not a real cost savings. i.e., there is not a heat pump chiller that has 5% less capacity than the one we are specifying with an associated savings of 5% first cost. We are limited by equipment size iterations that are available. Increasing the insulation in the building is easy and low cost.		
M2	Thermal Generation	\$ 2,715,000		X					11/25/25: GFT reviewed this option at the beginning of the project, to see if made sense to upgrade the entire building instead of just the addition. SBLSD would like a serious exploration of this option due to the amount of potential savings. 1/7/26: 1) This VE item requires a separate central plant/heat pump chiller that we would need to find a home for in the vicinity of or on the roof of the new addition. There will be impacts to structural and acoustics if the heat pump chiller is on the roof. If it is to be located somewhere on the site, there will be impacts to grading/site work. 2) Adding an electric boiler will impact the electrical service size that is currently planned. There is not enough capacity in the existing board to feed the boiler. We would need to extend the primary feeders, add a transformer and provide additional trenching on the site. Additional \$250K in electrical costs are expected. 3) Refurbishing the old Petra Chillers will provide some more years of service (estimated at up to 10 years if the equipment is brought back to factory specs based on conversations with the new rep), but a new warranty will not be available. The cost of the refurbishment work is unknown and depends on an existing equipment evaluation that would need to be contracted, scheduled and performed. Further information is expected from the Petra factory on this. If this evaluation is desired, it will need to be completed ASAP to keep the project on schedule. 4) With this VE approach, there will not be a reduction in fossil fuel consumption since the existing gas fired boilers will still be providing the heat for the building (minus the addition).		
M3	Energy Source	\$ 117,000		X					This option adds geothermal under the field. For a horizontal loop, you need double the occupied floor area (for each floor). Do we have the budget to do this site work? There is not space for the loop on the site.		

CLIENT:	Sumner-Bonney Lake School District									
PROJECT:	Bonney Lake High School Expansion									
DATE:	1.7.2026									
Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION	
E1	Transfer Power	\$ (49,000)				X		\$ (49,000)	If the existing generator can accommodate the building loads, we don't need a new transformer. The team has been collecting metering data from the existing building during the month of December. The existing generator is very close to being able to accommodate the total building loads, including the new addition building and the existing MDF and IDF rooms.	
E2	Store Power	\$ (45,000)		X					Battery storage systems can be really expensive. They don't have a long life span. The units are large and need a location, depending on the amount of storage needed.	
E3	Electrical Service	\$ 126,000		X					This creates two different voltages between the new and existing building. Better to have a single voltage for the whole building. Changing will trigger a larger transformer which adds more cost. Savings is not as large as expected.	
R1	Add Traffic control at two intersecting driveways in front of school (stop signs and paint).	\$ (8,000)	X				\$ (8,000)		Our intention is to do this anyway.	
R2	Designated carpool only parking.	\$ (5,000)		X					WSSP: Did not take the point. The district doesn't need carpool spots.	
R3	Place excavated soils from the addition basement at the two sports fields; place the sports field drainage network at existing grade and cover with the excavated basement soil.	\$ 180,000		X					This will raise the walls around the fields. We need to remove soil to create the underdrainage system and leveling. Need to strip out the grass to avoid organics under the field. Is there another place to store the excavated soils? Can turn the outfield into a berm. The field drains towards the outfield.	
R3-A	Place excavated soils from the addition basement in the existing discus area.	Design team add		X					The school still uses this area for discus and javelin. Team to review the survey and see if the team can build up the grade in this area. For a studied the grading. Only 1,000 cy can be accommodated before there would be grading issues with the water easement.	
R4	Spread excess earth on sports field (no haul off site and dump).	\$ 40,000		X					See reasons above for R3.	
R5	Add more seating in plaza entrance area.	\$ (15,000)	X				\$ (15,000)		Fora has reviewed and added more seating to the plaza. Much of the additional seating is designed as seat walls.	
R6	Provide smart irrigation controls with weather sensors.	\$ (30,000)		X					Removing fields from irrigation system. This isn't a big value add. Most landscaping is existing to remain.	
R7	Temporary irrigation in beds ILO permanent.	\$ 15,000		X					The district can still turn it off. Already at most of the cost of installing the system when you trench it down into the bed. Temporary just doesn't last.	
R8	Drip irrigation systems ILO high volume heads.	\$ (25,000)		X					This will likely cost more and be more plastic in the ground. This is a cost add.	
R9	Install tree bags for new tree establishment irrigation ILO permanent irrigation system.	\$ 10,000		X					Most trees are on the slope where the tree bags don't work so well.	
R10	Add banner supports to light poles along the entry drive.	\$ (5,000)	X				\$ (5,000)		This is already planned for the project. There are two light poles that will stay. The team will need to add a banner support to the existing poles.	
R11	Add storage sheds at ball fields.	\$ (50,000)		X					There are existing storage sheds that will remain near the field. It is likely that any equipment could fit in these sheds once they are reorganized. The grooming equipment is already within five minutes of the site. A new shed is not needed as part of this project.	
R12	Add prefab concession stand at baseball field + Hydration station (for parity with SHS fast pitch).	\$ (150,000)		X					There is an existing DF at the field that has broken a few times. Should it be replaced? Maintenance to review.	
R13	Bid drainage and artificial turf fields as separate ballfield add alternates.	\$ (1,884,383)		X					Fields to be in the contract. Coordination with the sub can be an issue. GC prefers to have it in contract to have control over the install. Korsmo has also run into quality (product) issues. (Wade to add to DT meeting agenda)	
R14	Eliminate landscape irrigation booster pump (see L3.05).	\$ 20,000				X		\$ 20,000	There are two existing pumps on site. One is mostly for the fields. The VE team may have assumed that one will be added back into the site. We may be able to irrigate the area around the building without a pump.	

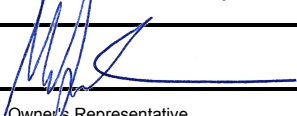
CLIENT:	Sumner-Bonney Lake School District								
PROJECT:	Bonney Lake High School Expansion								
DATE:	1.7.2026								
Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION
R15	Truncated dome and bollards ILO curb at Bus drop for enhanced barrier free access.	\$ (35,000)	X				\$ (35,000)		Accepted. This has already been incorporated into the most recent set of plans.
R16	Provide 1 parent drop zone at DLC.	\$ (45,000)		X					This creates potential traffic issues with parents able to circulate through the bus-only loop.
R17	Steel frame and storefront ILO curtain wall.	\$ 50,000		X					The team prefers to keep the curtainwall. The other system has a totally different look and feel. All west entry is storefront. Energy efficiency is higher.
R18	Omit holes in roof overhang for the planters.	\$ 10,000		X					If we take out the holes, we will take out the landscaping. There is not a great place to channel water from off of the roof to irrigate the bed in the winter.
R19	Consider TK Elevator or Skyline systems elevator as cost effective options (see https://www.primariuselevator.com/).	\$ 50,000		X					The cost is about the same for a machine-room-less (MRL) elevator. Controls are in the door jamb, so a thicker wall is needed. The savings are in the area that is not required for a machine room. This is a newer technology that tends to break down more often.
R20	Reduce window-to-wall area from 19% to 15% to reduce heating & cooling load.	\$ 100,000			X		\$ 50,000		The team will change from curtain wall to storefront at the stair tower. Team to use the tower to project light into the building. BCRA will raise the sill height to 30" at areas where there is curtainwall.
R21	Replace metal panels at accessible areas and use brick for improved impact durability.	\$ (50,000)	X				\$ (50,000)		On A-202, at stair down to existing exit doors. The team will install masonry at the base of the wall.
R22	Channel ILO downspout (see BCRA Olympia example on 2 story).	\$ (10,000)		X					With a U shape, nothing gets stuck in the downspout. It creates a waterfall. There is a long way to go down the wall (wind can blow water). There are envelope concerns.
R23	Add roof transfer ladder over the 2-hr parapet.	\$ (8,000)	X				\$ (8,000)		
R24	Reconfigure roof to eliminate large cricket on West End and allow single sheet flow roof plane .	\$ 25,000		X					There is not a great place for a downspout on the west end of the building.
R25	Install skylight cap over opening at entry precast concrete (thwart rain staining of expressed element).	\$ (16,000)		X					The opening in the roof at the column is an important part of the design. The District likes the gesture of the opening. An opening to the sky will be less desirable for birds to perch.
R26	Project images on the plaza under the main entry canopy.	\$ (10,000)		X					
R27	Install 3-M or other security film on all glass within accessible areas (to protect from scratch, acid, and other graffiti or breakage).	\$ (25,000)		X					3-M Film is already planned for the main entry vestibule. Where there is floor to ceiling glass, don't have glass all the way down to the ground. 30" off the ground is comfortable.
R28	Use exposed concrete in lieu of rubber flooring at stair treads.	\$ 18,000	X				\$ 18,000		
R29	Remove at grade exit door at main Stair in Hall 180B (is not required and adds security risk).	\$ 5,000	X				\$ 5,000		The team likes the idea of removing this door, it eliminates a hallway that is hard to supervise.
R30	Provide mechanical equipment access panels in both drop and hard-lid ceilings. Sesame Exposed Grid Ceiling Hatches are a proven product.	\$ (20,000)	X				\$ (20,000)		The system looks like a 2x4 tile grid and they flip down on hinges. This is a cost add. Team to review locations and costs. Maintenance will like this in areas that need maintenance above classrooms. Fan coil units are above the ceiling, mostly filters that get changed out. The units have control valves and condensate drain at the unit. There is one fan coil per zone. They are just recirculating units, the dampers are not at the unit. Can also go to return grille filters to minimize what needs to be accessed above the ceiling.
R31	Enhance acoustics of teacher prep room assemblies for better confidentiality and less disruption.	\$ (22,000)		X					This space would not be tailored for private conversations, just work stations.

CLIENT:	Sumner-Bonney Lake School District									
PROJECT:	Bonney Lake High School Expansion									
DATE:	1.7.2026									
Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION	
R32	High efficiency equipment and fixtures.	\$ (120,000)		X					GFT is specifying the highest efficiency products that they can. This may relate to 2024 code updates.	
R33	Open-source HVAC controls ILO sole-source (Honeywell).	\$ 70,000		X					The district would like to stay with the existing system.	
R34	Non-metallic (PEX) ILO metallic hydronic distribution run-out piping.	\$ 30,000	X				\$ 30,000		Piping would be kept rigid by installing rigid saddles. Can be some cost savings with installation costs.	
R35	Fan coil units in corridor ceiling space ILO classroom ceiling space to facilitate maintenance.	\$ 15,000		X					There is a lot of services that are already running in the hallways. The cable trays take up a lot of space.	
R36	Semi ILO fully-ducted ventilation (DOAS) air distribution (transfer grill to corridor).	\$ 100,000		X					The intent is to transfer air in each zone into the corridor and have a central return to the DOAS units. It is more work to balance the system and make sure the controls work. The District doesn't want a system that has potential comfort issues.	
R37	High-efficiency DOAS ventilation system ILO code minimum (C406).	\$ (75,000)				X		\$ (75,000)	This requires the efficiency to be 80% or greater. The units can meet the efficiency, but will get larger. The cost of the larger units doesn't seem to be accounted for in the cost. According to the product manufacturer, the high efficiency DOAS units are approximately a 17% cost increase. A payback period may be difficult to estimate without running an energy model comparing the two types of DOAS.	
R38	High-efficiency air-source heat pumps ILO code minimum at central plant.	\$ (150,000)		X					The proposed air source heat pumps are already more efficient than code minimum.	
R39	High-efficiency science room fume hoods with reduced exhaust air requirement.	\$ (20,000)		X					The high efficiency options require a 50-60 fpm velocity. The high performance hood can operate at a reduced velocity, but the code minimum according to the health department is 80 fpm. The manufacturer provided efficient options that will be used as the basis of design.	
R40	Increased ventilation (by 30%) throughout ILO code minimum ventilation.	\$ (75,000)		X					Increased ventilation decreases the efficiency of the building. The BOD has CO2 sensors.	
R41	Pre-programmed special ventilation modes including: 1) Airborne contagion (more ventilation), 2) Wild-fire smoke (less ventilation), and 3) Attack (no ventilation).	\$ (30,000)			X			\$ -	The emergency HVAC shut-down was planned in the original basis of design, which would fall under item 3 (no ventilation). The District does not want to incorporate items 1 and 2. Increased ventilation may affect the DOAS sizing as we are currently sizing the equipment for ASHRAE required ventilation values. Existing zones with air-side economizer could potentially have their programming modified to engage a manual economizer over-ride when desired. The equipment is not sized to condition that air to comfortable levels if over-ventilation during non-mild outdoor conditions.	
R42	New small generator just for addition emergency lighting + other loads including MDF room etc...(leave existing to continue to serve existing building).	\$ (80,000)				X		\$ (80,000)	If the existing generator can accommodate the building loads, a new generator will not be needed. The team has been collecting metering data from the existing building during the month of December. The existing generator is very close to being able to accommodate the total building loads, including the new addition building and the existing MDF and IDF rooms.	
R43	Ultra-violet (UV) air treatment to eliminate indoor airborne contagion.	\$ (40,000)		X					This is not required.	
R44	Radiant floor heating at main entry/welcome center public space.	\$ (60,000)		X					Is that a high priority space to keep comfortable? No. Team to spend money in the learning spaces. Ductwork needs are the same. It will just keep people more comfortable during heating season.	
R45	Velux or similar auto venting skylights (with integral wind/rain lockout controls) for fresh air.	\$ (25,000)		X					The district doesn't want skylights with operable parts.	

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Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION	
R46	Energy dash-board at main entry lobby for education.	\$ (20,000)			X		\$ (20,000)		The project will plan to have an energy dashboard as a website. Utilizing a website will make the information more accessible for students and teachers to use as part of the curriculum. The team will already have the metering infrastructure in the project.	
R47	Use flat plate fixed core heat recovery ILO heat wheel for air quality (avoid cross contamination).	\$ (40,000)		X					The fixed core provides less efficiency than the heat wheel. Heat wheel provides both sensible and latent recovery. Science has a heat recovery core to prevent cross contamination. Small motor, but units don't produce very much noise.	
R48	Use demand-controlled ventilation (DCV): Integrate CO ₂ sensors to adjust fresh air intake based on occupancy, reducing energy use.	\$ (50,000)		X					Already included in the project. Required by the energy code.	
R49	Two of four air-source heat pumps to be simultaneous heating & cooling capable ILO just one, for increased redundancy.	\$ (30,000)				X		\$ (30,000)	GFT is still working out the total load of the building to see what makes the most sense. There will most likely be two simultaneous units.	
R50	Provide fast-acting air flow control valves (dampers) to Science rooms to match performance of the fume hood and general exhaust fast-active valves (dampers).	\$ (25,000)		X					Already included in the project.	
R51	PEX ILO copper water pipes.	\$ 35,000	X				\$ 35,000		The intent was to allow PEX. GFT to clarify in the documents.	
R52	Non-metallic (ABS or PVC) drain, waste & vent (DW&V) piping ILO cast-iron (wrap with carpet pad for sound deadening).	\$ 45,000			X		\$ -		At floor one and down, this is acceptable to SBLSD. On upper floors, cast iron is preferred. GFT will allow PVC for waste on upper floors only in a chase. This approach was captured in the original estimate, so there is no cost change.	
R53	Eco-power faucets (charge hands-free lav. battery by water pressure in-line micro turbines).	\$ (12,000)		X					The district prefers faucets that do not require batteries.	
R54	Relocate restrooms and other plumbing fixtures for back-to-back or immediately adjacent space configuration to simplify the plumbing system and reduce cost.	\$ 22,000		X					There is not enough information to evaluate this item. The bathrooms are currently located based on programmatic requirements.	
R55	Provide automatic trap primers at all floor drains.	\$ (5,000)		X					This is already part of the project and is included in the estimate.	
R56	Aluminum feeders ILO copper.	\$ 10,000	X				\$ 10,000		The district is not opposed to aluminum feeders.	
R57	Re-Routing of power conduits outside in lieu of through kitchen area (and avoid internal disruption with pull boxes, (2) 4" conduits and (2) 2" conduits, and avoid risk and fit and conflicts inside. Finish with painted sheet metal on exterior to hide utilities.	\$ (30,000)		X					GFT will propose a better routing after the last site visit. Most likely, the routing will be inside the building.	
R58	Add PV on covered walkways and canopies in addition to the building roof .	\$ (125,000)		X					Team is planning for 20% PV. The district doesn't see the benefit in adding more PV to the project at this point.	
R59	Add low pedestrian lights in ramp stairway area.	\$ (20,000)	X				\$ (20,000)		The existing poles look good, but the ballasts need replacing. Team to review lighting in this area.	
R60	Increase solar panel energy generation by 20%.	\$ (70,000)		X					The district doesn't see the benefit in adding more PV to the project at this point.	
R61	Show future PV zone and provide capped conduit pathways routed to space reserved for additional future inverters.	\$ (20,000)		X					The existing roof structure doesn't have the capacity to add PV panels. The likelihood of the district ever installing some on the existing roof is very low.	
R62	Add large format video/digital wall display in Welcome area (like Sumner SH in commons but smaller).	\$ (45,000)		X					The digital display wall installed on past projects breaks continuously. The district has already had to replace five different panels. This technology will not be used on future projects until it becomes more reliable.	
R63	Upgrade existing fire alarm system to all voice.	\$ (120,000)	X				\$ (120,000)		The District would like to add this as part of the project. The District and the Fire Department want consistent alarms throughout the building to minimize confusion.	

CLIENT: Sumner-Bonney Lake School District									
PROJECT: Bonney Lake High School Expansion									
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Prop. #	COMPONENTS AND SYSTEMS	PROJECTED COST REVISION (Rough Order of Magnitude)	ACCEPT	REJECT	MODIFY	MAYBE	ACCEPTED VALUE OF PROPOSAL	VALUE OF 'MAYBE' COLUMN	COMMENTS / DISCUSSION
R64	Add conduit to existing building intercom system.	\$ (10,000)			X		\$ (10,000)		The district has a separate tech levy to do some of this work. This will likely be part of the bid for this expansion project.
	GRAND TOTAL ALL PROPOSALS	\$ 720,617					\$ 46,000	\$ (214,000)	

The owner has reviewed each of the Value Analysis proposals and recommends the responses contained herein. by  Matt Guilanians title Owner's Representative date 1/8/26		GENERAL COMMENTS REGARDING THIS VALUE ANALYSIS STUDY:							

LIST OF GENERATED FUNCTIONS

Civil / Landscape				
Remove Material	Protect Earth	Protect Structure	Enhance Entrance	Convey People
Retain Material	Communicate Direction	Protect People	Pedestrian Access	Drainage Challenge
Support Utilities	Support Building	Create Traction	Convey Fluid	Connect Nature
Balance Material	Support Vehicles	Treat Fluid	Simulate Lawn	Beautify Site
Support Pavement	Repel Weather	Supply Fluid	Reduce Water	Reduce Maintenance
Move Material	Circulate Vehicles	Remove Material	Soften Impact	Enhance Esthetics
Convey Fluid	Queue Vehicles	Maintain Material	Capture Storm	Access Grades
Increase Space	Segregate Circulation	Enhance Durability	Treat Storm	Create Space
Support Leveling	Store Vehicles	Encourage Safety	Support Players	People Comfort
Transfer Load	Load People	Collect Fluid	Circulate Pedestrian	Rest Pedestrian
Prevent Flood	Circulate Pedestrians	Control Erosion	Separate Pedestrian	Reduce Congestion
Convey Fluid	Convey Liquid	Organize Access	Direct Circulation	Ease Discomfort
Reuse Existing	Transport Liquid	Save Salmon	Protect Student	Contain Rubbish
Convey Waste	Reroute Utility	Save Wildlife	Indicate Route	Support Student
Enhance Existing	Clear Existing	Convey Waste	Access Site	Promote Health
Filter Fluid	Protect Environment	Pressurize Waste	Slip Issue	Rest Pedestrian
Release Fluid	Enhance Wildlife	Promote Sport	Comfort People	Access ADA
Treat Surfaces	Oxygenate Environment	Organize Recreation	Convey Fluid	Resist Loads
Queue Vehicles	Reduce Maintenance	Support Recreation	Support Community	

Structural				
Resist Fire	Resist Shear	Soften Footfall	Connect Components	Support Load
Resist Lateral	Resist Load	Level Ground	Support Ceiling	Resist Degradation
Support Roof	Resist Deflection	Resist Bounce	Resist Cupping	Support Occupant
Transfer Seismic	Resist Seismic	Support Enclosure	Support Equipment	Stiffen Assembly
Resist Wind	Resist Lateral	Support Distribution	Support HVAC	Transfer Load
Prevent Moisture	Endure Time	Attenuate Noise	Protect Occupants	Divide Volume
Impede Fluid	Span Space	Support Equipment	Frame Fenestration	Occupy Space
Resist Fire	Span Opening	Protect Shoes	Enhance Stability	Define Space
Strengthen Structure	Transfer Load	Impede Moisture	Stiffen Plane	Define Shape
Transmit Light	Distribute Load	Thwart Pests	Absorb Energy	Separate Area
Support Roof	Spread Weight	Provide Surface	Store Energy	Enable Egress
Prevent Sway	Distribute Load	Level Floor	Brace Wall	Support Enclosure
Resist Impact	Obstruct Passage	Protect Steel	Prevent Settling	Define Massing
Resist Explosion	Support Floors	Embody Carbon	Protect Egress	Prohibit Collapse
Resist Deflection	Resist Impact	Bond Materials	Embody Carbon	Accept Finish

Architectural				
Protect Interior	Protect Structure	Contain Heat	Supply Privacy	Secure Occupants
Insulate Space	Resist Elements	Store Snow	Protect Walls	Clarify Entry
Regulate Temperature	Protect Privacy	Contain Students	Enhance Aesthetic	Soften Footfall
Attenuate Noise	Create Excitement	Establish Character	Ease Maintenance	Soften Impacts
Conserve Energy	Exclude Weather	Express Interiors	Endure Time	Increase Load
Contain Environment	Define Aesthetic	Soften Footfall	Enhance Durability	Express Character
Define Aesthetic	Express Aesthetic	Conserve Energy	Resist Wear	Regulate Entry
Shed Moisture		Shed Rain	Absorb Impact	Resist Stain
Define Space	Absorb Noise	Protect Interior	Block Sound	Promote Wayfinding
Endure Time	Welcome People	Access Equipment	Order Function	Resist Wear
Match Existing	Please Community	Shelter Students	Resist Graffiti	Enhance Accessibility
Restrict Air	Encompass Community	Access Space	Access Space	Transport People
Resist Graffiti	Enhance Learning	Control Noise	Restrict Access	Enhance Aesthetic
Define Space	Reflect UV	Restrict Access	Ensure Safety	Direct Occupants
Arrest Light	Protect Structure	Support Equipment	Ease Custodial	Promote Privacy
Transmit Light	Support Items	Resist Fire	Define Space	Facilitate Efficiency
Resist Fade	Contain Occupants	Resist Damage	Access View	Convey Message
Endure Time	Create View	Block Smoke	Conceal Systems	Enhance Accessibility
Shed Rain	View Surroundings	Enhance Security	Absorb Sound	Conceal Systems
Resist Noise	Reflect Heat	Isolate Sound	Diffuse Sound	Define Aesthetic
Resist Forces	Seal Envelope	Reflect Light	Support Lights	Convey People

Architectural				
Resist Moisture	Transmit Light	Separate Space	Promote Navigation	Convey Objects
Enlighten Experience	Access Views	Control Noise	Resist Soiling	Facilitate Maintenance
Accent Environment	Reflect Radiation	Reflect Light	Complicate Access	
Waste Energy	Create Interest	Absorb Light	Reflect Sound	

Mechanical				
Convert Energy	Ventilate Space	Condition Space	Heat Fluid	Endure Time
Generate Heat	Comply Requirements	Recirculate Air	Resist Freeze	Control Systems
Generate Cool	Recover Energy	Exhaust Fumes	Sense Air	Fluid Flow
Consume Energy	Obstruct Space	Reduce Particulates	Maintain Quality	Accept Waste
Conserve Energy	Filter Particulates	Clean Air	Monitor System	Dilute Acid
Renew Existing	Exhaust Odors	Supply Temperature	Customize Temperature	Repair Existing
Support Totality	Create Heat-Island	Support Comfort	Adapt Climate	Replace Existing
Supplement Heat	Convey Waste	Protect Occupants	Adjust Setpoint	Detect Fire
Generate Noise	Protect Occupants	Protect Equipment	Comfort Occupant	Protect Occupants
Congest Area	Cool Fluid	Modulate Airflow	Resist Transfer	Resist Corrosion
Heat Fluid	Circulate Air	Reduce Pathogen	Attenuate Noise	Insulate Systems
Suppress Fire	Improve Smell	Filter Air	Congest Space	Neutralize Waste
Convey Fluid	Oxygenate Environment	Promote Health	Accommodate Health	Convey Fluid
Cool Air	Maintain Health	Support Existing	Vent Odors	Pressurize Fluid
Pollute Surroundings	Adjust Demands	Control Humidity	Validate Performance	Protect Structure

Mechanical				
Comply Code	Vary Capacity	Integrate Existing	Convey Fluid	Repair Convenience
Distribute Fluid	Deliver Climate	Manage System	Recirculate Fluid	Facilitate Maintenance
Mitigate Humidity	Heat Space	Accommodate Existing	Vent Gas	Condition Science
Force Fluid	Cool Space	Modulate System	Hygiene Occupants	Balance Distribution
Obstruct Passage	Protect Structure	Sense Environment	Convey Waste	Improve Tenancy
Convey Gas	Hygiene Occupants	Respond Condition	Access Water	Obstruct Systems
Heat Air	Ventilate Space	Circulate Fluid		

Electrical				
Deliver Power	Power Devices	Utilize Spectrum	Interlock Mechanical	
Transform Power	Distribute Energy	Awaken Occupants	Reactive Response	
Meter Power	Protect Distribution	Reduce Maintenance	Engage Involvement	Reduce Disruptions
Conduct Energy	Pathway Of Ions	Compliment Daylight	Transmit Signal	Generate Pollutants
Energize Devices	Reserve Future	Code Compliance	Suspend Activity	Conserve Energy
Comply Code	Provide Pathway	Adjust Intensity	Detect Breach	Coordinate Groups
Transmit Energy	Delivering Power	Occupancy Sensing	Enhance Safety	Store Fuel
Measure Power	Block Access	Support Change	Support Learning	Power Emergency
Convey Power	Define Route	Save Power	Allow Access	Illuminate Recreation
Distribute Energy	Enhance Mood	Detect Luminance	Transmit Signals	Illuminate Travel
Generate Power	Enhance Learning	Measure Luminance	Enhance Surveillance	Produce Energy
Move Electrons	Illuminate Space	Detect Motion		Switch Energy

Electrical				
Transmit Light	Improve Focus	Increase Safety	Access Space	Supply Redundancy
Generate Cost	Supply Function		Restrict Access	Enhance Learning
Generate Heat	Enable Visibility	Cause Noise	Notify Authorities	Reduce
Control Systems	Enhance Safety	Centralized Control	Control Access	Downtime
Protect Occupants	Create Focus	Notify Responders	Access Signal	Informing Public
Connect Devices	Ensure Health Of Eyes	Alert Responders	Capture Picture	Control Energy
Connect Energy	Control Lighting	Sense Smoke	Provide Information	Illuminate Recreation
Allow Lighting	Support Learning	Sense Heat	Transmit Signal	Define Path
Signal Processing	Enhance Environment	Emergency Control Activation	Capture Information	
Power Equipment	Illuminate Surfaces	Supervision And Monitoring	Support Communication	Project Imagery
Replenish Storage	Secure Area	Alert Occupants	Facilitate Distractions	Support ADA
Enhance Flexibility	Generate Glare	Secure Access	Charge Devices	Enhance Visual
Connect Existing	Dispel Dark	Cause Lights		Increase Accessibility
Protect Surges	Conserve Energy	Audible Illuminance	Facilitate Learning	Broadcast Voice
Ground Power	Energy Efficiency	Announce Emergency	Educational Tool	Enhance Learning
Capture Energy	Modulate Illumination	Facilitate Training	Amplify Sound	

CREATIVITY IDEA AND EVALUATION SHEETS

The following creativity idea and evaluation sheets are a record of options discussed and considered during the value workshop. They are included here to illustrate the range of options considered during the study for key project elements. Note that there are numerous ideas that were generated that did not make the cut to be included as official VE proposals. However, there may be a number of ideas that may still benefit the project, and all ideas should be reviewed and considered for possible incorporation into the project.

Idea Lists:

- | | |
|--|------------|
| • The total number of ideas generated: | 508 |
| • Total number of VE proposals: | 12 |
| • Total number of Revision proposals: | 64 |

See the top of the first page of each component for the top prioritized functions used to brainstorm ideas.

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Construction Management



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Control Costs	5	Communicate Information
	2	Ensure Safety	6	Organize Workflow
	3	Control Schedule	7	Document Construction
	4	Reduce Risks	8	Reduce Waste

CURRENT CONCEPT	\$ ROM		\$ ROM
[List current elements in these lines]	\$150k		

# leave blank	votes	ALTERNATE PROPOSALS
*		1 Generate phasing & relocation plan for staff and students during construction
		2 Use some of the existing classrooms as storage and site office to reduce logistic costs
*		3 Allow contractor to use bonded and secured warehouse for equipment delivery (considering occupied tight site)
*		4 Use field export to backfill retaining walls
*		5 Schedule major construction activities during summer break to utilize that time efficiently.
		6 Allocate part of student parking lot for construction crew vehicle parking to save on off site parking and shutting expense.
.		7 Utilize BIM to identify MEP and structural clashes in advance and mitigate them
.		8 Standardize design elements, classroom sizes, finishes, fixture types to reduce customization and procurement complexity.
.		9 Simplify architectural detailing (e.g., fewer façade transitions, standard window sizes)
.		10 Optimize site layout and access routes to reduce material handling and staging costs.
.		11 Minimize temporary works (e.g., fencing, utilities) through smart sequencing.
*		12 Use just-in-time delivery for weather sensitive items to reduce storage needs and theft risk.
		13 Bundle trades and scopes to reduce mobilization costs and improve coordination.
		14 Use drones or 360° cameras for site monitoring and stakeholder updates.
.		15 Identify laydown area.
		16 Design-build MEP systems.
*		17 Develop logistics plan.
*		18 Pre-purchase long-lead items
*		19 Early site prep package (demo canopy and begin scope in footprint of new)
*		20 Develop a utility phasing plan for service continuity to existing school.

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Construction Management



FUNCTIONS:	2-word: Active Verb / Measurable Noun	Active Verb / Measurable Noun
	1 <u>Control</u> <u>Costs</u>	5 <u>Communicate</u> <u>Information</u>
	2 <u>Ensure</u> <u>Safety</u>	6 <u>Organize</u> <u>Workflow</u>
	3 <u>Control</u> <u>Schedule</u>	7 <u>Document</u> <u>Construction</u>
	4 <u>Reduce</u> <u>Risks</u>	8 <u>Reduce</u> <u>Waste</u>

CURRENT CONCEPT	\$ ROM		\$ ROM
[List current elements in these lines]	\$150k		

# leave blank	votes	ALTERNATE PROPOSALS
*		21 Plan for security clearance, surveillance and badging requirements during construction for crew
*		22 Develop phasing and relocation plan with GC/CM and show detailed step by step phase by phase in DD set
*		23 Identify truck route, loading unloading zone, traffic control
*		24 Establish a bidder outreach program to get the word out to contractors
*		25 Develop Safety plan with GC/CM and show detailed step by step phase by phase in DD set.
*		26 Develop a schedule for contractor/worker access times to not coincide with student drop off or pick up time
*		27 Enhanced turn-over plan including O&M Manuals, warranties, supplier contacts, tabulated,
*		28 Air barrier test plan in set; including air barrier test by independent third party.
*		29 Develop safe walking area from parking lot around construction area
*		30 Implement envelope commissioning
*		31 Conduct a commissioning planning meeting during the design phase to facilitate development of the commissioning plan
		32 Preplan for purchase of pricing-volatile materials
*		33 Conduct air barrier test as soon as air barrier is complete and HVAC, but before installing ceiling tiles (to allow better access for any adjustments).
*		34 Provide owner provided equipment schedule to team
*		35 Plan for storage of reused structural steel beam
*		36 Conduct a constructability review (CR) to reduce change orders during construction.
		37 Develop a dirt/dust control and air quality monitoring strategy
		38
		39
		40

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Civil



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Move Material	5	Convey Fluid
	2	Support Building	6	Support Vehicles
	3	Treat Fluid	7	Supply Fluid
	4	Collect Fluid	8	Control Erosion

CURRENT CONCEPT	\$ ROM		\$ ROM
Earthwork	\$1,056k		
Storm Drainage	\$521k		
Pavement	\$370k		
Water utility	\$255k		
Demolition	\$150k		
Construction prep and TESC	\$228k		
Sanitary Sewer	\$70k		

# leave blank	votes	ALTERNATE PROPOSALS
*		1 Widen stairway at southwest corner to enable better two-way traffic
		2 Reduce 14' retaining walls with terracing.
.		3 Reuse at least 50% of excavated dirt in lieu of imported.
.		4 Pothole utilities during design ILO during construction
R1		5 Add Traffic control at two intersecting driveways in front of school (stop signs and paint)
		6 Above ground detention storage tank for roof drainage
		7 On-site waste water treatment system for new building
.		8 Use an edge drain ILO catch basins in entry plaza
?		9 Widen school drive queuing area to two lanes ILO one lane
		10 Heated sidewalk to eliminate ice hazard in winter
		11 Provide colored or textured walkway to facilitate wayfinding
		12 Use pavers ILO concrete in entry plaza
		13 Raise finish floor elevation to facilitate sanitary sewer connection in existing building
		14 Raise concrete crosswalks to slow traffic and improve safety
		15 Injection well for disposing stormwater
		16 Composting storm water vault filtration ILO biopods
		17 Designated flood area to deliberately allow flooding during heavy rain events
R2		18 Designated carpool only parking
		19 Offsite parking with shuttle
		20 Add covered bike parking (assume 8'w x 15'L)

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Civil



	2-word: Active Verb / Measurable Noun	Active Verb / Measurable Noun
FUNCTIONS:		
1	Move Material	5 Convey Fluid
2	Support Building	6 Support Vehicles
3	Treat Fluid	7 Supply Fluid
4	Collect Fluid	8 Control Erosion

CURRENT CONCEPT	\$ ROM	\$ ROM
Earthwork	\$1,056k	
Storm Drainage	\$521k	
Pavement	\$370k	
Water utility	\$255k	
Demolition	\$150k	
Construction prep and TESC	\$228k	
Sanitary Sewer	\$70k	

# leave blank	votes	ALTERNATE PROPOSALS
		21 E-bike and e-scooter program (Ride share company such as Lime)
		22 Designate official rideshare e-bike / scooter parking area
		23 PE credit for riding bike or walking to school
		24 Widen sidewalks from 6' to 10' to accommodate groups walking to school
W/C2		25 Raise grade of western drive 8" to reduce grade difference with school front entrance
.		26 Realign the eastern drive around existing "400" building to allow for extending queue lane
.		27 Follow existing topography to minimize grading and reduce need for retaining walls.
*		28 Limit over-excavation by refining geotechnical assumptions and soil stabilization methods.
.		29 Minimize stormwater pipe sizes and lengths by optimizing slope and layout. Sheet flow vs conveyance.
*		30 Use HDPE or PVC piping instead of concrete for non-structural drainage
		31 Switch DLC bus drop off location on the west side for parent drop off location on front and eliminate the ADA ramp and widen the stairs
		32 Standardize benches, bollards, and trash receptacles across campus.
		33 Use Polyethylene (PE) Turf with Minimal Infill instead of Synthetic Turf @ play areas
		34 Use permeable pavers instead of regular concrete pavers to reduce storm water management
		35 Use edge drains for flat areas in the entry plaza area to drain stormwater ILO yard drains
		36 Place excess soil adjacent to ballfields
		37 It appears there will be significant over excavation, show where plans indicate fill to balance site or show assumed haul and dump elsewhere.
R3		38 Place excavated soils from the addition basement at the two sports fields; place the sports field drainage network at existing grade and cover with the excavated basement soil.
		39 Drywells or French drains to infiltrate sports field ILO storm pipes connecting to existing storm.
*		40 Cartridge stormwater filters ILO Pods.

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
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		41 Bioswale ILO biopods for storm water treatment.
		42 Sand filter ILO of Biopods
.		43 Conduct a site-wide storm water management study; add additional storm water pondage as needed to accommodate the additional impervious surface area.
C1		44 Add a detention vault on east side to reduce pressure on existing conveyance system
!!!		45 Verify new artificial turf specification mandates zero PFAS chemicals
.		46 Verify heavy duty asphalt at Bus and Fire circulation areas
		47 Grasscrete parking at new
		48 New parking stalls with pervious pavers ILO asphalt
		49 Gravel parking lot and drive lane
\$		50 Revisit field netting price. Seems high
		51 Concrete ILO Asphalt
		52 Add roundabout in driveway midway with other driveway leg
		53 Move DLC bus stop to blend with parent drop off zone
C2		54 Shift DLC drop-off area further north and shift drive west to allow more room to design ADA ramp and stairs
R4		55 Spread excess earth on sports field (no haul off site and dump)
		56
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Landscape



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Support Recreation	5	Enhance Entrance
	2	Convey Fluid	6	Comfort People
	3	Reduce Maintenance	7	Enhance Esthetics
	4	Access ADA	8	Resist Loads

CURRENT CONCEPT	\$ ROM		\$ ROM
Baseball field	\$2,117k		
Multiuse Ball field	\$1,030k		
Softball field	\$943k	Irrigation	\$86k
Concrete Pavement Walkways	\$320k	Fencing	\$27k
Planting	\$244k		
Retaining wall	\$234k		
Site Improvements	\$195k		

# leave blank	votes	ALTERNATE PROPOSALS
W/C2		1 Alternative ramp design at front
.		2 Porous pavement ILO asphalt and concrete pavement
		3 Expand entry plaza area
		4 Extend northeast queue lane
		5 Plant more trees throughout the site
*		6 Plant more drought tolerant plants ILO grass in the entryway area
R5		7 Add more seating in plaza entrance area
		8 Add sculpture or other art to landscape areas
		9 Harvest rainwater for irrigation
		10 Add moveable planted pots entry area to improve esthetics
		11 Provide informational plaques that identify native plants
		12 Use soaker hoses ILO sprinklers for planted areas
		13 Add inspirational quotes imprinted in the walkways or stairs
R6		14 Provide smart irrigation controls with weather sensors
FFE		15 Add more tables in entry plaza area
		16 Add gas fire pit to entry plaza area
R7		17 Temporary irrigation in beds ILO permanent
R8		18 Drip irrigation systems ILO high volume heads.
R9		19 Install tree bags for new tree establishment irrigation ILO permanent irrigation system.
		20 repair and enhance natural turf ILO artificial turf for baseball field

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Landscape



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	<u>Support</u> <u>Recreation</u>	5	<u>Enhance</u> <u>Entrance</u>
	2	<u>Convey</u> <u>Fluid</u>	6	<u>Comfort</u> <u>People</u>
	3	<u>Reduce</u> <u>Maintenance</u>	7	<u>Enhance</u> <u>Esthetics</u>
	4	<u>Access</u> <u>ADA</u>	8	<u>Resist</u> <u>Loads</u>

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# leave blank	votes	ALTERNATE PROPOSALS
*		21 Relocate existing engraved monument stone within new entry plaza area
		22 Provide raised planted beds in entry plaza area
		23 Add imprint of leaves, ferns in walkway to add interest
R10		24 Add banner supports to light poles along the entry drive
		25 Indoor landscape to connect nature (living wall in the main entrance)
		26 Green ILO chain link fence
		27 Outside seasonal hydration stations - at ballfields
		28 Add canopy to bleachers at ballfields
		29 Reinforced earth walls ILO concrete retaining walls
		30 Use pavers ILO concrete for walkways
R11		31 Add storage sheds at ball fields
R12		32 Add prefab concession stand at baseball field + Hydration station (Parity with SHS fast Pitch)
w/C2		33 Eliminate stair and enlarge one ramp for all.
R13		34 Provide drainage and artificial turf as add alternate for the ballfields
		35 Self-healing risome turf technology ILO synthetic turf.
R14		36 Eliminate landscape irrigation booster pump (L3.05)
		37 Add precast or steel ping pong tables to main entry under roof cover
		38 Add fitness course and fixed equipment around multi purpose field
?		39 Elevated berm around sports fields for spectators using excess excavation material
		40 Ground cover to prevent grass that requires maintenance or weeds from growing between pavers or walkway surfaces.

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		41 Reduce pedestrian hardscape by 30%; replace with softscape.
		42 Standardize the size and design of pavers to reduce cutting and waste
		43 Decouple sidewalk from drive and provide meandering sidewalk
		44 Add Bike lane on drive (if room)
		45 Add outdoor classroom in conjunction with science program.
		46 Add green house in conjunction with science program.
R15		47 Truncated dome and bollards ILO curb at Bus drop for enhanced barrier free access
		48 Sheet piles
		49 Consider switching locations of parent drop off with DLC bus drop off area
R16		50 Provide 1 parent drop zone at DLC
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Structural



	2-word: Active Verb / Measurable Noun			Active Verb / Measurable Noun	
FUNCTIONS:	1	<u>Support</u>	<u>Load</u>	5	<u>Separate</u> <u>Area</u>
	2	<u>Transfer</u>	<u>Load</u>	6	<u>Enable</u> <u>Egress</u>
	3	<u>Stiffen</u>	<u>Assembly</u>	7	<u>Prohibit</u> <u>Collapse</u>
	4	<u>Resist</u>	<u>Deterioration</u>	8	<u>Accept</u> <u>Finish</u>

CURRENT CONCEPT	\$ ROM		\$ ROM
WF Steel Beams (Floor & Roof Framing)	967k	1.5" Metal Deck (roof framing)	143k
Reinforced concrete topping (Floor Framing)	279k	Layout/Equipment (Column Foundations)	125k
Steel columns (Floor & Roof Framing)	269k	Formwork (Basement walls)	122k
Concrete SOGs (Level 1+2 Floors)	220k		
3" metal deck (floor framing)	218k		
Misc. steel plates/connectors (floor framing)	199k		
Braced Frames	179k		

# leave blank	votes	ALTERNATE PROPOSALS
		1 <u>Reduce height/size of the large overhang</u>
		2 <u>Rigid foam fill at slab on grade adjacent to retaining wall</u>
		3 <u>Soldier piles for site retaining, so avoid excavating at parking lot</u>
		4 <u>Concrete shearwalls instead of braced frames</u>
		5 <u>Prepare a seismic review of existing building</u>
		6 <u>Make canopy support vertical ILO precast concrete wedge</u>
		7 <u>Use wood floor framing</u>
		8 <u>Mass timber ILO steel</u>
		9 <u>Use wood bearing walls</u>
		10 <u>Inflatable structure</u>
		11 <u>Cable-supported structure</u>
		12 <u>Exo-skeleton structure</u>
		13 <u>Eliminate any windows larger than three feet</u>
		14 <u>Gravity-resistant field</u>
		15 <u>Masonry ILO steel and concrete</u>
		17 <u>Eliminate the main entry overhang</u>
		18 <u>Fabric canopy at main entry ILO rigid structure</u>
?		19 <u>Opt for repetitive spans and uniform bay sizes to reduce complexity and material waste.</u>
		20 <u>Concrete structure in lieu of steel beams.</u>

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PROJECT: Bonney Lake High School Expansion & Intersection Improvements
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# leave blank	votes	ALTERNATE PROPOSALS
		21 <u>Light-weight (shallow) waffle slab floor system</u>
		22 <u>Post tensioned concrete ILO concrete on pan deck on Wide Flange</u>
		23 <u>Adobe!</u>
		24 <u>Pre-cast concrete panels for flooring system</u>
		25 <u>Design around standard dimensions to reduce cutting, waste, and lead times.</u>
*		26 <u>Minimize earthwork by balancing cut and fill, and align building placement with natural topography.</u>
		27 <u>Cross-laminated floor slab with light weight concrete topping ILO concrete on pan-deck.</u>
		28 <u>Move heavy mechanical equipment to grade ILO inside building</u>
		29 <u>Include demountable partitions or flexible structural grids to accommodate future changes in use</u>
		30 <u>Use suspended structure with cable system.</u>
S1		31 <u>open web steel joists to reduce wide-flange beams and provide additional space in floor cavity</u>
		32 <u>prestressed concrete beams ILO steel beams</u>
		33 <u>Gypcrete topping ILO all concrete floor (on steel pan-deck)</u>
		34 <u>fiberglass reinforced polymer beams ILO steel</u>
		35 <u>Eliminate basement</u>
		36 <u>3d Printed building modules</u>
*		37 <u>Steel stud bearing walls with steel joists</u>
		38 <u>Cast In place concrete floor and integral steel to eliminate wide flange and reduce depth.</u>
		39 <u>Insulated concrete form (ICF) basement construction</u>
		40 <u>Stacked containers</u>

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# leave blank	votes	ALTERNATE PROPOSALS
		41 <u>Tie-back earth anchors with shot-crete wall for basement</u>
		42 <u>Structural Insulated Panels (SIP) for above grade walls.</u>
		43 <u>Structural Insulated Panel (SIP) roof panels</u>
		44 <u>Prefabricated concrete floor panels</u>
		45 <u>Concrete masonry unit (CMU) structure</u>
		46 <u>Diagonal cables for stiffening</u>
		47 <u>Precast concrete insulated sandwich panel exterior wall form (i.e.... knife river)</u>
		48 <u>Replace Buckle Restrained Braces (BRB) with concrete shear walls.</u>
		49 <u>Dirt floors</u>
		50 <u>Indoor outdoor carpet</u>
		51 <u>Pressure treated wood on earthen floor</u>
		52 <u>Eliminate new stair near existing stair</u>
		53 <u>Concrete moment frames ilo steel braced frames</u>
		54 <u>Use the existing stairs in east instead of adding new stairs</u>
S2		55 <u>Steel stud shearwalls ILO steel braced frames</u>
		56 <u>Reduce basement size - move elevator and stair inboard, eliminate storage and one classroom to reduce retained soil height (would need to verify with topo map that would be enough reduction)</u>
		57 <u>Eliminate stair tower extension above roof level to reduce roof complexity</u>
		58 <u>Use buttressed walls in storage room to reduce reinforcing steel in concrete retaining wall</u>
		59 <u>Use canvas canopy ILO steel</u>
		60 <u>Increase roof slope to decrease snow load</u>

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# leave blank	votes	ALTERNATE PROPOSALS
		61 <u>Eliminate openings in canopy to increase material durability</u>
		62 <u>Eliminate skylights to increase materials durability</u>
		63 <u>Make walls CMU instead of light gage steel to provide a secondary gravity load support system</u>
*		64 <u>Coordinate framing with precast column & openings</u>
		65 <u>Sheet piles for retaining structures</u>
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Architectural Exteriors



		2-word: Active Verb / Measurable Noun			Active Verb / Measurable Noun
FUNCTIONS:	1	<u>Endure</u>	<u>Time</u>	5	<u>Isolate</u> <u>Sound</u>
	2	<u>Resist</u>	<u>Elements</u>	6	<u>Absorb</u> <u>Impact</u>
	3	<u>Reflect</u>	<u>UV</u>	7	<u>Clarify</u> <u>Entry</u>
	4	<u>Conserve</u>	<u>Energy</u>	8	<u>Define</u> <u>Aesthetic</u>

CURRENT CONCEPT	\$ ROM		\$ ROM
Exterior Wall Construction	\$1.16M		
Roof Coverings	\$707k		
Curtain Wall (Windows)	\$580k		
Storefront (Windows)	\$252k		
Exterior Doors	\$59k		

# leave blank	votes	ALTERNATE PROPOSALS
		1 Leave large canopy cover exposed to structure (no soffit).
		2 Reduce the amount of the large clerestory at entry
R17		3 Steel frame and storefront ILO curtain wall
		4 Upgrade roof to sheet metal
		5 Glass walls on the south(?) elevation are thermally inefficient. Reduce
		6 Vegetated roof
R18		7 Omit holes in roof overhang for the planters
A1		8 Reduce height of stair tower
		9 Increase pitch of roof and use asphalt shingles ILO membrane
		10 Exterior finish options - cost effective and repairable/ replaceable (such as Trespa panels??)
		11 Provide a small (secured) rooftop plaza for controlled access for special events (PTA fundraiser, staff socials, respite with mountain view, Astronomy club, etc...). Size to only require single stair egress.
*		12 Combine proposed NW stair tower with existing basement stair access.
*		13 Add exterior windows to west wall of Classroom 592 and Large Conference 190 by eliminating adjacent stair tower.
		14 Reduce height of large exterior canopy.
		15 Eliminate enclosed walls/windows at Welcome Center 181. Provide radiant heat devices for cold weather.
*		16 Collect roof drain runoff to water adjacent landscaping.
		17 Add ribbon windows at classrooms for more natural light.
		18 Eliminate window awnings
		19 Incorporate lower entry awning into covered area for exterior waiting area.
R19		20 Consider TK Elevator or Skyline systems elevator as cost effective options (see https://www.primariuselevator.com/)

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FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Endure Time	5	Isolate Sound
	2	Resist Elements	6	Absorb Impact
	3	Reflect UV	7	Clarify Entry
	4	Conserve Energy	8	Define Aesthetic

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# leave blank	votes	ALTERNATE PROPOSALS
		21 Remove the skylights to eliminate risk of leaks, moisture infiltration
?		22 Machine-roomless elevator ILO hydraulic to reduce cost and space requirements. - Verify code compliance
		23 Specify thermally broken fastener cladding
R20		24 Reduce window-to-wall area from 19% to 15% to reduce heating & cooling load.
		25 Switch to fiber cement panels instead of brick veneer or stone for a clean, modern look at lower cost.
R21		26 Replace metal panels at accessible areas and use brick for improved impact durability
		27 Clad building with EIFS
		28 Corten steel exterior wall for long-life with no maintenance.
		29 Add sun tubes to transmit daylight into interior spaces.
?		30 Graffiti-resistant coating where applicable.
		31 Use eave overhangs ILO parapet for enhanced weather protection of building envelope.
		32 Eliminate the new peaked roof over the existing building.
		33 Increase roof slope to resist ponding, leaks
		34 Consider polycarbonate panels in stairwells for daylighting with impact resistance and lower cost.
		35 Use "Bird Safe" glazing (either uv coat or frit pattern)
		36 Use storefront glazing only in key areas (e.g., entry, commons) and switch to punched windows elsewhere.
		37 Slope roof to south for improved photo-voltaic (PV) system performance.
		38 Use cool roof coatings to reduce heat gain and HVAC loads.
?		39 Install metal mesh or art screen elements on west glazing to reduce heat gain and glare
		40 Stainless-steel gutter & down-spout ILO sheet-metal for improved durability.

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COMPONENT: Architectural Exteriors



		2-word: Active Verb / Measurable Noun			Active Verb / Measurable Noun
FUNCTIONS:	1	<u>Endure</u> <u>Time</u>	5	<u>Isolate</u> <u>Sound</u>	
	2	<u>Resist</u> <u>Elements</u>	6	<u>Absorb</u> <u>Impact</u>	
	3	<u>Reflect</u> <u>UV</u>	7	<u>Clarify</u> <u>Entry</u>	
	4	<u>Conserve</u> <u>Energy</u>	8	<u>Define</u> <u>Aesthetic</u>	

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*		41 Use color blocking or murals built out of recycled material instead of expensive materials to create visual interest.
		42 Living wall introduction into the clerestory facade
R22		43 Channel ILO downspout (see BCRA Olympia example on 2 story)
		44 Follow ASHRAE recommendation on fenestration (window) shading, including vertical sun-shades to east & west.
		45 Active facade shading devices (motorized) for proactive solar gain control.
		46 Simplify stair tower to not visually compete with main entry, make it just large enough to provide roof maintenance access
		47 Skylight ILO full-story lantern at south grand stair tower.
		48 Use color blocking or murals made of recycled material instead of expensive materials to create visual interest.
?		49 Resolve roof geometry and continuity of drainage where new gable roof over the existing meets the new roof (see model image on S-000)/
		50 Eliminate roof openings for rain runoff to reduce waterproofing requirements
R23		51 Add roof transfer ladder over the 2-hr parapet
		52 Relocate the three large DOAS units from inside to outside above the existing roof and eliminate the high peaked roof above this area. Repurpose the large mechanical room as additional educational space.
R24		53 Reconfigure roof to eliminate large cricket on West End and allow single sheet flow roof plane
R25		54 Install skylight cap over opening at entry precast concrete (thwart rain staining of expressed element).
		55 Reduce floor-to-floor height from 14 to 13' - 6" to reduce building volume, cost and energy use.
		56 Photovoltaic array ballasting with tilt footprint along with maintenance access
		57 Add living wall (vines) at large column at SE corner of overhang.
.		58 Provide robust thermal break details at exterior wall, including floor intersection, and roof
		59 "Paint" the building exterior with color-controllable LED lighting with emphasis on the main entry.
R26		60 Project images on the plaza under the main entry canopy.



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PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Architectural Exteriors



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	<u>Endure</u> <u>Time</u>	5	<u>Isolate</u> <u>Sound</u>
	2	<u>Resist</u> <u>Elements</u>	6	<u>Absorb</u> <u>Impact</u>
	3	<u>Reflect</u> <u>UV</u>	7	<u>Clarify</u> <u>Entry</u>
	4	<u>Conserve</u> <u>Energy</u>	8	<u>Define</u> <u>Aesthetic</u>

CURRENT CONCEPT	\$ ROM		\$ ROM
Exterior Wall Construction	\$1.16M		
Roof Coverings	\$707k		
Curtain Wall (Windows)	\$580k		
Storefront (Windows)	\$252k		
Exterior Doors	\$59k		

# leave blank	votes	ALTERNATE PROPOSALS
R27		61 Install 3-M or other security film on all glass within accessible areas (to protect from scratch, acid, and other graffiti or breakage).
.		62 Provide backlighting for the large school BLHS graphic on the front of the building
		63 Skylights at the main entry canopy for improved natural daylighting both in the covered main entry courtyard and to the adjacent three-story store-front glazed spaces.
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Architectural Interiors



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Define Space	5	Reflect Light
	2	Protect Privacy	6	Resist Wear
	3	Enhance Learning	7	Secure Occupants
	4	Soften Footfall	8	Conceal Systems

CURRENT CONCEPT	\$ ROM		\$ ROM
Partitions	\$1.05M		
Wall Finishes	\$888k		
Interior Doors	\$564k		
Ceiling Finishes	\$443k		
Floor Finishes	\$368k		
Specialties	\$295k		
Elevators and Lifts	\$176k		

# leave blank	votes	ALTERNATE PROPOSALS
?		1 Eliminate skylights to Classroom 592.
		2 Replace interior stair tower at main entry (south) with exterior stair under large entry roof.
		3 Eliminate stair access from stair tower at main entry (south).
		4 Extend elevator to roof.
		5 Reduce ceiling height of Welcome Center 181.
		6 Eliminate basement and relocate program elements to on grade.
		7 Add additional floor to existing building to accommodate new classrooms.
		8 Use ground concrete floor in lieu of rubber tile.
		9 Add interior light shelves at classrooms in lieu of exterior awnings.
		10 Add precast partitions instead of drywall for changing rooms
		11 Operable wall between two classrooms on each floor for future flexibility
		12 Eliminate relight at classrooms for improved privacy and security
		13 Locate all classroom corridor relights to the same side of the door.
.		14 Phase Area 100 and Area 200 work during summer hours.
		15 Eliminate dropped ceilings in the hallways.
		16 Eliminate dropped ceiling in Welcome Center 181.
		17 Giant aquarium
		18 Eliminate Vestibule 520A due to low occupancy use at side entry.
R28		19 Use exposed concrete in lieu of rubber flooring at stair treads.
		20 Replace Rubber floor in Classroom 555 (Science Ready) with Carpet tile.

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Architectural Interiors



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Define Space	5	Reflect Light
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Ceiling Finishes	\$443k		
Floor Finishes	\$368k		
Specialties	\$295k		
Elevators and Lifts	\$176k		

# leave blank	votes	ALTERNATE PROPOSALS
R29		21 Remove at grade exit door at main Stair in Hall 180B (is not required and adds security risk)
		22 Add Primarius Elevator as acceptable subcontractor
*		23 Graphic wall art in hall to inspire based on program (Science)
R30		24 Provide mechanical equipment access panels in both drop and hard-lid ceilings. Sesame Exposed Grid Ceiling Hatches are a proven product.
		25 Colored pattern in floor to facilitate wayfinding
		26 High gloss paint for durability and resisting marking
		27 Revise finish pallet to lighter colors to better reflect light
		28 Add wainscot to corridor walls for durability
		29 Consider adding STC ratings to the walls instead of acoustical wall panels in the classrooms.
R31		30 Enhance acoustics of teacher prep room assemblies for better confidentiality and less disruption
		31 Gender neutral RRs
.		32 Relocate Meeting Room 554 to allow more daylight into Flex Space 550.
		33 Consider reducing or eliminating the Alcove 550C at the toilet rooms.
		34 Stretch toilet room wall with drinking fountains into alcove (making restroom area larger) and eliminate doors for no-touch circulation and improved safety.
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Planning



	2-word: Active Verb / Measurable Noun			Active Verb / Measurable Noun	
FUNCTIONS:	1	Educate Students	5	Reduce Energy	
	2	Increase Capacity	6	Be Modern	
	3	Improve Flexibility	7	Be Flexible	
	4	Identify Entry	8	Be Timeless	

CURRENT CONCEPT	\$ ROM		\$ ROM
[List current elements in these lines]	\$150k		

# leave blank	votes	ALTERNATE PROPOSALS
		1 2024 Building and energy codes in lieu of 2021
		2 Lower Energy Use Index (EUI) for future clean building compliance
		3 Allow alternate materials to District Standards
R32		4 High efficiency equipment and fixtures
.		5 Use surplus budget to modernize more existing
.		6 Increase target credits to provide a Healthy cushion of WSSP points
		7 Expand canopies over walkways
		8 Add canopy over ADA ramp
.		9 Roadmap to Net-zero energy
.		10 Upgrade existing building to meet Clean Bldgs EUI Target
?		11 Pursue energy resilience/microgrid grants in 2026 - provided by the Department of Energy
		12 Plan for future net zero building.
		13 WELL or Fitwell to improve staff & student health & well-being beyond WSSP minimums
		14 Designate 5, 10 and 15-year walls for improved long-term adaptability
		15 Collaborate with the Police Department for a CPTED study to enhance passive security
		16 Lower entry roof canopy to improve rain protection
		17 Replace existing school ILO partial modernization
		18 Infill spaces between sub-building corridors with new classrooms to make use of available space and systems
		19 Two-story building with no basement
		20 Adopt innovative ways to involve the stakeholders in the development program to encourage sense of ownership.



CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Planning



FUNCTIONS:

	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
1	Educate	Students	5	Reduce Energy
2	Increase	Capacity	6	Be Modern
3	Improve	Flexibility	7	Be Flexible
4	Identify	Entry	8	Be Timeless

CURRENT CONCEPT	\$ ROM		\$ ROM
[List current elements in these lines]	\$150k		

# leave blank	votes	ALTERNATE PROPOSALS
A2		21 On the third floor from the hallway, provide "Study Bar seatings" including windows looking down into the welcome center for quiet or pair study. Provide outlets for 1:1 device charging.
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvement
COMPONENT: HVAC



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Generate Thermal	5	Control Systems
	2	Distribute Fluid	6	Insulate Conveyance
	3	Condition Space	7	Condition Science
	4	Ventilate Space	8	Balance Distribution

CURRENT CONCEPT	\$ ROM		\$ ROM
Central Plant (four AWHPs); pumps, piping	\$3M	Science AHUs	175K
Ductwork	\$1M	TAB	\$150K
Fan coil units, 4-pipe	\$350K	VAV Terminal Units (VAV boxes)	\$125K
DOAS Units	\$300K	Exhaust Fans	\$30K
Hydronic 4-pipe distribution	\$300K	Electric Heaters	\$20K
Controls (DDC)	\$275K	Fume hood dampers	\$20K
Insulation (ductwork and piping)	\$200K	Misc (split-Dx, Gas to Science, other)	\$25K

# leave blank	votes	ALTERNATE PROPOSALS
M1		1 Premium Thermal Envelope
T1		2 PSE Energy Grant
		3 Induction terminal unit ILO fan coil unit HVAC to match Sumner HS
T2		4 Implement Clean Bldgs Energy Audit Energy Conservation Measures (ECMs) for Existing; eliminates need for boiler plant due to reduced load.
M2		5 Stand-alone plant (producing chilled and heating hot water) ILO expanded central plant
		6 Variable refrigerant flow HVAC ILO 4-Pipe fan coil units
R33		7 Open-source HVAC controls ILO sole-source (Honeywell)
current		8 All new controls at existing building
		9 Improve insulation at existing building to reduce need to retain existing high-temperature boilers at central plant (allow new air-source heat pumps to carry full load).
M3		10 Ground ILO air-source heat pumps at central plant for improved energy efficiency
		11 Solar-thermal heat generation using high-efficiency vacuum tube technology
		12 Low-temperature solar-thermal absorption-chiller chilled water generation
		13 Radiant floor full heating and partial cooling at new (addition) building
R34		14 Non-metallic (PEX) ILO metallic hydronic distribution run-out piping
R35		15 Fan coil units in corridor ceiling space ILO classroom ceiling space to facilitate maintenance.
R36		16 Semi ILO fully-ducted ventilation (DOAS) air distribution (transfer grill to corridor)
R37		17 High-efficiency DOAS ventilation system ILO code minimum (C406)
R38		18 High-efficiency air-source heat pumps ILO code minimum at central plant
		19 All new controls throughout the entire high school, existing and new
		20 Non-metallic ILO galvanize sheet metal ductwork

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvement
COMPONENT: HVAC



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Insulation (ductwork and piping)	\$200K	Misc (split-Dx, Gas to Science, other)	\$25K

# leave blank	votes	ALTERNATE PROPOSALS
		21 Recycled-jean duct insulation ILO fiberglass
R39		22 High-efficiency science room fume hoods with reduced exhaust air requirement
		23 Micro-science curriculum to eliminate need for fume hoods in science rooms
?		24 Reverse return hydronic piping system to reduce need for balancing valves and simplify hydronic system Testing, Adjusting and Balancing (TAB)
?		25 Self-balancing devices in maintenance accessible locations on both air and water distribution to simplify TAB
		26 Ceiling fans in common areas
		27 Ceiling fans in classrooms
		28 Natural ventilation ILO DOAS as classrooms
R40		29 Increased ventilation (by 30%) throughout ILO code minimum ventilation
R41		30 Pre-programmed special ventilation modes including: 1) Airborne contagion (more ventilation), 2) Wild-fire smoke (less ventilation), and 3) Attack (no ventilation)
		31 Electric baseboard heat with space-by-space ventilation
		32 No air conditioning
		33 Radiant ceiling panels
(W/M3)		34 Horizontal geothermal exchanger laid down in loops under fields during install of new sub-drainage for ground-source heat pump ILO air source.
		35 Wood stoves
		36 Subterranean building
		37 Cold Fusion
		38 Only provide central plant capacity needed for new (addition) building; no upgrades or renewal of existing central plant capacity serving the existing buildings.
R42		39 New small generator just for addition emergency lighting + other loads including MDF room etc...(leave existing to continue to serve existing building).
		40 Battery backup emergency egress lighting for new addition, no new generator

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvement
COMPONENT: HVAC



		2-word: Active Verb / Measurable Noun			Active Verb / Measurable Noun
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# leave blank	votes	ALTERNATE PROPOSALS
(w/M2)		41 Rebuild existing Petra chillers - only supply (2) new heat pumps for the new addition
		42 Addition as high temp hydronic heating ILO having switch from existing system (smaller pipes)
		43 High-temperature system at new (addition) ILO low-temperature for uniform and simplified mechanical system, which simplifies maintenance and operation (traditional boilers and chillers)
		44 New pumps for new (addition) building; retain existing pumps as-is for existing building
		45 Active beams ILO fan coils
		46 Electrostatic filters ILO traditional particulate filters
R43		47 Ultra-violet (UV) air treatment to destroy indoor airborne contagion
		48 Displacement ventilation (DV) ILO traditional overhead for improved indoor air quality and reduced fan energy
R44		49 Radiant floor heating at main entry/welcome center public space
?		50 Natural ventilation for main entry atrium space
?		51 Stack effect auto-venting windows at top of high window bays at Atrium
R45		52 Velux or similar auto venting skylights (with integral wind/rain lockout controls) for fresh air.
R46		53 Energy dash-board at main entry lobby for education
R47		54 Use flat plate fixed core heat recovery ILO heat wheel for air quality (avoid cross contamination)
		55 Enhanced Commissioning
		56 Under-ground ductwork for self-insulated ductwork
		57 Include ductless fume hoods and filtered fume hood systems as the most flexible and cost-effective alternatives to traditional ducted exhausts
		58 Woven grass insulation
		59 Vertical hydronic four-pipe fan coil units for the new (addition) building, same as for the existing buildings to simplify maintenance and operation and provide improved indoor air quality with full economizer ventilation capability
R48		60 Use demand-controlled ventilation (DCV): Integrate CO ₂ sensors to adjust fresh air intake based on occupancy, reducing energy use

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvement
COMPONENT: HVAC



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Controls (DDC)	\$275K	Fume hood dampers	\$20K
Insulation (ductwork and piping)	\$200K	Misc (split-Dx, Gas to Science, other)	\$25K

# leave blank	votes	ALTERNATE PROPOSALS
*		61 Route the exhaust ductwork from Restrooms and Changing Rooms through shaft walls to reduce the ductwork quantity
R49		62 Two of four air-source heat pumps to be simultaneous heating & cooling capable ILO just one, for increased redundancy
Alt		63 Upgrade existing building fan coil units to be low-temperature (140 deg F) full-heating capable, hence no need for the existing gas-fired boilers
		64 Utilize class C chilled water pipes instead of class B: Lower pressure rating but sufficient for school applications.
		65 Utilize galvanized iron (GI) grilles in lieu of aluminum grilles: More cost-effective and durable.
R50		66 Provide fast-acting air flow control valves (dampers) to Science rooms to match performance of the fume hood and general exhaust fast-active valves (dampers).
		67 Packaged rooftop units ILO four-pipe fan coil and DOAS system.
?		68 Ventilation-only cooling for MDF, IDF, and EMR.
		69 Clarify scope of work at existing - the BOD says all new controls throughout, plus upgrade to existing fan coil units to variable capacity.
		70 Recondition existing central plant chillers ILO replacement with new air-to-water heat pumps.
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Plumbing



FUNCTIONS:	1	Distribute	Fluid	5	Pressure	Fluid
	2	Convey	Waste	6	Neutralize	Compound
	3	Hygiene	Occupants	7	Repair	Existing
	4	Heat	Fluid	8	Modernize	Existing

2-word: Active Verb / Measurable Noun

Active Verb / Measurable Noun

CURRENT CONCEPT	\$ ROM		\$ ROM
Water distribution, copper	\$250K	Acid waste	\$20K
Drain, waste & vent, cast iron	\$250K	Storm drainage pump	\$25K
Plumbing fixtures	\$200K	Bldg 500 piping repairs	\$50K
Water heater, heat pump	\$50K	Minor TI work	\$35K
Water heater, electric, for Science	\$15K		
Elevator sump pump	\$5K		
Sanitary Waste	\$25K		

# leave blank	votes	ALTERNATE PROPOSALS
R51		1 PEX ILO copper water pipes
		2 Store and reuse rainwater
		3 Grey water to flush toilets
		4 Replace existing 4" Sanitary sewer with 6" in existing building ILO Force main sewer
		5 Heat-traced and insulated hot water lines ILO recirculating loop
R52		6 Non-metallic (ABS or PVC) drain, waste & vent (DW&V) piping ILO cast-iron (wrap with carpet pad for sound deadening)
		7 Additional plumbing fixtures for occupant comfort, health and well-being
		8 Gas-fired hot-water heater ILO heat pump
		9 Traditional electric hot water heater ILO heat pump
		10 Water-conserving plumbing fixtures ILO code minimum
		11 Micro-chemistry ILO traditional to eliminate need for acid waste neutralization
		12 Insta-hot water heaters at ganged toilet areas
R53		13 Eco-power faucets
		14 Waterless urinal
		15 Add amenities to the new sport fields including restroom, concessions and drinking fountains
		16 Provide support for future CTE program, such as compressed air
R54		17 Relocate restrooms and other plumbing fixtures for back-to-back or immediately adjacent space configuration to simplify the plumbing system and reduce cost
		18 Provide coffee bar for students at each of the large Flex spaces.
?		19 Provide rough-in for future sinks at one classroom per floor.
		20 Provide coffee bar at the Welcome Center.

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Plumbing



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Distribute Fluid	5	Pressure Fluid
	2	Convey Waste	6	Neutralize Compound
	3	Hygiene Occupants	7	Repair Existing
	4	Heat Fluid	8	Modernize Existing

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Plumbing fixtures	\$200K	Bldg 500 piping repairs	\$50K
Water heater, heat pump	\$50K	Minor TI work	\$35K
Water heater, electric, for Science	\$15K		
Elevator sump pump	\$5K		
Sanitary Waste	\$25K		

# leave blank	votes	ALTERNATE PROPOSALS
R55		21 Provide automatic trap primers at all floor drains.
		22 Reconfigure the building and plumbing system for gravity drain sewer connection (not to ejector station with forced-main). For example no basement level plumbing fixtures.
*		23 Relocate plumbing fixtures away from outside walls to reduce freeze risk.
		24 Vacuum drain, waste & vent (DW&V) system.
		25 Rain water harvesting (RWH) for flushing water.
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Fire



FUNCTIONS:	2-word: Active Verb / Measurable Noun		Active Verb / Measurable Noun	
	1	Suppress Fire	5	Endure Time
	2	Protect Occupants	6	Resist Corrosion
	3	Protect Structure	7	Facilitate Maintenance
	4	Resist Freeze	8	Obstruct Systems

CURRENT CONCEPT	\$ ROM		\$ ROM
Fire sprinkler, wet-pipe	\$300K		
Fire sprinkler, dry-pipe for canopy	\$35K		

# leave blank	votes	ALTERNATE PROPOSALS
		1 Upgrade fire alarm system to voice throughout existing facility
		2 Fogging fire suppression ILO traditional sprinkler system
		3 3M fire suppression fluid ILO water
		4 Fire sprinkler design by design team ILO bidder-design
		5 Linear heat detectors
		6 Aspirating Smoke detector system (vesda)
		7 Dry sprinkler system with oxygenator
		8 PVC sprinkler piping
		9 Two-story building to simplify fire/life-safety
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CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Electrical



FUNCTIONS:	2-word: Active Verb / Measurable Noun	Active Verb / Measurable Noun
	1 <u>Distribute</u> <u>Energy</u>	5 <u>Illuminate</u> <u>Space</u>
	2 <u>Generate</u> <u>Power</u>	6 <u>Improve</u> <u>Focus</u>
	3 <u>Capture</u> <u>Energy</u>	7 <u>Complement</u> <u>Daylight</u>
	4 <u>Power</u> <u>Devices</u>	8 <u>Code</u> <u>Compliance</u>

CURRENT CONCEPT	\$ ROM		\$ ROM
Devices and Gear	\$1,039M	Site	\$1.27M
Conduit, Wiring & Feeders	\$963K	Emergency Distribution	\$100k
Lighting	\$598K	Intercom/Clock	\$117K
Lighting Controls	\$299K	Classroom AV	\$299K
Fire Alarm	\$365K	Misc Systems	\$92K
LV Systems (Intrusion, Access Control, CCTV)	\$326K		
Data	\$352K		

# leave blank	votes	ALTERNATE PROPOSALS
E1		1 Upgrade existing Automatic Transfer Switch (ATS) in lieu of adding another Automatic Transfer Switch
R56		2 Aluminum feeders
R57		3 Re-Routing of power conduits outside in lieu of through kitchen area (and avoid internal disruption with pull boxes, (2) 4" conduits and (2) 2" conduits, and avoid risk and fit and conflicts inside. Finish with painted sheet metal on exterior to hide utilities.
*		4 Use smaller 75 kVA ILO 300 kVA transformer (requires enlarging electrical closets)
*		5 Add uplighting of trees, and buildings to improve aesthetics
R58		6 Add PV on covered walkways and canopies in addition to the building roof
R59		7 Add low pedestrian lights in ramp stairway area
*		8 Produce more realistic PV array layout, including maintenance pathways between rows with walk pads shown
R60		9 Increase solar panel energy generation by 20%
R61		10 Show future PV zone and provide conduit pathways
		11 New service to building addition
?		12 Route new power to new addition in conduits under drive lane (exterior) ILO thru Kitchen
?		13 Route new power to new addition in conduits surface mounted to exterior ILO thru Kitchen
		14 Metal-clad (MC) cable ILO cable in conduit for circuiting
		15 Hydrogen fuel cells
		16 Provide central emergency power battery inverter to augment existing generator.
		17 Micro-wind turbines
		18 Glass roof
E2		19 Battery Energy Storage System (BES) system for all emergency and standby power (no new generator); for the addition.
		20 Glass floor panels

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Electrical



FUNCTIONS:	2-word: Active Verb / Measurable Noun	Active Verb / Measurable Noun
	1 <u>Distribute</u> <u>Energy</u>	5 <u>Illuminate</u> <u>Space</u>
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LV Systems (Intrusion, Access Control, CCTV)	\$326K		
Data	\$352K		

# leave blank	votes	ALTERNATE PROPOSALS
		21 <u>Students on treadmills</u>
		22 <u>Add biomass generator</u>
		23 <u>Electric vehicle (EV) integration with building power system</u>
		24 <u>Micro-grid power generation, storage and distribution system</u>
		25 <u>Piezo-electric floor at main entry (to generate demonstration power) by student and staff circulation</u>
		26 <u>PV generating glass wall enclosure (and reduce heat gain)</u>
		27 <u>Really long extension cords from neighboring building</u>
		28 <u>Rain water harvesting with micro-hydro power generation</u>
		29 <u>Native direct current (DC) power distribution to light-emitting diode (LED) lighting fixtures and other DC power loads</u>
		30 <u>Extension cords ILO hard-wired circuiting for flexibility</u>
		31 <u>Pull-down extension cord reals</u>
		32 <u>Install raised-canopy with PV array (to create shade for small rooftop gathering space)</u>
		33 <u>Building-integrated PV including vertical walls with good solar insolation (exposure)</u>
		34 <u>Use surface-mounted LED panels or LED troffers in lieu of 2'X2' RECESSED FLAT PANEL LED FIXTURE</u>
		35 <u>Centralized power poles</u>
		36 <u>Reduce power and data outlets.</u>
		37 <u>Switch to LED fixtures with selectable wattage and CCT: Reduces SKU count and allows post-installation tuning.</u>
		38 <u>wireless occupancy and daylight sensors</u>
*		39 <u>Standardize fixture types across zones: Simplifies design, installation, and future replacements.</u>
E3		40 <u>All 208V power at addition (new) building to simplify distribution and eliminate transformer loss</u>

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Electrical



	2-word: Active Verb / Measurable Noun	Active Verb / Measurable Noun
FUNCTIONS:		
1	Distribute Energy	5 Illuminate Space
2	Generate Power	6 Improve Focus
3	Capture Energy	7 Complement Daylight
4	Power Devices	8 Code Compliance

CURRENT CONCEPT	\$ ROM		\$ ROM
Devices and Gear	\$1,039M	Site	\$1.27M
Conduit, Wiring & Feeders	\$963K	Emergency Distribution	\$100k
Lighting	\$598K	Intercom/Clock	\$117K
Lighting Controls	\$299K	Classroom AV	\$299K
Fire Alarm	\$365K	Misc Systems	\$92K
LV Systems (Intrusion, Access Control, CCTV)	\$326K		
Data	\$352K		

# leave blank	votes	ALTERNATE PROPOSALS
		41 Natural gas fueled generator ILO diesel
		42 Use Spider boxes
*		43 Use cable trays instead of conduit where allowed: Easier to install and modify
		44 Use LED string lights under canopy
		45 Reduce quantity of lights - increase lumen output
		46 Add kinetic pavers to generate power for lighting.
		47 In pavement flashing lights at crosswalks
		48 Clap-on Clap-off light controls
		49 Add mirrors to reflect sunlight into the space
		50 Utilize a highly reflective floor surface
		51 Utilize battery operated desk lamps
		52 Add light shelves
		53 Add transom windows from the corridors
		54 Create an atrium 3 story space with windows
		55 Light tubes
		56 Coordinate with PSE and enter the addition only into the "Coordinated Demand Response" program and reduce energy bills. Voluntarily implement energy load reduction with intelligent load shedding system during predicted peak demand.
!!		57 Coordinate with PSE and enter the entire Campus into the "Coordinated Demand Response" program and reduce energy bills. Voluntarily implement energy load reduction with intelligent load shedding system during predicted peak demand.
*		58 Enhance pedestrian lighting with low level lighting along walkways
		59 Tunable lighting @ DLC
		60

CLIENT: Sumner-Bonney Lake School District
PROJECT: Bonney Lake High School Expansion & Intersection Improvements
COMPONENT: Low Voltage



FUNCTIONS:	1	Alert	Occupants	5	Educational	Tool
	2	Secure	Access	6	Amplify	Sound
	3	Enhance	Safety	7		
	4	Support	Learning	8		

CURRENT CONCEPT	\$ ROM		\$ ROM
Fire Alarm	\$365K	Intercom/Clock	\$117K
LV Systems (Intrusion, Access Control, CCTV)	\$326K	Classroom AV	\$299K
Data	\$352K	Misc Systems	\$92K

# leave blank	votes	ALTERNATE PROPOSALS
*		1 Add magnetic door hold opens on classroom doors tied to a panic button
		2 Wireless data only
		3 Battery operated clocks
		4 Bull horn for announcements
		5 Combine the intercom with the fire alarm for alerting
R62		6 Add large format video/digital wall display in Welcome area (like Sumner SH in commons but smaller)
*		7 Utilize the digital wall for warnings - fire, security, etc.
		8 Virtual glasses
		9 Augmented reality with mixed reality
		10 More wireless
		11 Guard dogs
		12 Digital tracking of visitors
R63		13 Upgrade existing fire alarm system to all voice
		14 Lighted text signs in every classroom for hearing impaired announcements
		15 Cell signal damping systems
		16 Vape sniffing devices in bathrooms and entry areas
		17 Facial recognition equipment for perimeter CCTV
		18 USB charging outlets
*		19 HDMI AV connection drops on both sides of classroom
R64		20 Add conduit to existing building intercom system